Preparation of Project Technical Reports for the IREC

Team 91 Project Technical Report for the 2026 IREC

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CU Up There **is a high-power rocket designed, built, and tested by Clemson University Rocket Engineering (CURE) to compete in the 10k COTS category of the 2026 International Rocket Engineering Competition (IREC). The vehicle features student researched and developed (SRAD) fiberglass and carbon fiber airframes, a triple redundancy, dual bay, dual deployment recovery system, a carbon fiber plate spar system locked in by radially mounted aluminum hardstops, and an SRAD flight computer equipped with live telemetry. The rocket supports three payloads: a ballast, an SRAD electronics bay, and a 3U CubeSat experiment demonstrating hemolysis on porcine blood. A successful test flight verified the comprehensive vehicle design/build and our assembly/integration process. Multiple ejection tests were performed for both drogue and main recovery systems to verify charge sizing. Telemetry distance tests were performed to verify downlink integrity. The launch of *CU Up There* at 2026 IREC will mark** the **team's continued steps towards modularity and comprehensive engineering design.**

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I. Nomenclature

f, g = generic functions
MARTHA = Miniaturized Avionics for Rapid Testing, Handling, and Assessment
PCB = Printed Circuit Board
USB = Universal Serial Bus
SPI = Serial Peripheral Interface
UART = Universal Asynchronous Receive Transmit
USART = Universal Synchronous/Asynchronous Receive Transmit
ADC = Analog to Digital Converter
SWD = Serial Wire Debug
GPS = Global Positioning System
EMI = Electromagnetic Interface
RF = Radio Frequency
W = Watts
V = Volts
dB = decibel
AHRS = Attitude and Heading Reference System

II. Introduction

Clemson University Rocket Engineering (CURE) is bringing forward a solid propelled rocket, *CU Up There*, to compete in the 10K COTS category. In light of the team's previous success in the form of its first ever podium rocket at IREC 2025, CURE aims to carry this momentum forward, making the key changes towards improving the design in *CU Up There* in terms of assembly, flight dynamics, and modularity of the recovery system.

CU Up There will be utilizing a COTS M2500T motor and nose cone, while the rest of the vehicle's aerostructures including air frames, couplers, bulkheads, and internal mounting structures are SRAD. The propulsion section features four fins that have a swept-back trapezoidal design. The rocket is propelled from a fin motor section and provides enough force to launch the vehicle's modular spar system. These spars stretch from the nose to the top of the recovery's main parachute bay, allowing for the mounting of the vehicles 3U payload, ballast, and non-essential flight electronics. Radially mounted SRAD aluminum ring "hard stops" were developed to lock in a bulkhead to the vehicles air frames supporting the base of this spar system. A threaded rod extending from the top of the spar system allows for simple integration of the vehicles' nose cone through mechanical fastening of the nose cone tip. This allows for very simple removal of the vehicles nose cone providing easy access to any items or payloads mounted to these internal spars.

To support future modularity of the CURE's competition and research vehicles, a dual bay recovery system was developed in *CU Up There* (as opposed to our 2025 IREC entry, *Pocket Tens',* single bay nose cone design) and was integrated just above the rockets motor section. The bay itself houses a triple redundant design, featuring 3 COTS altimeters where redundant deployment triggers are staggered alongside a GPS that is also located in the bay. On the pad, the recovery system is energized via key switches that can be easily actuated with keys from outside of the rocket.

As previously mentioned, the internal spar system is utilized to mount *CU Up There's* main bay: ballast, non-essential flight electronics, and payload. The ballast weight section allows for CURE to manipulate the total mass *CU Up There* on launch day to reach the desired stability and apogee. The non-essential flight SRAD electronics features CURE's SRAD data collection printed circuit board (PCB) titled Miniaturized Avionics for Rapid Testing, Handling, and Assessment (MARTHA). MARTHA collects various data of the vehicle's flight characteristics including attitude, altitude, velocity, inertial, and magnetometer data as well as connecting to any payload-specific sensors. *CU Up There* will be transmitting from a COTS long range (LoRa) transceiver placed in this bay alongside a redundant GPS. An SRAD ground station comprised of a matching LoRa modem and a computer is configured to receive this transmitted data. Live flight data that is received is then displayed by the ground station to the user through a custom user-interface which features a 3-D orientation visualizer, altitude and acceleration graphs, and a downlink integrity meter.

CURE aims to leverage its allotted 3U Payload section to complete research on the effects that high-G flight has on the characteristics of blood. With the demonstrated interest in conducting transfusion procedures in Low Earth Orbit (LEO), medical technologies will need to adapt to be able to handle the harsh high vibration and high acceleration environments that come with reaching space[1]. CURE will be completing this investigation through the launch of various samples of porcine blood, a known transfusion supplement for humans, along with surrounding data collection electronics, all of which will be fully enclosed within the machined aluminum 3U allocation.

The success of CURE's previous 2025 IREC 10K COTS vehicle, *Pocket Tens*, which placed third in the category, paved the way for the teams' goals for 2026 IREC. These amassed to five major objectives: improve ease of vehicle integration, utilize a functional ballast, reduce vehicle mass, redesign the recovery bay to be smaller and dual split bay, incorporate SRAD live telemetry, and facilitate a successful test launch (Figure 1).



Figure 1 *CU Up There* Test Launch March 7th, 2026.

Clemson University Rocket Engineering is a primarily undergraduate, delegated student organization (DSO) at Clemson University. All members of the club participate as an extracurricular activity. We operate in Clemson's Mechanical Engineering Student Shop in Cook Engineering Laboratory, which provides the necessary tooling for most of the team's manufacturing processes. We launch in Dalzell, South Carolina, as part of the Rocketry South Carolina.

Our primary financial contributors include several Clemson organizations, including the Clemson University Student Funding Board, Department of Mechanical Engineering, and College of Engineering, Computing, and Applied Sciences. Outside of Clemson, we receive material donations from Chomorat, who have gifted us several rolls of carbon fiber. Additionally, our largest sponsorship category is software, for which we receive discounts and free licenses. Our software sponsors include: AirTable, Slack, Altium, Ansys, and Onshape. Finally, our flagship sponsor is Aeronix, Inc., an embedded systems company based in Melbourne, Florida.

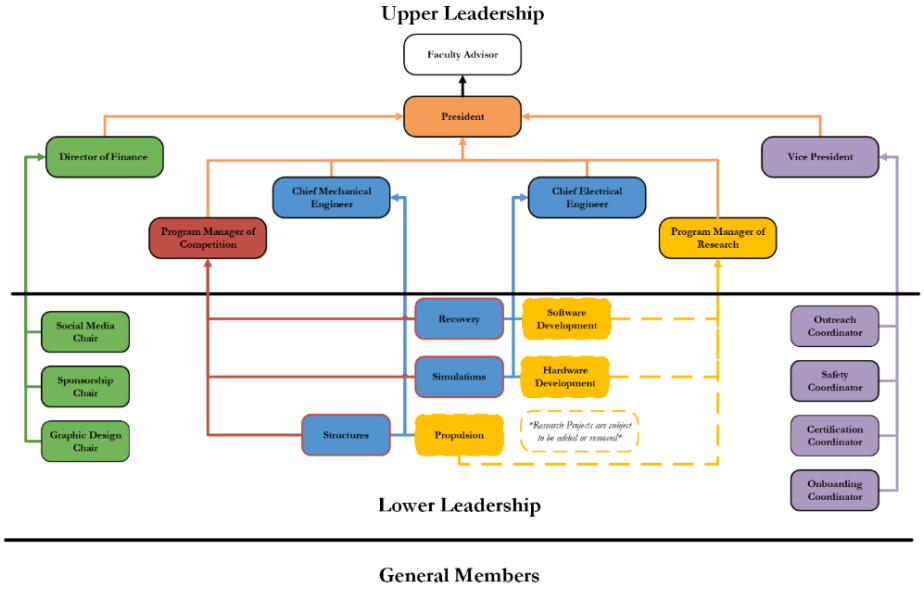


Figure 2 Organization Chart

The team structure, as presented in Figure 2 consists of three tiers: Upper Leadership, Lower Leadership, and General Members. Upper Leadership each runs their own division and is overseen by the President. The Vice President oversees the administrative division, which operates our onboarding, certification, safety, and outreach programs. The Director of Finance oversees the financial division, which manages our social media presence, procurement process,

and sponsor portfolios. The engineering division is by far the largest and thus is overseen by both Chief Engineers and Program Managers. The Chief Engineers operate with a focus on pushing forward the engineering initiatives and goals throughout the year. Program Managers are designed to handle administrative tasks pertaining to the engineering division, acting as the interface between Upper Leadership and Lower Leadership. Each division meets weekly to discuss progress and direct focus on specific initiatives. Engineering Lower Leadership holds 1-3 weekly meetings for their subteam and Administrative Lower Leadership holds meetings as they see fit.

The technical report was prepared by 13 members of the organization over the course of 4 weeks. The President and subteam leads oversaw the completion of the report and organizing delegation of sections of the report. The technical report followed the workflow as shown in Figure 3. Internal reviewers are members selected by the President and serve to review the correctness of the technical, grammatical, and formatting of the report. External reviewers are alumni of the team selected by the President and serve to review the quality of analysis. As sections of the report are completed by subteam members, they progress through the stages of review until satisfactory.

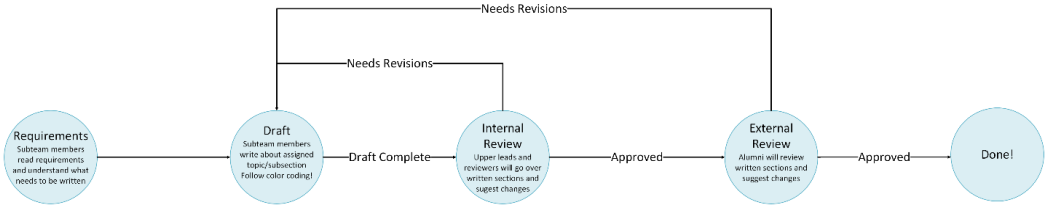


Figure 3 Technical Report Workflow.

III. System Architecture Overview

CU Up There is a 6" diameter launch vehicle standing at a height of 116 in and lift off mass of 62.2 lbs. Its mission is to successfully reach a target apogee of 10,000 ft, successfully carry out a biomedical inspired research experiment as its payload, and support a live telemetry channel of flight data to an SRAD ground station. Our SRAD flight computer, MARTHA, both logs and transmits multiple flight parameters to both record and provide live flight data. The ground station has a custom live telemetry visualization to provide 3D orientation and flight profile graphs during the duration of the flight. *CU Up There* will enter the 10K COTS competition with an AeroTech M2500T motor as its single stage. A CAD render of *CU Up* can be observed in Figure 4 below.

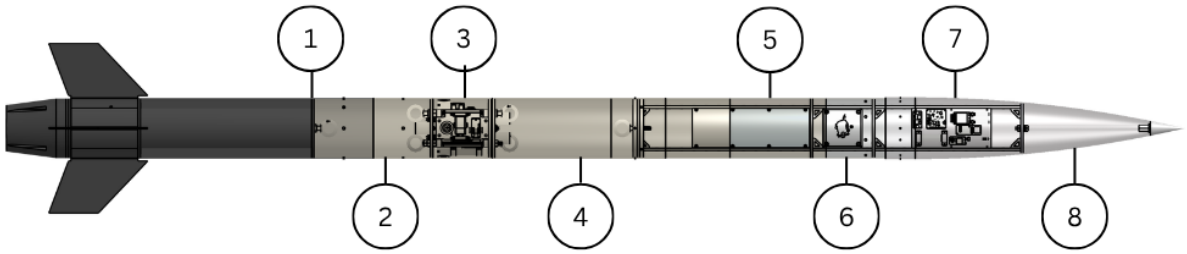


Figure 4 Full Section View of *CU Up There*

As annotated by numbers in Figure 4, the sections of *CU Up There* from left to right are as follows: a motor section (1), a drogue parachute bay (2), recovery electronics (3), a main parachute bay (5), a 3U CubeSat payload (6), a ballast (7), SRAD avionics and telemetry, and a nose cone section (8). *CU Up There* has already been mission tested with a lower impulse motor, due to the site's waiver, validating its flight critical capabilities and live telemetry streaming.

A. Aero-structures Subsystems

Objectives and Overview

The aero-structures subsystem is designed and manufactured by our structures sub team. It houses and protects the recovery of hardware, payload, and SRAD electronics from external forces during launch. The system permits integration of all projects within an internal modular spar system. The aerostructure of the vehicle is composed

primarily of composite material. The airframe, and couplers are all 120 E-glass fiberglass. The bulkhead, spars, and fins are constructed of quasi-isotropic carbon fiber plate, cut by water jet. The motor section airframes are constructed of hand layup carbon fiber. Composites are selected for their superior strength-to-density ratio as compared to other teams' selections such as aluminum, validated by Ashby material selection [2], with all sections containing recovery electronics additionally being restricted to fiberglass construction for RF transparency, as required by DTEG 8.2.1 and 8.3.1. To innovate from last year, the aero-structures primary objectives are as follows:

1) Reduce weight in the airframes by reducing structure thickness from .125" to 0.094" by changing the number of fiber wraps from 4 to 3.
2) Increase ease of integration through fabrication of continuous spars without breaks between sections.
3) Protect all recovery hardware from external elements throughout the duration of the flight.
4) Properly constrain the M-2500 motor during flight such that no components are damaged due to thrust.
5) Ensure effective separation of rocket into three components during drogue and main parachute deployment such that zippering is avoided.

Motor Section

The motor section is tasked with protecting the M2500 motor from damage during the flight and ensuring that thrust is delivered properly to the vehicle throughout the duration of the rocket's ascent. A cutaway view of this motor section can be found in Figure 5 below.

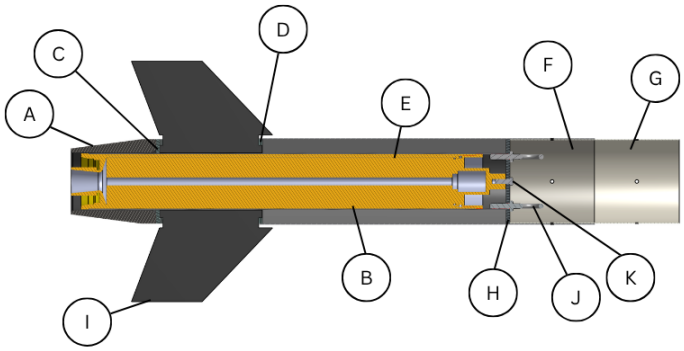


Figure 5 Cutaway View of the Motor Section

Figure 5 details the motor section architecture. The motor is friction fit within a carbon fiber airframe: the inner motor tube (E). Two carbon fiber centering rings (D) are epoxied around the inner motor tube such that 4 1/8" carbon fiber fins (I) could be placed at 90° from each other, which are in turn epoxied to the centering rings. A 6" outer diameter carbon fiber bulkhead (H) is epoxied to the top of the inner motor tube, such that two 7/16" eyebolts (J) are permanently installed for shock chord attachment for the drogue parachute. A 3/8" bolt secures the motor into the inner motor tube. A bulkhead (C) is placed beneath the bottom centering ring such that the nylon boattail (A) can be secured during flight, delaying flow separation and curtailing the profile drag. A 6" inner diameter carbon fiber airframe named the outer motor airframe is placed over all the internal components to protect them from the elements and make section flush with other airframes (F). To ensure a final 1/4" fin thickness with increased factor of safety against fin flutter, a tip-to-tip layup of additional carbon fiber in sequential 0/90, 45/45, 90,0 orientations is added. When loading the motor section of the rocket, the loaded casing slides into the base section and is held by forward retention in the form of a bolt through the top motor bulkhead. Two rail guides are placed on the exterior of the airframe to allow for proper dismount from the rail. The first rail guide is placed 73 inches below the tip of our rocket. This is to ensure it is no less than 1 inch above our center of gravity to prevent a moment upon launch. The second rail guide is placed 34 inches below the previous which centers it relative to the fin root chord.

Fins

The fins are responsible for the aerodynamic stability of the rocket. Four 1/8” quasi-isotropic carbon fiber plate modified trapezoidal fins are utilized in the motor section. As mentioned in the previous section, the fins are subject to tip-to-tip layups to increase the total fin thickness to 1/4". This grants the fins a factor of safety of 3.85 for flutter and 2.13 for aeroelastic divergence. The fin geometry design from previous years changed drastically with respect to overall fin size for *CU Up There*. *CU Up There's* fins were drawn up after the major design decisions including the rocket's diameter, distribution of mass, and rockets height were almost completely finalized. This allowed for fin design to be utilized for “fine tuning” of the simulations and ultimately leading to a much more efficient and light weight design than the previous year. Figure 6 below shows a size comparison of the fins between *CU Up There* and *Pocket Tens* (CURE's IREC 2025 vehicle).

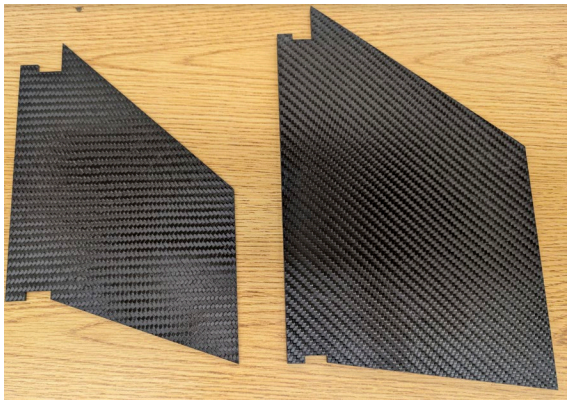


Figure 6 Fin Side By Side Comparison, *CU Up There* (Left) and *Pocket Tens* (Right)

The fin design process was the last physical component of the rocket. This allowed for an iterative process between stability, apogee, and fin strength allowing for the best possible design. Both OpenRocket and our Fin Flutter Analysis calculator were used in this process. The target requirements for the parameters listed above were a stability between 1.8-2.2 cal, an apogee between 9950-10250 ft, and a fin factor of safety above 2 for both flutter and divergence.

Recovery

A designated recovery section is deemed critical by the team to ease integration and protect recovery electronics from forces during launch. Having a specific section of coupler and airframe where the recovery bay resides creates logical areas for parachute bays directly above and below these electronics, thus achieving the team's goal of a dual bay, dual deploy recovery system. Achieved through precise cutting of the over-lapping composite wall, the radially facing key switches of the recovery bay are accessible from the outside. One iteration on last year's design was the height of the bay now being low enough in the rocket such that key switches are accessible without a ladder. A carbon fiber ring was fabricated to function akin to a hard stop such that the recovery bay bulkheads can be clamped onto this ring and the 5.8” inner diameter coupler spanning exactly 6” distance. The external diameter of the coupler fits within a fiberglass recovery airframe. Through flight, the load path travels through the outer airframes. After parachute deployment, tensile load from the harnessing system is dissipated through the eye bolts, bulkheads, and into the threaded rods of the recovery assembly. Figure 7 is provided to visualize the stages of recovery section integration, with first the carbon fiber ring being secured, the recovery bulkheads being installed, and then finally the recovery electronics being put into place, key switches facing outwards.

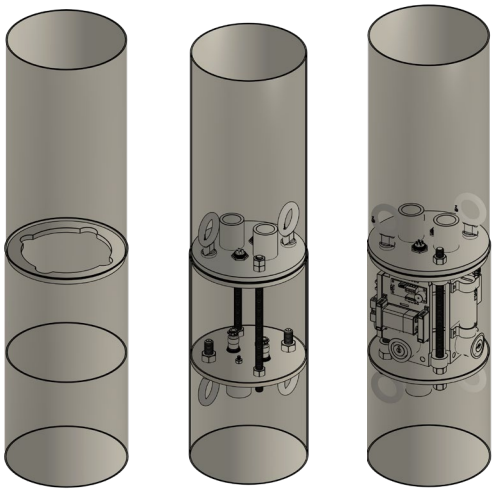


Figure 7 Recovery Bay Functional Views: No Recovery Bay(left), Threaded Rod Assembly (Middle), and Fully Mounted Recovery Bay (Right)

The layout of having two threaded rods to support the structure of the recovery bay was chosen based on maximizing the internal space and bulkhead area. The axial force on the minor diameter of the threaded rod and the pull-out force on the threads of the rod was considered in the selection of the McMaster-Carr 3/8”-16 x 8 inches long Grade B7 Medium Strength Steel threaded rods and Zinc-Plated Medium Strength Steel Hex Nut assembly. Through those calculations, noted in Appendix A, it was found that the max allowable force considering the minor diameter is 7,806.7 lbf and the max allowable force for the threads is 5,659.7 lbf. With our expected max force on the recovery bay being at main deployment calculated at 1,091.11 lbf, calculations in Appendix A, the factor of safety is greater than 5 times the allowable force on the threaded rods.

Hard Stop Section

The hard stop section is designed such that force from main parachute ejection does not damage the payload, ballast, or SRAD projects above it. Furthermore, the bulkhead in the middle of the section serves as the base to mount the internal spars system. The mentioned “hard stops” specifically refer to the two machined aluminum rings that are placed on either side of a carbon fiber bulkhead, then radially bolted into the airframe. Two more 3/8” eyebolts are also secured into the bulkhead but facing the recovery section such that shock cords can be attached to them. Both rings and the bulkhead are 5.8” in outer diameter and are mounted within the overlap fiberglass coupler and airframe fastening the assembly together. Shear pins are placed at the bottom of this hard stop airframe such that separation is smooth upon main deployment. Below, Figure 8 shows the assembled hard stop section and disk-bulkhead system respectively.

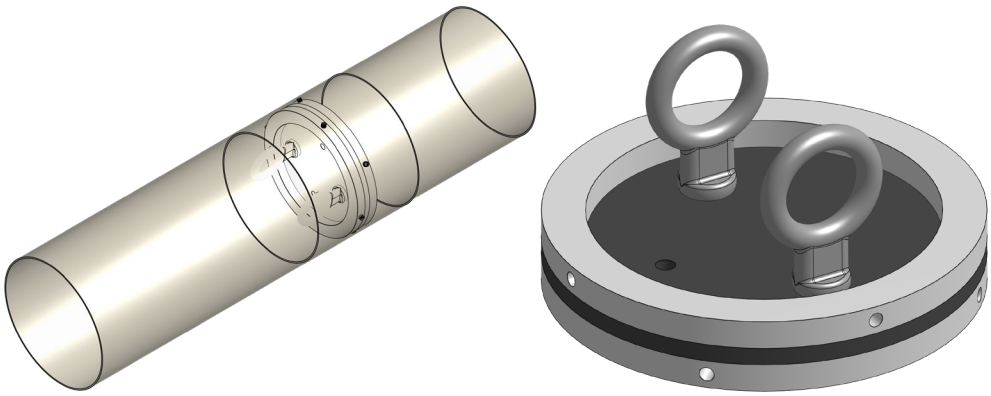


Figure 8 Aluminum Ring Hard Stop Section

Another key function of this hard stop assembly is also to seal gasses of the main parachute bay from the payload bay of the vehicle, thus allowing a proper seal for ejection charges. To guarantee a sufficient seal a custom gasket, cut from 1/8" rubber sheet, with matching profile of the lower hard stop ring placed between the lower hard stop ring and the bulkhead between. This prevents gasses from navigating through the section into the payload bay. To evaluate the design of the hard stop section and ensure that it could withstand the ejection forces of the launch vehicle, FEA was performed.

Internal Spar System

The internal aerostructures of the rocket consist of a modular spar system, such that the ballast, payload, and SRAD electronics can be securely fastened within the airframes of the vehicle during flight. Each project is mechanically fastened into the spars, such that assembly within this section is not permanent and highly modular. The spars are carbon fiber, lined with holes for ¼-20 bolts. The spar system begins at the hard-stops and terminates at a bulkhead with an attached threaded rod which is dedicated to mounting the nosecone. Each project, ballast, payload, and SRAD are separated by a carbon fiber bulkhead with an outer diameter matching the inner diameter of the airframes. The bulkheads and spars are fastened through 1"x1"x1" aluminum gussets. The spar-bulkhead-gusset design allows for custom heights of each section, as well as simple removal of any component for pre-launch preparation. In total, the internal spar system consists of two spars spanning 38.5", four bulkheads, and 8 gussets. Figure 9 and Figure 10 demonstrate the unloaded and fully integrated spar system respectively.

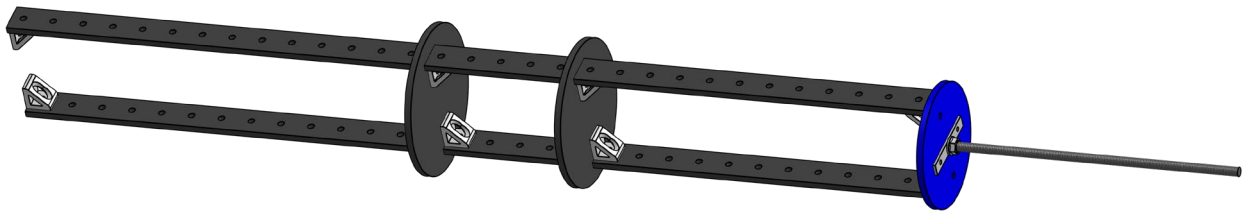


Figure 9 Internal Spar System

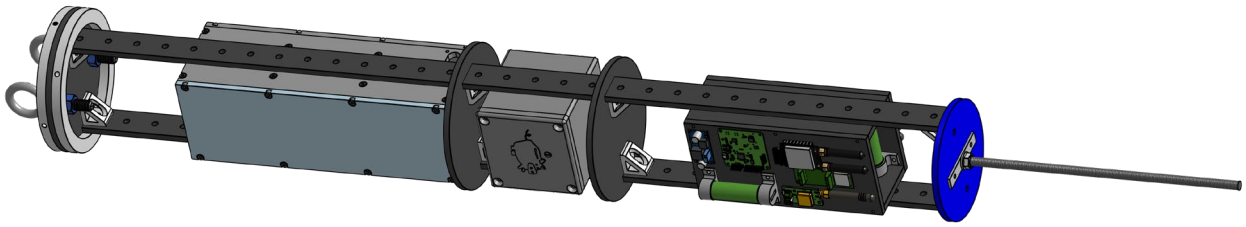


Figure 10 Spar System with attached Payload 3U CubeSat, Ballast, and SRAD Electronics/Telemetry

The spar system creates unmatched modularity for *CU Up There's* payloads by allowing for any payload that can adapt to the spar hole spacing and the bay diameter to be carried by the vehicle. Furthermore, the team's utilization of a threaded rod assembly extending from the spar system creates a clever method of attaching the nose cone of the rocket.

Nose Cone

As previously mentioned, the nose cone of *CU Up There* is mechanically fastened to the rest of the vehicle's aerostructures through a threaded rod that extends from the top of the internal spar system. The nose cone tip has a tapped threaded hole allowing for the tip of the nose cone to be mechanically fastened to this threaded rod. The nose cone itself will press against the top bulkhead of the spar system from force provided by the threads of the nose cone tip. Figure 11 displays the 3D rendering of the nose cone alone as well as a visual of its fastening to the spar system via this threaded rod.

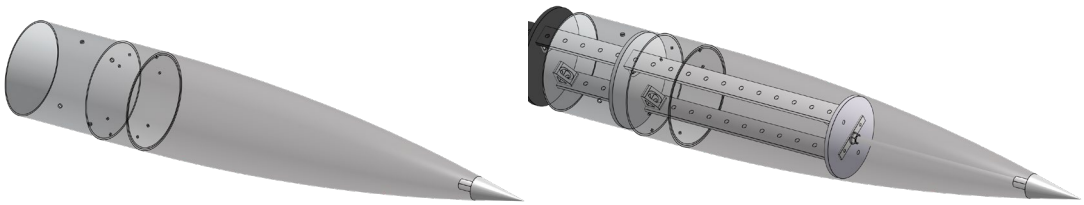


Figure 11 3D Model of COTS 4:1 Nose Cone (Left) and Assembly of Nose Cone Attachment (Right)

The nose cone itself is a COTS 4:1 Tangent Ogive nose cone, the ogive being fiberglass while the tip is aluminum. The nose cone and its assembly provide a couple of other key benefits, especially regarding space. By utilizing a smaller bulkhead and the threaded rod assembly, the spar system can extend much further into the nose cone of the vehicle allowing for much more efficient utilization of the spar systems mounting range. Additionally, the nose cone and main body tube can be removed by simple fastening of the nose cone tip allowing for ease of access to the payload bay and electronics. This drastically increased day of launch integration capabilities. While the team did not utilize the space this year, with iteration of this design, the thread rod in the nose cone creates another optimal mounting point for electronics or other payloads with various capabilities.

B. Propulsion Subsystems

The propulsion system is a COTS 98mm solid motor from AeroTech's RMS-98/10240 series, classified as the M2500T by AeroTech. The total impulse is 9,055.9 Ns, which is approximately 88% of a maximum M motor. This motor was selected due to the team's experience designing around it and creating an airframe to withstand the forces. The thrust curve of the motor provides the team with a comfortable 2,488 N average thrust, as well as a peak thrust of 4,080.5 N. With the weight of our rocket as 45.44 lbs weight with a full ballast, it provides a comfortable margin for a thrust to weight ratio of 13.9 TWR. Last year, the team used the M2500T motor and found it to be weight restrictive, creating a small margin of error for safe weight variations. This year, the team downsized the mass of the vehicle to prevent this issue from reoccurring, allowing the same motor to be used on a lighter rocket to provide a larger margin of error for weight variations. In the future, the team plans to move to the SRAD category as experience and testing of hybrid propulsion systems grows.

C. Avionics

The avionics system is developed by our hardware and software subteams. It gives the flight vehicle the ability to collect data during flight and supports all the other research and test vehicles the organization works on.

The hardware development team designs and builds the data collection hardware for the flight vehicle and our other vehicles. They work closely with the software development team on joint projects and help other subteams with custom electrical systems. The team created the MARTHA avionics platform, which handles in-flight data collection for the flight vehicle and telemetry transmission to a ground station. They have now completed the Just an Expanded MARTHA (JEM) board to move to a more advanced flight platform. The team has also started working on a new communication board for telemetry connections to the MARTHA or JEM, a power distribution board to power our electronics, and a parachute deployment board that tests continuity. This year the team developed the ability to do in-house assembly, which helps them build and test hardware faster and at a cheaper cost. This process involved ordering all components, the PCB, and a stencil. Solder paste is then applied, components are placed, and then a heat source such as a heat gun or reflow oven is used to solder components. There were multiple issues with the process, and we hope to improve upon it next year.

The main function of the flight computer is to both log and transmit flight data. A full flight profile is saved to onboard storage, including altitude and acceleration curves. This data is also transmitted to a ground station with a LoRa radio modem. The telemetry link allows commands to be sent to the computer, such as restarting the telemetry stream. The ground station receiver also displays live flight profile data, including a live 3D orientation of the vehicle and live altitude/acceleration graphs. The base station also includes a signal integrity readout. This system allows us to analyze vehicle performance both in real time and post flight, providing valuable insight into engineering decisions.

D. Recovery Subsystems

*CU Up There'*s recovery bay has marked many large developments in CURE's overall recovery design. CURE wants to fly different motor section designs, including ones that adapt to SRAD motors, for either competition or research. While all rockets have a recovery bay, they may not have the same motor or payload subsystems. Therefore, to improve future modularity, qualification, and decouple the recovery section of the vehicle from the propulsion and payload subsystems, we sought to develop an insular modular recovery bay that is agnostic to the greater system of the vehicle. An agnostic recovery bay means that a specific propulsion configuration, whether that is COTS solid or SRAD hybrid, does not require co-design of an upper payload section, and vice versa. Additionally, an insular recovery bay improves recovery reliability and assurance, as it physically and electrically isolates recovery systems from other COTS and/or SRAD systems. CURE continues to promote modular design to further allocate resources to other areas of research.

As alluded to, the main and drogue parachute bays exist directly above and below an electronics bay that contains three COTS altimeters and one COTS GPS module completing this full dual bay, dual deploy design. Two 8-Pin aviation connectors are utilized to transfer charge signals through each of the upper and lower bulk heads, where each connection terminates a primary or secondary charge located in charge wells that are mounted on the opposite side of the bulkhead in each parachute bay. This year, the design leads requested the recovery electronics to fit into a bay of 6" or less with the goal of optimizing packaging. This is nearly an 8" decrease from the previous year's design. To accommodate this much smaller space and still support the use of redundant altimeters, CURE's recovery team decided to create a modular recovery bay. By utilizing the structural threaded rods passing directly through the bay to mount to, an internal structure can be designed to mount on these rods. Specifically, one that houses the electronics and can be fully swapped without altering the structural integrity of the bay. This internal structure will provide nothing to the overall load path of the rocket, but the swappable nature of this bay is where this modularity is derived.

Internal Bay Design

The internals of the recovery bay specifically refer to the organizational pieces that mount to the threaded rods within the bay (visualized in Figure 12 in section A). Again, these internals provide nothing to the overall load path of the rocket and are used to mount the necessary electronics in the bay. After two design concepts featuring plates that were oriented laterally to the body of the vehicle (and then stacked within the bay), the team decided that wiring would be significantly easier if the plates were oriented axially with a central pathway for the wiring. Due to weight considerations and the desire to make the recovery bay as modular as possible, the team decided to manufacture the structure with nylon 3D-printed parts. The material chosen for the internal design was decided to be nylon due to its heat-resistant properties and durability when compared to other 3D printable materials. The final design is a hollow bay with protective plating surrounding the hardware. The structure has four panels to hold each of the devices with the wiring residing between the panels. Each of these plates are mechanically fastened and can be removed to access the inside wires between the panels. Since the GPS module does not have wires attached to the aviation connectors, its entire circuit is isolated on its respective plate, meaning it's easily removable to access the inside wires. The switches mount to the plates as well allowing for the key face to be accessed radially. A CAD visualization of these four panels with selected electronics can be observed in Figure 12 below.

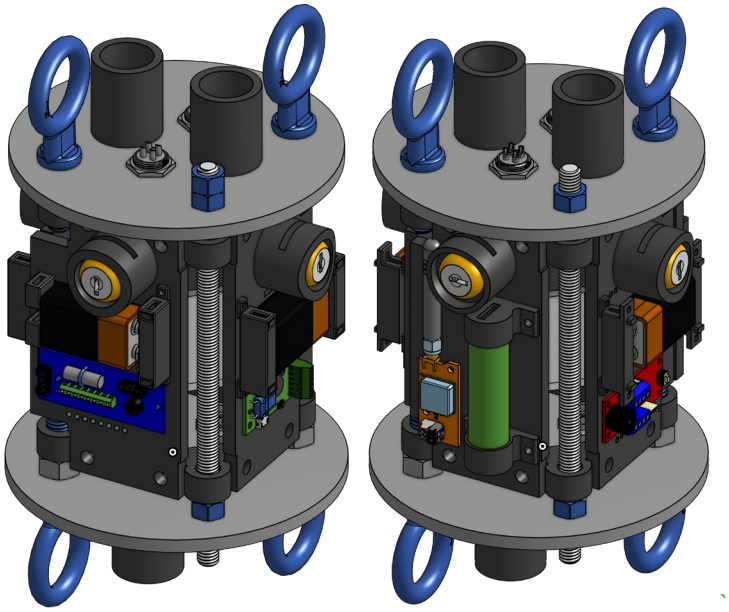


Figure 12 Recovery Bay, Front Panels View(Left) Back Panels View(Right)

The batteries for the bay were secured with standard 9V clips, but to constrain movement further and reduce vibrations, an upper and lower brace was printed onto the plate on either side of the battery. This ensures that the altimeters wouldn't lose power from disconnection during the flight, which was determined as an issue for previously utilized battery connectors in previous years. Finally, the batteries were locked into place with zip-ties constraining each of the battery movements radially. Each of the three altimeters and GPS were mounted to the panels with heat set inserts. The team prepared the wires for the primary and secondary wiring systems ahead of soldering by making "twisted pairs" of negative/positive leads for each charge. This was done by twisting the wires tightly together and securing them with heat shrink tape. The twisted pairs make it easier to assemble or disassemble the recovery bay within the bulkheads. Twisted pairs also are helpful in reducing electromagnetic interference (EMI) noise on the signals on each wire. Each of these twisted pairs leave its specific altimeter, exiting behind its respective panel, and connect to its specific aviation connector that is attached to either upper or lower bulkhead. When installing the bay into its section of coupler and air frame, the upper bulkhead is removed leaving behind hanging wires and aviation connectors. Following, this recovery assembly (with removed upper bulkhead) is slotted into the section of airframe as one piece until it is resting on the coupler inside. The top bulkhead would be later installed from opposite side to the airframe and coupler assembly till it rests on the mounted ring as visualized in Figure 12. Given there is a 15-inch-tall section of the airframe coupler, the previously mentioned hanging wires would need to extend long enough so that the aviation connectors fastened to the top bulkhead can be connected prior to installation of this top bulkhead. These long, twisted pairs were identified by the designers to be a possible issue upon assembly because these wires could snag or bump into other wires when being pushed into the bay upon installation. The solution to this was for there to be a cuplike structure built into the bay that would hold the excess wire length and prevent it from interfering with other systems. This serves as a reservoir to house excess wiring and separate it from the main cavity. This also restricts movement, so no wires come loose or interfere with previously installed components. Figure 13 below visualizes the cup structure with the panel removed.

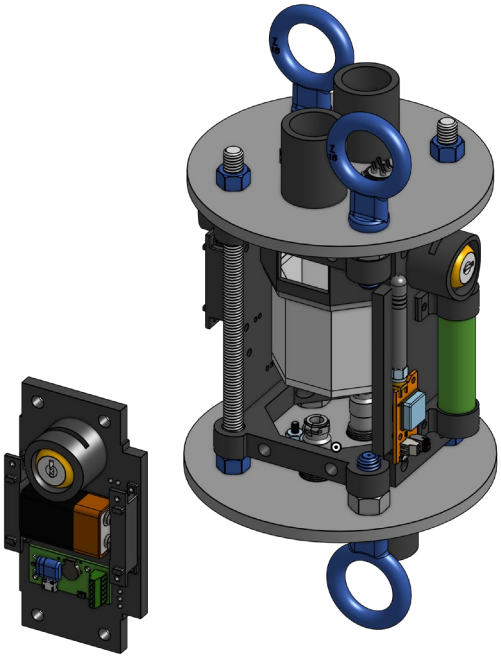


Figure 13 Bay with Panel Removed, Revealing Wire Catch Cup

The selected wires for these twisted pairs are stranded core 22 American Waire Gauge (AWG) for twisted pairs. These were selected over solid core wire for extra flexibility and commercial availability. It also should be noted that each of the panels has a raised mounting point for the key switches. These serve to raise the face of the key switch to the edge of the airframe. With holes cut into the overlap of the coupler and airframe at the location of each key switch, arming of all electronics can be completed simply with unmodified keys from the outside of the rocket.

Recovery Wiring

CURE adopted a recovery electronics architecture for IREC 2025 that closely follows its previous design, with the primary similarity being triple-redundant altimeter-based deployment. While the team has not previously integrated the Blue Raven altimeter from Featherweight Altimeters, it was selected as the tertiary system to enhance overall redundancy. The primary altimeter is the RRC3+ from Missile Works, and the secondary altimeter is the Stratologger CF from PerfectFlite, both of which have demonstrated reliability in prior competition flights.

As shown in Figure 14, the primary altimeter (RRC3+) is connected via an e-match to the primary drogue and main black powder charges, programmed for deployment at apogee and 1000 ft, respectively. The secondary altimeter (Stratologger CF) is connected to independent drogue and main charges via e-matches, with deployment set to occur one second after apogee and at 950 ft. The tertiary altimeter (Blue Raven) is connected to a second e-match tied to the primary black powder charges, providing an additional deployment trigger at two seconds after apogee and at 900 ft.

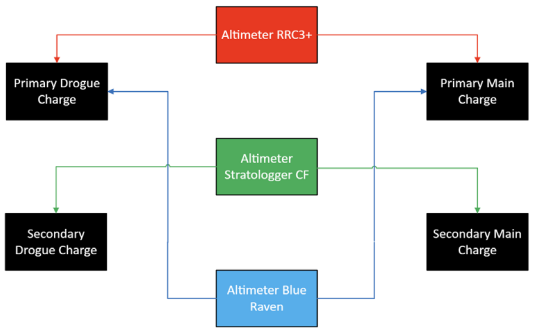


Figure 14 Simplified Ejection Charge Diagram



Figure 15 Simplified Power Diagram

A simplified power distribution diagram is shown in Figure 15. Each altimeter operates on an independent series circuit consisting of its own battery and key switch. In addition to altimeter redundancy, the system incorporates redundant wiring for each deployment channel. Using WAGO splitters, each altimeter output is divided into two independent current paths to the charges. This configuration ensures that a single wiring or connection failure will not prevent charge ignition, significantly improving overall system reliability.

Parachutes, Harnessing, and Charges

The recovery harness utilizes eyebolts, quick links, swivels, and shock cord to ensure that the airframe sections and parachutes will remain attached during the recovery stage. The harness attachment points at each of the rocket airframe sections are made up of two eyebolts, as seen in Figure 16. While each of the eyebolts are rated to withstand the entire force of parachute deployment, as shown in Appendix A, the dual attachment method adds a layer of redundancy by distributing the load from the parachute across both attachment points. The nylon shock cord is chosen for both its strength and its elastic properties, reducing the impulse caused by parachute deployment. To connect the shock cord to the attachment points, the cord is hand-sewn into a loop at the ends, using aramid thread. The cord is then looped through itself around the eyebolts, allowing for a secure connection. By tension testing the loops, the team verified that the sewn aramid threads are strong enough to resist the force of parachute deployment. Furthermore, at each attachment point, a short section of two shock cords from each of the eyebolts is attached to a quick link, which is then connected to the longer length of cord. This can be seen in Figure 16. The drogue and main parachutes are then attached with a quick link to the harness. While both parachutes are affixed to the quick links via a swivel, three

additional swivels are added to the harness, with their locations shown on Figure 16. The purpose of the swivels is to mitigate the chance of the shock cord twisting and tangling with itself during descent. Similarly to the previously discussed eyebolts, both the quick links and swivels were chosen because their rated load exceeds the maximum calculated load that could occur during parachute deployment.

The lengths of each section of shock cord are chosen to ensure the distance between each of the airframe sections is at least four times the total length of the rocket. Additionally, the placement of the main and drogue parachutes within the sections is selected to prevent the airframe sections from making contact during descent.

The recovery deployment system employs black powder charges to eject the parachutes and separate the airframe sections during descent. To do so, two charges are placed in charge wells on either side of the recovery bay, which are then triggered by the altimeters. To protect each of the parachutes, parachutes are placed in fire-resistant deployment bags. These bags will not only prevent the parachutes from receiving damage from the ejection charges but also help unfurl them properly during deployment. Charges are 4F black powder. The main charges are 3g primary and 6g secondary. The drogue charge is 2.5g and 5g.

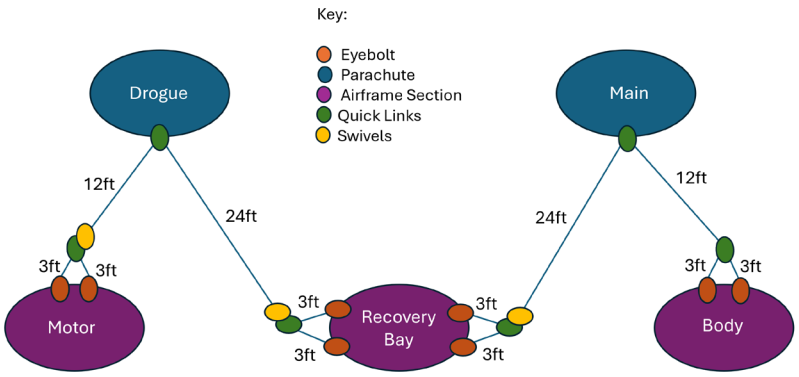


Figure 16 Recovery System Harness Schematic

E. Telemetry

Our SRAD telemetry system provides a real-time communication link between the rocket and the ground station, allowing us to monitor vehicle and avionics status throughout the flight while also supporting a servicing and diagnostic remote command shell. Through this link, we can track critical information such as battery voltage, internal temperatures, and key sensor measurements, giving the ground team immediate insight into the health of the onboard electronics. The telemetry stream also improves mission awareness by indicating that key events, such as drogue parachute deployment, have occurred and by providing an initial estimate of max altitude at apogee. In addition, transmitting data to the ground preserves valuable flight information for post-flight analysis in the event onboard avionics are damaged or unrecoverable. The system has already been validated on two flights, including an L2 certification flight and the team's test flight, demonstrating its ability to operate in real launch environments. More broadly, the telemetry system establishes a flexible foundation for future development by enabling more advanced monitoring, diagnostics, and support for increasingly complex payloads and vehicles.

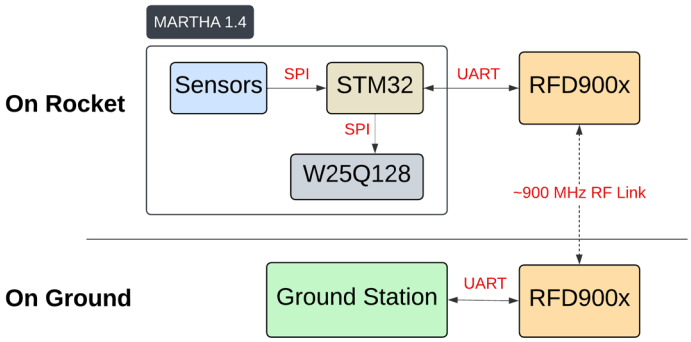


Figure 17 Telemetry hardware architecture for the MARTHA 1.4 flight computer and ground station

Hardware Configuration

The telemetry system is composed of four primary components: one MARTHA flight computer, two RFD900x modems, and one ground station laptop. We selected the RFD900x modem because it was widely used by other IREC teams last year, we had prior experience working with it, and we already had two units in our inventory. The modem supports ranges of up to 40 km and provides sufficient data throughput for live telemetry streaming. To comply with the 500 mW power limit specified in the band plan, the radios were configured to a 27 dBm transmit power. The MARTHA PCB interfaces with the separate onboard RFD900x modem over UART through a 6-pin JST connector. The overall hardware architecture is shown in Figure 17.

Software Configuration

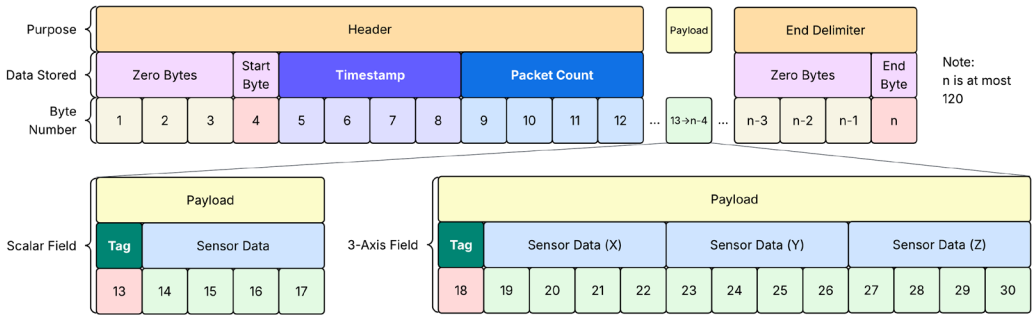


Figure 18 MARTHA Telemetry Stream Format

1. Telemetry Stream Composition

The telemetry stream is composed of 120-byte packets, each containing a header, payload, and end delimiter. The header begins with three zero bytes and a start byte, which helps the ground station decoder realign in the event of a previous partial or malformed packet. Following these alignment bytes are the timestamp and packet count fields, which allow dropped packets to be detected. The sensor data payload is composed of tag bytes followed by either four or twelve bytes of data, depending on whether the field represents a single value or a 3-axis vector. Every 10 ms, a packet is constructed by iterating over each sensor stream and checking whether enough time has elapsed since that data was last transmitted. If so, the data is added to the packet as long as doing so does not cause the packet to exceed the 120-byte limit, which matches the configured maximum transmit window size of the RFD900x and helps limit latency by preventing the modem from waiting to accumulate a larger packet before transmission. The structure of a packet can be seen in Figure 18.

2. Data Rates

We transmit different sensor streams at different rates based on their value for ground monitoring and post-flight analysis. For example, battery voltage and temperature data do not change rapidly, so they are transmitted at a lower rate. Acceleration and gyroscope are both 3-axis streams, so each update requires 13 bytes: 1 tag byte and three 4-byte floating-point values. Altitude is a single-value stream, requiring 5 bytes per update: 1 tag byte and one 4-byte floating-point value. Since all three streams are transmitted at 5 Hz, their combined payload rate is (13 + 13 + 5) bytes × 8 bits/byte × 5 Hz = 1.24 kbps. Those three streams make up the bulk of our telemetry payload data rate. An additional nine lower-priority telemetry streams are transmitted at 1 to 2 Hz, contributing approximately 0.46 kbps of payload data. When packet overhead, including the header and end delimiter, is included, the total telemetry data rate in the current configuration is approximately 2.33 kbps. The RFD900x is configured for an air data rate of 64 kbps, which provides a substantial margin over the required telemetry bandwidth.

3. Ground Station and Remote Shell

At the ground station, we receive the data using a paired RFD900x which delivers the stream via a USB port into our laptop running a custom graphical user interface (GUI). The ground station GUI parses, stores, and displays the data, allowing for real-time updates on rocket and flight-computer status. The ground station constantly checks for missed packets and calculates a signal integrity metric that is shown visually. The ground station is also designed to automatically reconnect to the RFD900x after the stream is interrupted, improving fault tolerance. Besides live telemetry streaming, the radio link is also used to host a remote shell on MARTHA. When the command-entry sequence of "ccc" is received, the telemetry system enters "command mode". In this mode, the outgoing telemetry stream is paused, and all inputs are redirected to MARTHA's command handler. The command handler echoes incoming characters back to the user and buffers them until a newline is received, at which point the completed command string is parsed and dispatched along with any provided arguments. This allows an operator to issue predefined remote commands over the radio connection, such as restarting the system, checking subsystem status, and dumping onboard flash data. It is not used for active flight control. After the operator issues the "exit" command or a 10 second timeout elapses, the telemetry system exits "command mode" and resumes normal operations.

Payload Subsystems

4. CubeSAT

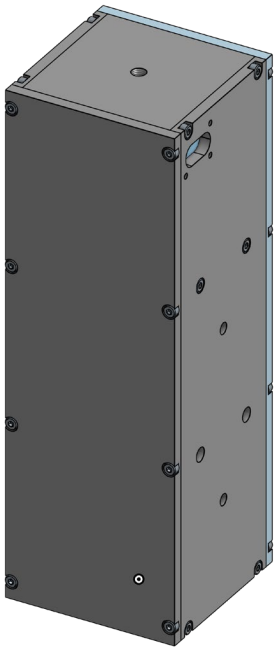


Figure 19 Aluminum Payload CubeSAT

The CubeSat is a machined aluminum box tasked with housing the payload. The payload fits the 3U form factor (5.25”), designed with shelving to allow internal modularity. Empty, it has a mass of 4.4lb, such that even unloaded, it meets the minimum weight required as determined by the IREC rules and requirement (2.3.1). Furthermore, the shelving and walls inside the CubeSat restrict the axial motion of the payload, preventing damage. It is presented in Figure 19.

5. Ballast

The ballast of *CU Up There* is simple in nature and exists in one of the three payload sections. The design is a 3D printed cube shaped housing and lid that bolts into the spar systems between the 3U CubeSat payload and the SRAD electronics and telemetry. The ballast’s main purpose is to allow rapid redistribution of the center of gravity by adding or removing rolls of US pennies into the container. The empty ballast weighs 1.96 lbs and holds up to twelve rolls, each adding increments of 0.25 lbs. The full ballast weighs 5.37 lbs after all rolls are placed inside. Once the correct weight is satisfied, the container and lid assembly are fastened together with 1/4-20 screws to prevent the pennies from falling out and placed into the vehicle. To determine the necessary weight, simulations are run in OpenRocket with the current wind speeds at location with stability and apogee being iterated upon until stability is between 1.9-2.1 cal and the apogee is as close to 10,000 ft as possible. Figure 20 demonstrates the housing for ballast weight.



Figure 20 “Piggy Bank” Ballast Housing

6. SRAD Electronics/Telemetry Housing

The SRAD bay housing is a 3D printed structure that mounts all SRAD electronics, the secondary GPS, a radio modem, and the batteries to power said equipment. The housing mechanically fastens the internal spar system from holes in the side of the housing. A 3D rendering of this housing can be found in Figure 21 below.

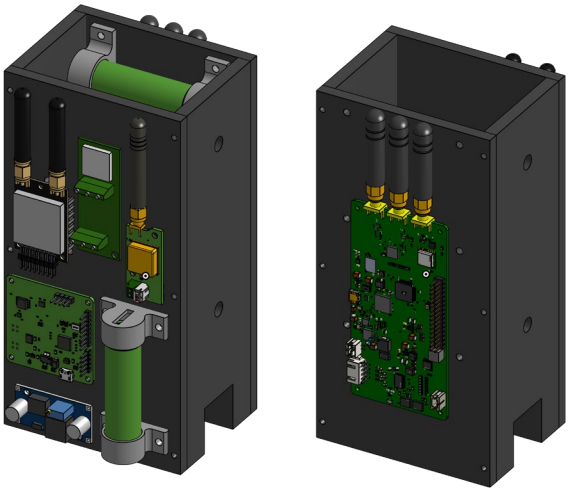


Figure 21 SRAD Electronics Bay

The SRAD flight computer is powered directly by 2 9V batteries. The connected LoRa radio modem is powered by a buck converter module to regulate the 9V down to 5V. The featherweight GPS is powered by a single 18650 cell and is turned on by an egg timer Wi-Fi switch. The other side contained a custom board from one of our sponsors that was flown on the test launch, and this board was powered by a battery of 18650 cells.

MARTHA v1.4 Hardware Development

Miniaturized Avionics for Rapid Testing, Handling, and Assessment (MARTHA) is the team's custom avionics platform designed to collect flight data including acceleration, rotation, magnetic field, pressure, temperature, and altitude. MARTHA v1.4 builds on the v1.3 design with a full layout redesign, improved power management, and updated components to better align with the team's Printed Circuit Board (PCB) and Electrical Design Standards document. The standards document includes details on best practices on documentation, design process, schematics, and layouts, as well as component selection, the manufacturing process, and testing after fabrication. Additionally, next year the subteam plans to expand on MARTHA by creating a Just an Expanded MARTHA (JEM) project that seeks to provide a more comprehensive flight platform. JEM will have redundancy for each sensor, incorporate GPS, and have more connections for external devices and sensors.

Design Changes from v1.3

Version 1.4 was a complete redesign of the PCB layout to address electromagnetic interference (EMI) and routing issues identified in v1.3 that did not meet the team's newly established PCB design standards. While the core architecture remains the same, an STM32F103CBT6 microcontroller interfacing with sensors and flash memory over Serial Peripheral Interface (SPI), but several key changes were made to improve performance and expand functionality. The most significant electrical change was adding a buck converter into our power system. The buck converter provides higher efficiency and better supports future power demands, since we intended to attach an RFD900x to the board, which can pull over 1W of power. Additionally, power ORing circuitry was added to allow the board to switch between Universal Serial Bus (USB) and battery power sources. The magnetometer was changed from the LIS3MDL to the LIS2MDL, as the LIS3MDL had been deprecated by the manufacturer. The LIS2MDL is the newest iteration, even though the name has a 2 in it. STMicroelectronics does not follow a strict numbering convention for iterations. A 5V output was added to the Universal Asynchronous Receive Transmit (UART) connector to support powering external telemetry radios such as the RFD900x. The SPI and UART connectors were swapped from simple male pins to 6-pin JST connectors, allowing for a much more secure connection. The Serial Wire Debug (SWD) connectors were also updated, and the Analog to Digital Converter (ADC) pin assignment was changed to allow for battery voltage monitoring.

Manufacturing and Assembly

Five PCBs were ordered for the v1.4 production. Two boards were ordered fully assembled from the manufacturer and arrived without issues; all sensors, the flash memory, and the buck converter functioned correctly on initial testing. The remaining three boards were ordered as bare PCBs along with a solder paste stencil so the team could attempt in-house assembly for the first time. The in-house assembly effort was ultimately unsuccessful. The team encountered significant difficulty placing the smaller sensor packages, specifically the LSM6DSOX, LIS2MDL, and BMP390, due to their small footprints. Inconsistencies during the stenciling process led to uneven solder paste application, and manual placement of these components proved too challenging for the team's first attempt. As a result, none of the three in-house boards achieved functional sensor connections. This experience highlighted the skill and equipment gap between professional assembly and the team's current in-house capabilities. Moving forward, the team plans to invest in assembly training and improved tooling, such as using reflow oven, to improve in-house success rates in future iterations.

Software Development

The software for MARTHA and other SRAD electronics is developed by the software development team with three key principles in mind: reusability, reliability, and performance. These principles are extremely important to reduce wasted time when redeveloping features, improving the success rate of data collection and state detection, and remaining maintainable for future students.

Avionics Library Integration

MARTHA's flight software is built around our team's Avionics software library, which acts as an interconnected toolbox for supporting general flight-computer functionality. The library was started in 2024 and has continued to expand as new vehicles, hardware, and payloads have demanded novel and more robust versions of features. It provides reusable tools for launch detection, apogee detection, vertical velocity estimation, coast-phase apogee prediction, orientation estimation, ground level estimation, data logging via multiple interfaces, command handling, telemetry streaming, and power monitoring. By developing each of these tools in a generic, modular fashion, we are able to use them across a large array of microcontrollers as well as run them on our own laptops for rapid unit testing.

New Software Features from this Year

Over the course of this year, we have added several major tools to the Avionics toolbox which are utilized in our latest MARTHA firmware. The telemetry stream tool dynamically combines data from different sensors based on a priority system into 120-byte packets. These are then sent to our radio to be transmitted to and then decoded by our ground station as detailed by the telemetry section above. It also allows for remote command execution for diagnostic and servicing reasons.

Another major addition was the onboard orientation estimator, or AHRS. In previous designs, rocket orientation was estimated on the ground station by interpreting the raw IMU data. That produced sub-optimal results due to the relatively slow data rate of our telemetry data compared to the onboard collection speeds (5 HZ vs 100HZ). The new onboard orientation estimator uses a rocketry-adapted Madgwick filter [3]. While the rocket is on the pad, it leverages data from an accelerometer, gyroscope, and magnetometer to estimate orientation. After liftoff, it shifts to a gyroscope-only approach because the accelerometer is no longer measuring gravitational forces.

The Avionics library also gained a fast launch detector. This detector provides a low-latency detection of the vehicle's launch, allowing the data logging and telemetry streaming systems to capture the high-priority flight data as soon as possible. Because the flight detector is intentionally oversensitive, the standard launch detector must confirm the event within a short window (500 ms in MARTHA's case). If the launch is not confirmed, the system rolls back to a pre-launch state. The standard launch detector has successfully detected launch within 500 ms for all the past 11 test flights a MARTHA has flown on (it typically detects within 250 ms).

Two smaller new features are the power management tool and ground level estimator. The power management tool can be configured to monitor a specific analogue pin and convert the voltage read into an estimate of the board's supply of battery voltage. With this new system, we can isolate faults caused by low battery and monitor the battery level remotely by including these estimates in the telemetry stream. The ground level estimator continuously samples altitude above sea level and applies an exponential moving average until launch, providing a stable ground-level reference for calculating above-ground-level altitude. Unlike simply recording altitude at power-on, this allows the estimate to adjust if the rocket is moved before liftoff.

Reliability Improvements

Reliability also saw major improvements this year through changes to the team's development and testing process. The Avionics library already had unit tests, but they were isolated in a separate repository, causing the tests to become out of sync with the actual software. To fix this, unit testing has been merged into the Avionics repository to keep the tests synchronized with the code they verify. Additionally, the power of the tests was strengthened with address and undefined behavior sanitizers, which caught multiple out-of-bounds accesses and integer overflows during testing. Compiler warnings were also made much stricter with flags such as -Wall and -Werror, ensuring that minor warnings are treated as critical errors and reducing the amount of unreliable code that can merge into the main branch. Several systems in the Avionics library, such as the launch detector and command handler, previously relied on dynamically allocated memory. These systems have been refactored to use fixed-size, compile-time-allocated buffers instead, reducing runtime allocation and improving reliability on our heavily RAM-constrained embedded hardware.

To enforce these new reliability standards, the team has created multiple GitHub Actions workflows that automatically run the above-mentioned unit tests along with linters such as Clang-Tidy on each pull request. The automatic workflows have prevented warning debt from building back up, which is critical after multiple weeks dealing with the previous warnings. Additionally, when new members put up a pull request, they are given immediate feedback and can start correcting the issues. The team has also added Doxygen-style docstrings on every public

function in each class's header file. These comments are then compiled into a website that can be found at https://curocketengineering.github.io/Avionics/. Making it much easier for students that are developing flight computer software to understand how and when to use each of the tools provided in the Avionics library.

Validation

The team flew two MARTHA's this academic year. A MARTHA 1.3 flew in November on an L2 certification vehicle and most recently a MARTHA 1.4 during the team's test flight. In both cases, the new Avionics tools were tested and validated successfully. The November flight was our first test of the live telemetry system, and multiple quality-of-life issues were discovered on the ground station GUI, including test contrast, non-automatic reconnects, graph layouts, and more. In addition to flight tests, we perform validation using our data injection system that allows us to stream previous flight data into the MARTHA board via UART where that data is then used in place of real sensor data to effectively test the entire MARTHA firmware minus the sensor drivers. Data injection testing is always performed prior to launches to validate state detection and data logging.

8. Payload

The objective of the payload is to analyze the effects of vibrations and acceleration in a rocket launch on samples of stored, refrigerated blood. From this experiment, the team aims to examine the viability of transporting blood into orbit for medical procedures. Porcine blood was chosen because its structure and properties are similar to human blood, allowing it to be representative of the blood's integrity under extreme conditions. The blood must be kept at a temperature between 0 and 6 degrees Celsius during the entirety of the experiment to avoid hemolysis. This temperature will be maintained through the use of styrofoam and frozen gel packs to limit heat transfer to the samples. The main body of the payload itself consists of a styrofoam cube, measuring 3.437 x 3.437 x 3.562 inches in length, width, and height, respectively. A 3D rendering of this assembly can be found in Figure 22 below.

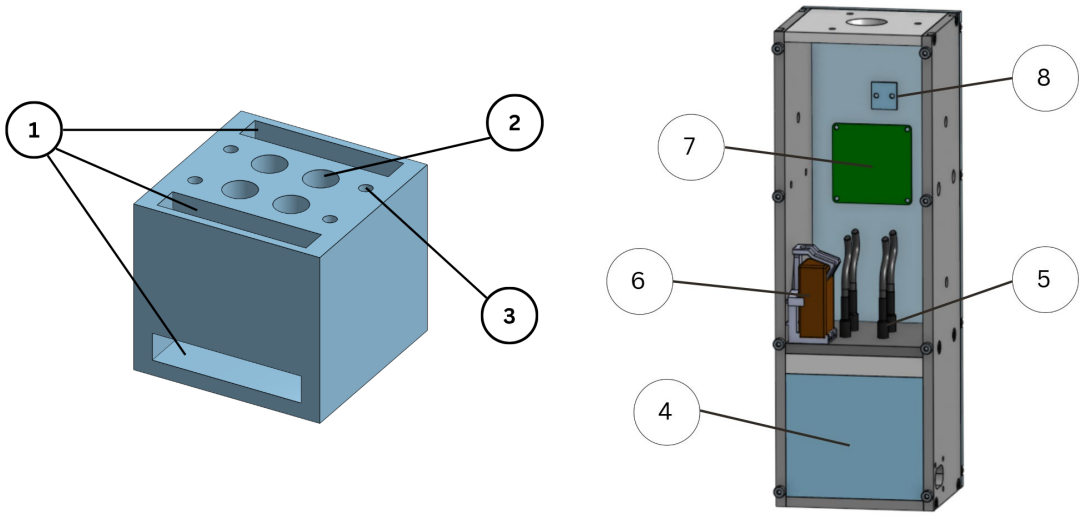


Figure 22 Body of styrofoam cube and CubeSat with payload assembled

When fully constructed with its contents, it will be secured in one of the three sections of the pre-constructed aluminum CubeSat payload box (4). Styrofoam was chosen as the material due to its insulating properties. The styrofoam cube is divided into a lid and a body, with the body containing 4 cylindrical holes for the blood vials (2). and 3 rectangular slots for gel packs (1). The slots for the gel packs are spaced so that two are on each side of the vials, while one lies underneath all four. This helps ensure even distribution of temperature as the ice packs keep the blood within the required temperature range. The lid of the styrofoam cube will be secured to the body using a dowel joint system. The four cylindrical holes inside the body of the styrofoam cube will each contain one 5-millimeter vial of porcine blood. To ensure temperature compliance, the use of DS18B20 temperature probes (5) will be implemented.

The probe will be inserted into the blood to measure the internal temperature. The signal from the probe will be quantified by an ADC (8) and communicated over a SPI channel on a MARTHA board (7) to log temperature measurements. The systems will be powered by a 9-volt battery (6). Upon collection of the samples, this data can be used to verify that the samples did not exceed their temperature range. If the range is exceeded, the magnitude and duration of the temperature can be analyzed to determine if the samples are still viable for testing. Samples will be tested for hemolysis through centrifuge-based microhematocrit testing, where separation of the plasma layer, which is visually determined, denotes hemolysis, and therefore unusable blood.

IV. Mission Concept of Operations Overview

The CONOPS Overview is a comprehensive section detailing the sequence of phases that the vehicle's mission will undergo, from pre-assembly to landing. The overview of the CONOPS phases is described in Figure 23.

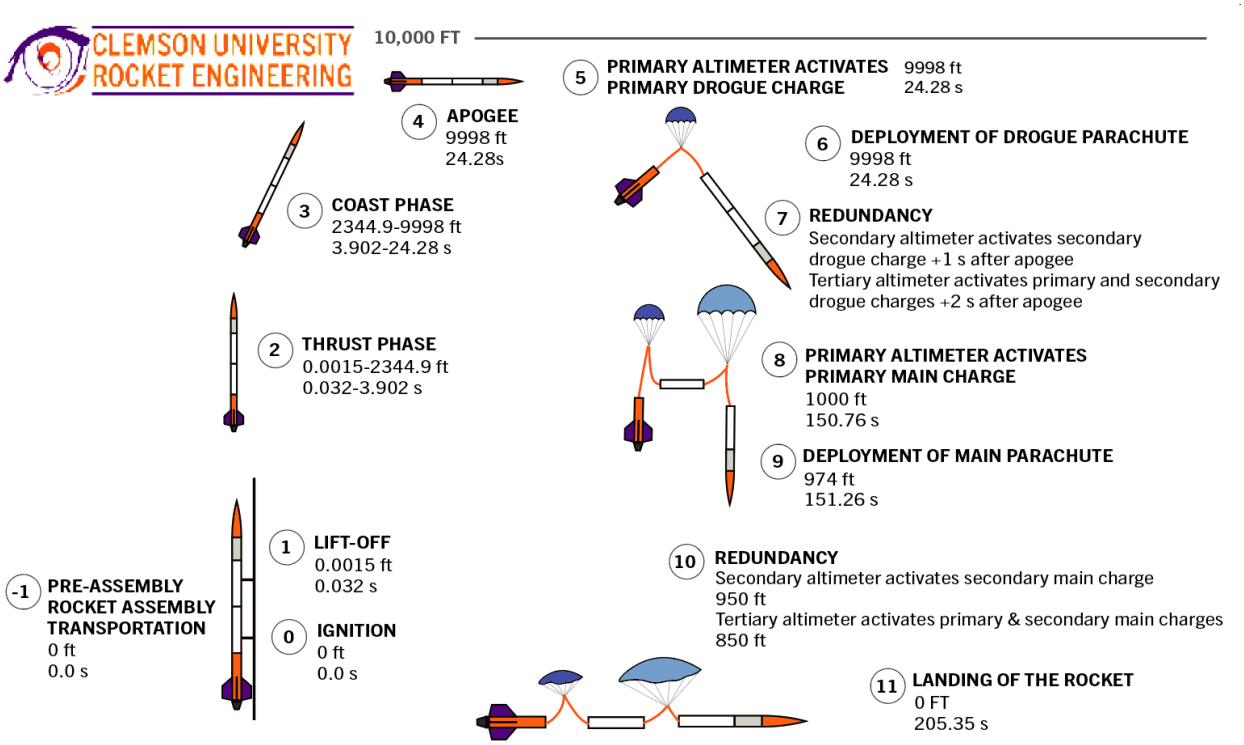


Figure 23 CONOPS overview of nominal flight for *CU Up There*

The CONOPS Overview illustrates these phases, outlining the nominal operations and interactions of all subsystems during each stage. Each phase is described, highlighting the role of various subsystems such as avionics, propulsion, and recovery systems and defining the critical events that signify transitions between phases. These transitions are triggered by specific mission events such as the ignition signal, motor burnout, reaching apogee, and altitude milestones for parachute deployment, all culminating in the safe recovery of the vehicle post mission. This structured approach ensures clarity in mission planning and execution, allowing for precise coordination and monitoring of each phase in relation to the mission's overarching goals and objectives.

Phase –1 Pre-Assembly, Rocket Assembly, Transportation

During the pre-assembly phase, all members of the team will work to assemble the payload and avionics bays. Once both the avionics and payload system are assembled and the electronics for payload is turned on, both bays will be secured into the vehicle body. Once the bays are secured in their respective body tubes, each body tube will be secured together, and the launch vehicle will be carried and loaded on the pad.

Phase 0 Ignition

Once the vehicle is loaded onto the pad, the SRAD electronics will be turned on, the recovery bay will be armed and tracking verified, and the e-matches will be loaded into the motor. After a confirmed avionics signal from the spectator area, Spaceport officials will ignite the e-matches, beginning motor burn, and the vehicle will begin accelerating off the launch rail.

Phase 1 Lift-Off

After the motor ignition, the motor will burn for 3.9 seconds. According to motor information, the vehicle should leave the 17 ft launch rail at a velocity of 103 ft/s. Lift-off occurs at T+0.03 seconds at an altitude of 0 ft. Each SRAD flight computer detects the launch at this point, affirming departure.

Phase 2 Thrust Phase

According to simulations, the M2500T motor will provide an average thrust of 2,500 N. Since each COTS altimeter is altitude based, launch is detected at this point. The thrust phase of the flight will take place between T+0.03 and T+3.903 seconds, propelling the vehicle from 0 ft to approximately 2,344.98 ft.

Phase 3 Coast Phase

Following motor burnout at T+3.903 seconds, the vehicle will coast until apogee. This phase will occur between T+3.903 and T+24.27 seconds from 2,344.98 ft to 9997.57 ft.

Phase 4 Apogee

According to OpenRocket simulations, the predicted apogee is 9997.57. ft. Apogee will be reached at T+24.27 seconds. This altitude and timing will be confirmed post-flight by altimeters in the payload and SRAD avionics bays.

Phase 5 Deployment of Drogue Parachute

At apogee, the RRC3+ will attempt to ignite the primary drogue charge to deploy the drogue parachute from the drogue parachute bay. The StratoLogger CF, the secondary flight computer, will ignite the secondary drogue charge to attempt to deploy the drogue parachute one second after apogee has been reached. Finally, the BlueRaven, the tertiary flight computer, will attempt to deploy both the primary and secondary drogue parachute charges two seconds after apogee. The primary recovery event occurs at T+24.28 seconds where the secondary event occurs at T+25.28 and the tertiary event at T+26.28. The drogue will inflate and decrease vertical velocity to a maximum of –76.69 ft/s, and will slowly increase to a vertical velocity of –67.38 ft/s.

Phase 6 Deployment of Main Parachute

At 1000 ft, the RRC3+ will fire the primary main deployment charge, ideally deploying the main parachute. The StratoLogger CF, will then fire the secondary main deployment charge at 950 ft, or approximately 1.25 s after a failed main deployment attempt. At 900 ft AGL, the BlueRaven will attempt to deploy both the primary and secondary main deployment charges. This occurs approximately 1.25 s after a failed main deployment attempt of the StratoLogger, assuming a failed main deployment attempt by the RRC3+ as well. The nominal deployment is predicted to occur at T+151.26 seconds, while a secondary and tertiary deployment would occur at T+152.51 seconds and T+153.76 seconds respectively. Main will inflate and increase vertical velocity to –17.8 ft/s.

Phase 7 Landing of the Rocket

After the vehicle has landed at T+205.35 seconds, the recovery team will travel to and recover the launch vehicle. To accomplish this, the recovery team will use the onboard Featherweight GPS tracker and in-hand ground station to track the vehicle's location.

V. Conclusions and Lessons Learned

Conclusions

The creation of *CU Up There* marks many lessons learned across the board for the team but also highlights a step further into modularity and more comprehensive design. The hurdles face challenges each of the team leads in various ways and helped them learn various new skills which will be discuss in the following subsections. CURE avidly awaits the launch of *CU Up There* at IREC 2026.

Admin/Financial

Leading a team is not an easy task and this year has been educational. Effective delegation is perhaps the single most important skill on an engineering project. It is not adequate to simply assign tasks – upper leadership must always be available for help / advice, and regular progress checks are a must to ensure completion. Fostering a good team culture is also a difficult but necessary task. An environment where members are friends leads to better project collaboration and aids the exchange of ideas. Leaders must see members' strengths and weaknesses, delegate accordingly, and work to grow skills. Finally, timelines never last. Delays are almost inevitable due to a range of factors. Projects must be budgeted enough time to account for delays.

Structures

The subteam has worked towards reducing the weight of all components, issues arose with insufficient strength in fiberglass airframes and couplers in the form of drastically increased malleability. From this error, an opportunity arose to ensure consistency between hand layup components. The frequency of abandoning a layup when an epoxy's quality was unsatisfactory, due to suboptimal atmospheric temperature has increased, resulting in a temporary setback of materials and time, but producing higher quality end products such that the team is not at a detriment. Secondly, during manufacturing, an issue arose when two different types of fiberglass were used for separate tubes, as necessitated for running out of a previous stock. This resulted in inconsistent wall thicknesses, preventing a proper interference fit during assembly. The team resolved the issue by double-wrapping the thinner fiberglass section and sanding down the excess material to achieve uniformity. Although effective, this solution was both time consuming and resource intensive. To avoid similar complications in future builds, it will be essential to standardize material selection.

Recovery

Through this past year of leading the recovery subteam, many new lessons were learned helping to guide our team to successfully integrating and test launching *CU Up There*. A major goal this year was to drastically decrease the size of the recovery bay's design. This proved to be a major hurdle for the subteam forcing us to grow in many new areas related to the design itself. While the task was large since CURE has not created recovery bay this small in the past, the recovery bay design was successfully downsized from previously being 14 inches down to 6 inches. This goal forced the subteam to consider all ends of the design from high level integration of the bay down to detailed optimization of space within the bay. The team's design proved successful when tested during flight and highlighted key growth that was made by the subteam this year. While the final bay was printed in nylon for the actual launch, the team utilized PLA to rapid test and prototype fitment of different bay internals. Multiple iterations of these bay internals were idealized using this method. This consequently taught the team the importance of this process when designing and allowed the team to make major design improvements such as the precise fitment of the key switches that are accessible from the outside of the vehicle as well as the retaining cup for the twisted pair wiring. The last main obstacle the recovery team overcame was designing the structural element of the recovery bay. This entailed the selection of the correct threaded rods, nuts, and eyebolts to withstand the forces on the recovery bay during the parachute deployments. Since the previous year's recovery bay was placed below both the drogue and main parachute bays, the forces due to the parachutes were previously considered a task of the structures subteam to design rather than that of the recovery subteam. The dual bay, dual deploy architecture this year forced the recovery team to learn to manage the load path of the bay itself and calculate the forces to ensure the bay does not fail. As a result of the year, even though the design challenges for the team were high, the team rose to the challenge of designing a bay capable of successfully recovering *CU Up There* while learning many new skills for future bay research and development.

Simulations

Over the past year leading the simulations team, several lessons stuck out that defined our year of progress. First, we completely redid our mass estimation method by creating a comprehensive list of all physical items on our rocket. This was due to an oversight last year in mass, where our original prediction was incorrect during the final weight in. This year, the subteam tracked every component that will be on the rocket with a corresponding weight. With this, throughout the year while changes were made and manufacturing began, it was possible to track and compare between different components and see what most of our feasible rocket mass could be. Along with this, the simulations sub team was also able to track weights of various sections of *CU Up There* and properly enforce other subteams to stay within mass limits of their sections design. Continuing this budget into next year will be beneficial to stay accountable with overall rocket mass. The consequence of this is the timeline I had planned for incorporating some initial computational fluid dynamic (CFD) testing; however, this has gotten postponed due to the mass budget taking priority.

I hope in the next couple years of the simulations sub-team we can break into CFD and obtain some estimates for our coefficient of drag, pushing us one step closer to an active air breaking systems and other exciting developments.

Hardware

Leading the Hardware Development Team this year provided several valuable lessons that shaped both the team's output and my own growth as a leader. The most impactful change was establishing a semester plan and schedule before the first team meeting. Before starting the semester, I planned to have a series of lectures on the basics of electronics and microcontrollers to bring a wide range of skill levels to a common foundation. After this, the team took theory into practice by building their own MARTHA breadboards. Once the team had hands-on experience, we moved into developing a demo project in Altium. After this, everyone had the capabilities to start working on real avionic challenges. Having a clear roadmap from the start gave the team direction and made it easier to track progress, prioritize tasks, and keep projects moving forward. Another key takeaway was the importance of documentation: recording design changes provided reasoning behind decisions and the solutions to problems encountered along the way, ensuring knowledge is not lost and making it easier to build on past work. This lesson directly motivated the creation of the PCB and Electrical Design Standards document, which now serves as a living reference for the team. Perhaps the most personal lesson was learning to delegate effectively. As someone who naturally preferred to handle tasks independently, I had to recognize that taking on everything myself prevented team members from developing their own skills and left me spread too thin. Delegating work not only allowed others to learn through hands-on experience, but also helped build a team capable of operating independently.

Software

The biggest lesson learned this year was that reliable flight software depends not only on the code's quality, but also on the processes used to build, test, and maintain it. Since the beginning of the Avionics software library, the native unit tests and flight software lived in two separate repositories, making it more confusing for new members to manage their local environment and causing multiple synchronization issues. By merging them into one, we have massively simplified our testing process, paving the way for easier code quality and unit testing automation. We learned that simple repository structure and automation are critical for maintaining code quality and allowing new members to contribute fearlessly. Another lesson is that flight validation is necessary to verify the individual Avionics tools and the MARTHA system as a whole. Real flights catch issues around timing, telemetry, radios, power, and usability on site. For example, an undiscovered bug in MARTHA's code caused every other altimeter reading to be dropped, which caused the vertical velocity estimator to reuse a duplicate altitude value every other cycle. As a result, the vertical velocity estimator began estimating that the rocket was moving slower than it actually was, causing the apogee predictor to underestimate the final apogee. Our data injection testing and other unit tests based on past flight data used interpolation, which masked the missing altitude data and caused the apogee prediction tests to appear successful. The lesson we learned from eventually identifying and fixing this issue is that there is no true replacement for flight testing. Deviations from simulation results should be investigated aggressively, and the results of those investigations should be used to improve our simulations and tests so they can catch similar problems more effectively in the future. The team also learned that graphical software used in the field must be built differently than software used indoors, as the ground station GUI needs high contrast, large fonts, and clear status indicators to remain usable in direct sunlight. From a team management perspective, the software team learned that onboarding needs to be more structured in the future. Tools such as Git and PlatformIO, along with the overall software architecture of MARTHA and Avionics, need a formal introduction to help new members get on their feet faster and understand how their projects fit into the larger picture. Additionally, bringing software members to launches is important so that they can see how their code is actually used and connect it to the end goal of the process. Overall, the software team learned that sustainable flight software is not based on just working code, but also clear architecture, automated verification, frequent flight testing, and intentional documentation writing.

Propulsion

In the past, propulsion systems have been an afterthought for the team once a motor has been selected and safe weights have been calculated. However, due to a test launch this year, the team has learned two lessons for the future when approaching propulsion with regards to a test launch. The first lesson the team learned is to conduct analysis on the motor for the test flight, weighing the rocket and the motor before and after flight to be able to ensure that the motor is not a cause for discrepancies in performance during a test flight. Last year, it was discovered that the M2500T had been previously altered by AeroTech, and that thrust curves for our simulations were inaccurate, so the team now uses a thrust curve that more accurately matches the performance of the M2500T. In the past years, the team has used

weaker motors for test flights to limit the apogee to prevent drifting at our test site. This year, the team had a discrepancy in flight performance on the weaker motor and neglected to conduct such a test to determine if the motor performed as expected. Therefore, the team has learned to conduct motor analysis before and after the flight. This motor analysis will be good practice as the team moves forward with research motors. For the second lesson, if the team can acquire two of the same motor, the team will try to use the same motor for test launch and competition to allow for a higher level of confidence in matching flight performance through propulsion. Additionally, as the team gains knowledge and experience in hybrid motor systems, higher control over motor performance will pose a new opportunity as the team works towards a functional SRAD hybrid motor to be used at a future IREC.

Test Launch

While the test launch was a major success and had minimal issues, one of the most important points was working in parallel. At the launch, the team performed each step to prepare the rocket in succession. For example, the parachutes weren't folded until the recovery charges were installed, when these steps could occur at the same time. The team could be ready to launch significantly sooner if it had more trained hands that were working simultaneously. Overall, each of the integration steps on launch day went overall smoothly with minimal issues meaning bottle necks from physical integration steps were minimal. Since this was the case, the lack of efficiency of integration due to not taking these steps became very apparent. With this gained knowledge of *CU Up There's* integration capabilities in mind, the team plans to enact a parallel integration process further boosting the team's efficiency and ease of integration on launch day of IREC 2026.

The test launch highlighted some general improvements to the recovery bay that will be factored into next year's designs. Overall, the recovery bay worked as planned successfully deploying charges in compliance with the planned CONOPS, but the necessity of having a triple redundancy bay was put into question. With added complexity of a third set of leads connecting to the primary and secondary charges on each bulkhead, this made prepad integration much more difficult than if only dual redundancy w2qcvw 1as utilized. Dual redundancy is considered sufficient by IREC rules, and CURE will incorporate this into future recovery designs.

The test launch also used an M1800FJ, which provided some uncertainties regarding our overall flight when compared to the M2500T. The first lesson learned during our test launch is the importance of our ballast for stability. With lower power motor being utilized, it made sense from an apogee standpoint to remove the ballast. Since the *CU Up There's* recovery system was moved lower in the vehicle, there is a lacking mass towards the top causing some instability when the ballast is removed and no mass within the rocket is shifted. Currently, our rocket utilizes a full ballast which provides a stability of 2.04 cal. With *CU Up* There's internal spar system, ballast weight and other payloads can be shifted in this section to improve stability of flight. Improving our simulations to incorporate various ballast locations as well as weights allows us to understand the implications of the ballast better was a major lesson learned of the test flight and will better CURE when integrating for competition.

VI. Acknowledgements

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VIII. Appendices

Appendix A—System Weights, Measures, and Performance Data

Vehicle and Flight Dynamics Data

CU Up There is powered by a COTS AeroTech M2500T solid motor. Displayed below in **Table 1** are specifications of the M2500T. Data for the team's simulations was pulled from a sourced thrust curve[4].

Table 1 Motor Specification [4]

Description	Value	Units
Motor Diameter	98	mm
Casing Length	732	mm
Propellant Weight	4711	g
Motor Weight	8064	g
Average Thrust	2488.0	N
Maximum Thrust	4080.5	N
Total Impulse	9055.9	Ns
Burn Time	3.6	s

Table 2 Basic rocket vehicle information

Description	Variable	Units
Number of Stages	1	
Length	116	in
Airframe Diameter	6	in
Number of Fins	4	in
Fin semi-span	5.25	in
Fin tip chord	5	in
Fin root chord	8	in
Fin thickness	0.25	in
Weight on launch rail	63.2	lbs

Propellant weight	4.711	kg
Empty motor case/structure weight	3.35	kgs
Payload weight	5.29	lbs
Liftoff weight	63.2	lbs
Center of pressure	82.83	In
Center of gravity	70.57	in
Launch rail length	5	m

Table 3 Propulsion information

Description	Variable	Units
Motor Type	Solid	
COTS/SRAD	COTS	
Manufacturer and designation	Aerotech	
Motor letter classification	M	
Average thrust	2500	N
Total impulse	9055.8	Ns
Motor burn time	3.9	2

Table 4 Predicted flight data

Description	Variable	Units
Liftoff thrust-weight ratio	12.4	[-]
Rail departure velocity	103	ft/s
Minimum static margin	1.5	cal
Maximum acceleration	14.7	G
Maximum velocity	285.96	m/s
Fin flutter velocity	3850	ft/s
Target apogee	10,000	ft
Predicted apogee	9997.52	ft

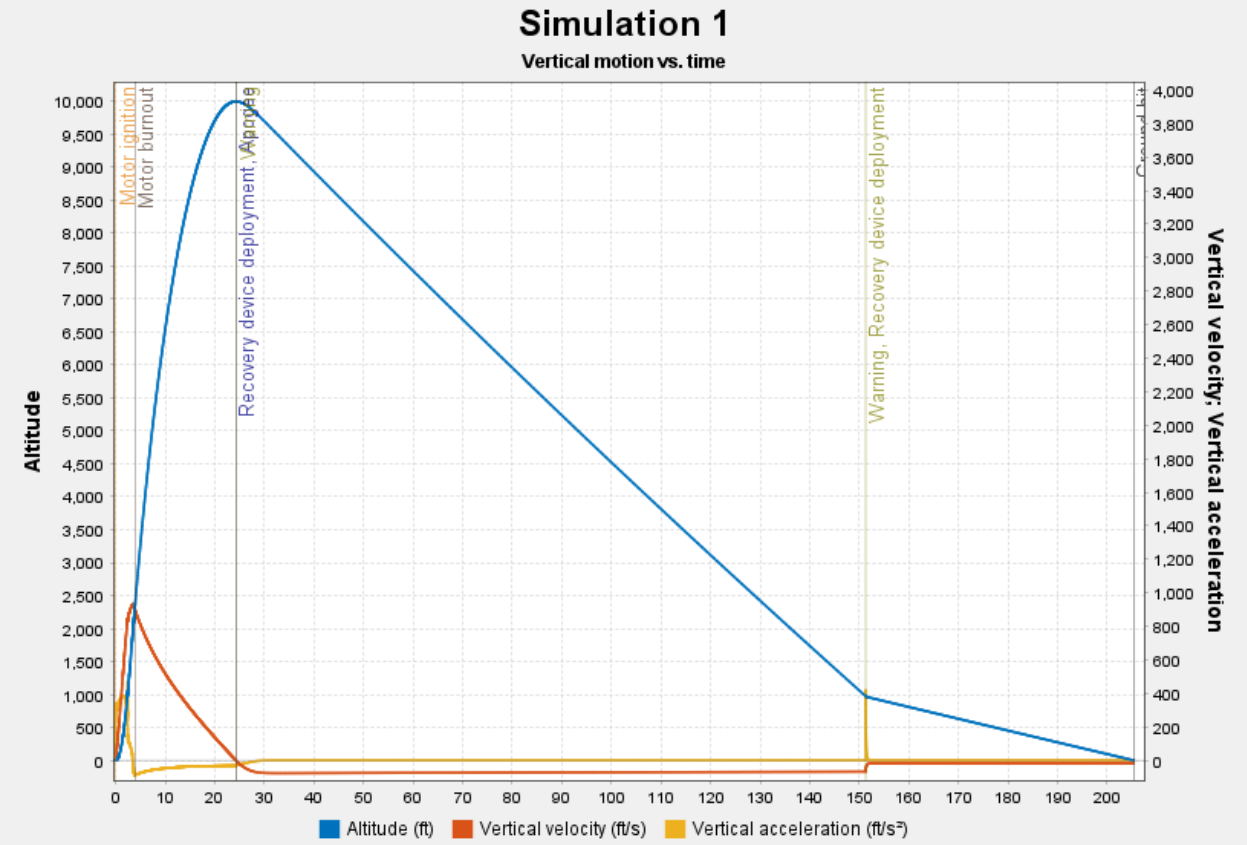


Figure 24 Flight Profile Graph

Recovery Information

Within the recovery system the design includes triple redundancy of the recovery electronics including the primary altimeter RRC3+, the secondary altimeter Stratologger CF, and the tertiary altimeter Blue Raven. Both the RRC3+ and the Stratologger have been used previously in past competition flights. Each altimeter has its own independent series circuit of a key switch and power source of a 9V battery. The primary altimeter is connected with an ematch to the primary drogue and main black powder charge which will be programmed to deploy at apogee for the drogue parachute and at 1000 ft for the main parachute. The secondary altimeter is connected to the secondary drogue and main charge with an ematch and is programmed to eject at a one second delay from apogee for the drogue parachute and at 950 ft for the main parachute. Lastly the tertiary altimeter connects to a second ematch connected to the primary black powder charge which deploys at a two second delay from apogee for the drogue parachute and at 900 ft for the main parachute. The reason for three different flight computers is that even in the event of an altimeter failure during flight the dual redundancy is still maintained additionally this is the first competition launch with using the Blue Raven, so its capabilities are only just being tested which is why it is the tertiary altimeter.

Table 5 Recovery information

Description	Variable	Units
Primary altimeter	RRC3+	--
Secondary altimeter	Stratologger CF	--
Tertiary altimeter	Blue Raven	--
Drogue parachute size	30	in^2
Drogue parachute primary deployment charge	2.5	g
Drogue parachute backup deployment charge	5	g
Drogue parachute primary deployment altitude	9984	ft
Drogue parachute secondary deployment altitude	9936	ft
Drogue parachute tertiary deployment altitude	9855	ft
Drogue parachute descent rate	-67.337	ft/s
Main parachute size	96	in^2
Main parachute primary deployment charge	3.0	g
Main parachute backup deployment charge	6.0	g
Main parachute primary deployment altitude	1000	ft
Main parachute secondary deployment altitude	950	ft
Main parachute tertiary deployment altitude	900	ft
Main parachute deployment descent rate	-17.8	ft/s

For the drogue parachute, CURE selected a *Rocketman High Performance CD 2.2 Parachute 30 inch*. Using 4F Black powder, a primary 2.5 g charge was chosen through use of initial calculations, ejection ground testing, and proof of the test flight which can all be found in Appendix B. Using 4F Black powder, a secondary 5 charge was chosen through use of initial calculations, ejection ground testing, and proof of the test flight which can all be found in Appendix B.

Table 6 Drogue Parachute and Charge Information

Description	Variable	Units
Drogue Parachute CD	1.5	-
Drogue Parachute Diameter	30	in
Max Drogue Parachute Force	347.16	lbf
Primary Charge Size	2.5	g
Secondary Charge Size	5	g

Calculations were done to predict the maximum possible force due to the drogue parachute which the calculations are provide below in Figure 25.

$F_D = \frac{1}{2}\rho C_D A_D V_D^2$ drogue: 30 in elliptical

$\rho = 0.905$ kg/m^3 @ 10000 ft

$C_D = 1.5$ (based off fruity chutes)

$D = 30$ in $= 0.762$ m

$A_D = 4.7124$ ft^2 $= 0.4378$ m^2 (from fruity chutes)

$V_D =$ worst case drogue velocity @ apogee + 3 s when tertiary altimeter fires & + 3 s for drogue full inflation ∴ apogee + 6 s

$\Rightarrow S_f = S_o + V_o t + \frac{1}{2} a t^2$

$S_f = \frac{1}{2}(9.81 \text{ m/s}^2)(6 \text{ s})^2 = 176.58$ m

$\frac{1}{2} MV^2 = mgh$

$V_D = \sqrt{2gh} = \sqrt{2(9.81 \text{ m/s}^2)(176.58 \text{ m})} = 58.86$ m/s

$F_D = \frac{1}{2}(0.905 \text{ kg/m}^3)(1.5)(0.4378 \text{ m}^2)(58.86 \text{ m/s})^2$

$= 1{,}029.5 \text{ kg m/s}^2 = 1{,}029.5$ N

$F_D * SF = 1{,}029.5 \text{ N } (1.5) = 1{,}544.25$ N

$F_D = 347.16$ lbf

Figure 25 Calculated Max Drogue Parachute Forces

For the main parachute, CURE selected a *96 inch iris ultra standard from fruity chutes*. Using 4F Black powder, a primary 3 g charge was chosen through using initial calculations, ejection ground testing, and proof of the test flight which can all be found in Appendix B. Using 4F Black powder, a secondary 6 g charge was chosen through use of initial calculations, ejection ground testing, and proof of the test flight which can all be found in Appendix B.

Table 7 Main Parachute and Charge Information

Description	Variable	Units
Main Parachute CD	2.2	-
Main Parachute Diameter	96	in
Max Main Parachute Force	1091.11	lbf
Primary Charge Size	3	g
Secondary Charge Size	6	g

Calculations were done to predict the maximum possible force due to the main parachute which the calculations are provided below in Figure 26.

$F_D = \frac{1}{2}\rho C_D A_m V_m^2$ Iris Ultra Standard 96in

$\rho = 1.177\ \text{kg/m}^3$ @ 1000 ft

$C_D = 2.2$ (based off fruity chutes)

$D_m = 120\ \text{in} = 3.048\ \text{m}$

$A_m = 48.7084\ \text{ft}^2 = 4.52516\ \text{m}^2$

$V_m =$ worst case main velocity when the rocket is under drogue @ terminal velocity

F_D ↑

•

mg ↓

$F_{net} = ma \Rightarrow 0 = F_D - mg \Rightarrow F_D = mg$

$m_{rocket} = 48\ \text{lbm} = 21.77\ \text{kg}$

$F_D = 21.77\ \text{kg}\ (9.81\ \text{m/s}^2) = 213.6\ \text{kg}\cdot\text{m/s}^2 = 213.6\ N$

$F_D = \frac{1}{2}\rho C_D A_m V_m^2$ $\rho = 1.177\ \text{kg/m}^3,\ C_D = 2.2,\ A_m = 4.52516\ \text{m}^2$

$$V_m = \sqrt{\frac{2F_D}{\rho C_D A_m}} = \sqrt{\frac{2(213.6\ \text{N})}{1.177\ \text{kg/m}^3\,(2.2)(4.52516\ \text{m}^2)}} = 23.51\ \text{m/s}$$

$$F_D = \frac{1}{2}(1.177\ \text{kg/m}^3)(2.2)(4.52516\ \text{m}^2)(23.51\ \text{m/s})^2$$

$F_D = 3,235.69\ \text{kg}\cdot\text{m/s}^2$

$F_D = 3,235.69\ \text{kg}\cdot\text{m/s}^2 \times SF = 3,235.69\ \text{N}\ (1.5) = 4,853.53\ \text{N}$

$\boxed{F_D = 1,091.11\ \text{lbf}}$

Figure 26 Calculated Max Main Parachute Forces

Recovery Harnessing and Mechanical Links

The full diagram for recovery harnessing was provided in the main body of the report, which is visualized in Figure 16. The following tables outline all the components of the recovery system

Table 8 Parachute Shock Cord Lengths

Description	Variable	Units
All Y Harness Lengths	3	ft
Drogue - Motor	12	ft
Droge – Recovery Bay	24	ft
Main – Recovery Bay	24	ft
Main - Body	12	ft

Table 9 Harnessing Mechanical Links Amounts

Description	Amount
Eyebolts	8
Quick Link	6
Swivel	3
Threaded Rod	2

Table 10 Harnessing Mechanical Total Forces

Description	Amount	Units
Eyebolts	1800	lbf
Quick Links	3000	lbf
Threaded Rod	5659.7	lbf

The worst-case failure planned was if one of the mechanical links took the entirety of the force before the forces were in equilibrium in the rocket. Therefore, all links were planned to withstand the max forces of the main parachute. Eyebolts were selected with an extra factor of safety of 1.65 rather than just 1.5 account for angled loading since the force capacity of the eyebolts is given in vertical loading only. Since the threaded rods of the recovery bay also withstand the forces of the harnessing, they are also considered a linkage. To obtain max force the calculations can be found in the figures below.

$\frac{3}{8}$" – 16 threaded rods

$\Rightarrow$ Minor Diameter: 0.3006 in

$$A = \frac{1}{4}\pi d^2$$

$$A = \frac{1}{4}\pi\ (0.3006\ \text{in})^2 = 0.07097\ \text{in}^2$$

Grade B7 Medium Strength Steel Threaded rod

tensile strength = $P = 110{,}000\ \text{lbf/in}^2$

$$P = \frac{F}{A} \quad \Rightarrow \quad F = P \times A$$

$$F = (0.07097\ \text{in}^2)\left(110{,}000\ \frac{\text{lbf}}{\text{in}^2}\right) = \underline{\overline{7{,}806.7\ \text{lbf}}}$$

Figure 27 Confirming Selected Diameter of Threaded Rods [5]

$\frac{3}{8}$" – 16 threaded rods

$\Rightarrow$ Minor Diameter: 0.3006 in

$$A = \frac{1}{4}\pi d^2$$

$$A = \frac{1}{4}\pi\ (0.3006\ \text{in})^2 = 0.07097\ \text{in}^2$$

Grade 5 Medium Strength Steel Hex Nut

Shear strength = τ = 72,000 psi

A = thread shear area = $\frac{\pi}{4}(D - 0.938194\ P)^2$

D = nominal diameter = 0.375 in

P = thread pitch = $\frac{1}{16}$ in

$$A = \frac{\pi}{4}\left(0.375\ \text{in} - 0.938194\ \left(\frac{1}{16}\ \text{in}\right)\right)^2 = 0.07861\ \text{in}^2$$

$$F_{po} = \tau A$$

$$F_{po} = (72{,}000\ \text{lbf/in}^2)\ (0.07861\ \text{in}^2) = 5{,}659.7\ \text{lbf}$$

Figure 28 Validating Pull out Force of Threaded Rods [5]

Appendix B—Project Test Report

Overall Summary

The recovery system was extensively tested in the assembly phase, ground test, and flight test. During the assembly of the bay, wiring was checked for both correct assembly and functioning redundancy with a multimeter. Before live charges were tested, voltage was tested with a multimeter to verify correct deployment function of the altimeters. Multiple ground tests were performed to verify chute deployment. Several tests on both main and drogue were completed until ejection occurred at sufficient velocity to guarantee successful deployment during flight. The charge sizes started with calculated values and were stepped up until adequate. During the test flight, both parachutes deployed at the correct altitude and redundant function was verified as all charges both primary and secondary were ignited.

Ground Test Demonstration [DTEG 6.13.4]

This section describes the ejection of ground testing conducted prior to the test flight to verify that both parachute bays would reliably separate using the selected black powder charges. Multiple iterations of testing were performed to determine the optimal charge of masses, and the results of those trials are documented below.

Prior to ground testing, initial black powder charge estimates were calculated based on parachute bay volume using an online black powder calculator that has also been utilized in previous years. The calculated starting values were 2.5 g for the primary drogue charge and 5.0 g for the backup drogue charge, along with 3.0 g for the primary main charge and 6.0 g for the backup main charge. These estimates were also influenced by the charge construction method, as all black powder charges were tightly packed in rubber glove fingertips and contained two e-matches per charge to improve ignition reliability.

The e-match wiring configuration is as follows:

- Primary drogue charge connected to the primary altimeter (AIM 3.0) and tertiary altimeter (RRC3+)
- Backup drogue charge connected to the secondary altimeter (Stratologger CF) and tertiary altimeter
- Primary main charge connected to the primary altimeter and tertiary altimeter
- Backup main charge connected to the secondary altimeter and tertiary altimeter

Additional details can be found in the recovery wiring diagram. Each parachute bay contains two black powder charges: one primary charge and one backup charge.

The primary drogue bay charge calculations were based on a tube with a 6-inch diameter and 9-inch length, with a target pressure of 15 psi. The primary main bay calculations were based on a 6-inch diameter and 15-inch length tube, also with a target pressure of 15 psi. Backup charges were designed to contain approximately twice the mass of their corresponding primary charges to provide sufficient redundancy.

Type Deployment/ Ejection Testing
Date March 4 2026
Description
Deployment testing of flight computers lighting e-matches on primary drogue charge. Fully integrated test confirming proper wiring, adequate charge weights, mechanical construction, integration, and proper body tube separation.
Result
One test completed for primary drogue charge. Successful firing of approximately 1.56-gram black powder charge from Blue Raven flight computer, unsuccessful separation of body tubes from one another.
Action items Increase amount of black powder used

Type Deployment/ Ejection Testing
Date March 4 2026
Description
Deployment testing of flight computers lighting e-matches on the primary charge for drogue. Fully integrated test confirming proper wiring, adequate charge weights, mechanical construction, integration, and proper body tube separation.
Result
One test completed for primary drogue charge. Successful firing of approximately 2.5g black powder charge from Blue Raven flight computer, successful separation of body tubes from one another. Additionally a chosen back up in the secondary charge for the main parachute bay was 5g

Action items None needed

Type Deployment/ Ejection Testing
Date March 5 2026
Description
Deployment testing of flight computers lighting e-matches on primary main charge. Fully integrated test confirming proper wiring, adequate charge weights, mechanical construction, integration, and proper body tube separation.
Result
One test completed for primary main charge. Successful firing of approximately 2.5g black powder charge from Blue Raven flight computer, unsuccessful separation of body tubes from one another.
Action items Increase amount of black powder used

Type Deployment/ Ejection Testing
Date March 5 2026
Description
Second deployment testing of flight computers lighting e-matches on primary main channel. Fully integrated test confirming proper wiring, adequate charge weights, mechanical construction, integration, and proper body tube separation.

Result

Successful firing of approximately 3 g gram black powder charge from Blue Raven flight computer, successful separation of body tubes from one another. Additionally a chosen back up in the secondary charge for the main parachute bay was 6g

Action items None needed

Flight Test Demonstration [DTEG 6.13.5]

The flight test successfully demonstrated proper parachute deployment and validated the performance of the recovery system. It also confirmed that the selected black powder charge masses were appropriate for the parachute bay volumes. Additionally, the successful recovery of the rocket validated the structural integrity of the aerostructure throughout flight and recovery operations.

Type Flight Test

Date March 7 2026

Description

Result

Successful drogue ejection at appogee and main ejection at 1000 feet. The rocket was recovered in premium condition with little to no repairs necessary. All the black powder had already gone off with no live energetics.

Action items Small fixes to body tubes and cleaning of the recovery bay. New drogue needed and additional dog barf necessary in the parachute bay. Additional swivel added to the drogue end y harness.

Dual Redundancy of Recovery Electronics [R&R 2.6.2.10.a.]

The recovery system incorporates triple-redundant recovery electronics consisting of the primary altimeter (RRC3+), secondary altimeter (Stratologger CF), and tertiary altimeter (Blue Raven). Utilizing three different flight computers provides dissimilar redundancy and helps mitigate common failure modes. In the event of a single altimeter failure during flight, dual redundancy is still maintained to ensure reliable deployment capability.

Each altimeter operates on an independent power and arming circuit consisting of a dedicated 9V battery and key switch. The batteries are connected using battery cap connectors, secured in clip holders, and additionally restrained to the avionics board using cross-pattern zip ties. All components were selected with consideration for the high-acceleration and shock environment experienced during launch and recovery operations.

To reduce susceptibility to electromagnetic interference and improve cable organization, all wiring within the recovery bay is routed as twisted pairs and labeled with cable flags for identification. Additional reliability measures include the use of waterproof heat shrink and potting material on all soldered connections, as well as ferrules on all wires terminated in screw terminals.

Both parachute bay bulkheads utilize 8-pin aviation connectors to provide secure locking electrical connections throughout the vehicle. The connector pin assignments support redundant wiring paths for each altimeter, allowing multiple independent current paths to the deployment charges in the event of a wiring fault or terminal back-out.

The recovery system also includes two independent Featherweight GPS trackers for vehicle tracking and recovery. One tracker is mounted within the recovery bay, while the second is located in the nose cone. Each GPS tracker has an independent battery and arming switch. The trackers are powered by 2.6 Ah 1S Li-ion 18650 battery packs, providing an estimated operational duration of approximately 29 hours per tracker. This estimate was calculated using the Featherweight GPS tracker datasheet specification that a 400 mAh 1S LiPo battery provides approximately 4.5 hours of runtime.

Figure 29 and Figure 30 showing the complete recovery bay wiring diagram are provided on the following pages.

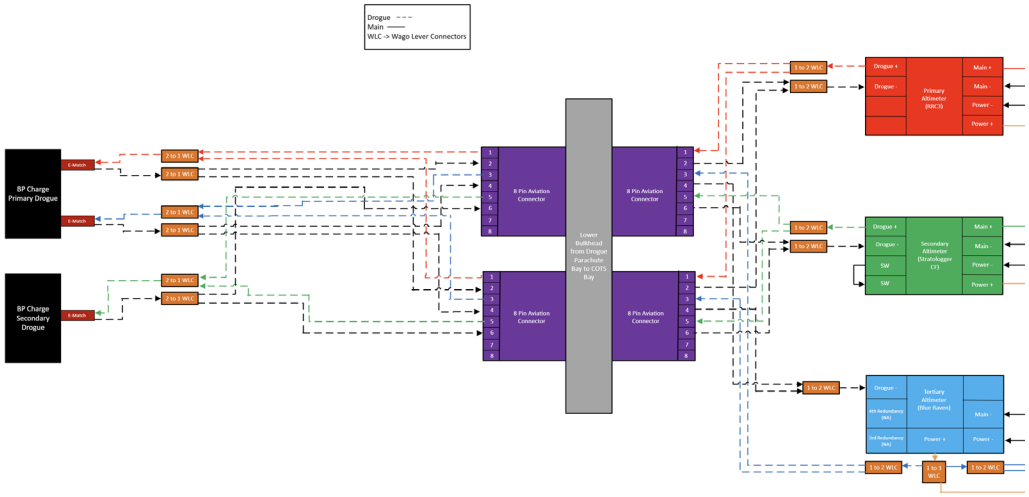


Figure 29 Full Signal Wiring Diagram

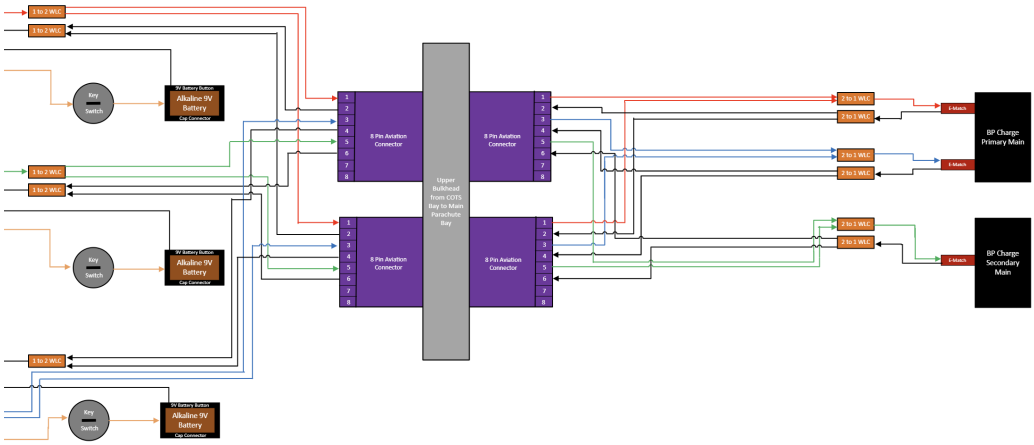


Figure 30 Full Power Wiring Diagram

Stored-Energy Devices – Energetic Device Safing and Arming [DTEG 6.14]

In order to have the energetics considered stored and safe we have two different necessary events before the black powder could be ignited. The first is that our flight computers are programmed for the correct altitudes for the line to the ematch to go off. The second is the key switches that keep our circuit open until the rocket is ready to be armed. Our circuit have been ensured with a multimeter that when the key switches are in the off postion there is no voltage running through the charges. As a last consideration a battery drain test was done to ensure throughout the launch the batterys will be working effectively so no mishaps run through our circuit.

The battery drain test

Type	Battery Drain
Date	14 February 2025

Description

Test flight computers' battery drain characteristics at flight temperatures to gauge longevity.

Result

All three flight computers sustained ready for launch state for over six hours at launch conditions (~105°*F*) with an indistinguishable decrease in battery voltage.

Action items None needed.

Arming Devices [DTEG 6.15]

The method for arming the energetics is key switches which simply requires one member of the pad team to insert a key and turn it clockwise when the rocket is vertical to activate the altimeters. This was tested throughout the integration process with continuity testing to ensure the circuits would properly power when the key switches closed the circuit.

Type Continuity Testing on Recovery Bay

Date 27 Febuary 2026

Description

Verification of the continuity test done on the recovery bay, this test was done on the fully finalized wiring of the recovery system which checked the continuity from the terminal on the altimeter to both the bulkheads AV connectors. This was done for main and drogue, positive and negatives lines.

Result

Successful continuity was found between all connection points by use of a digital multimeter.

Action items

None needed.

Arming Device Verification [DTEG 6.16]

The arming of the flight computers is only done once the rocket is vertical and is done with key switches that close the power circuit to the altimeters. Once this is done each altimeter will read out a series of beeps to prove it is flight ready. Throughout the design and integration phases these arming verification noises from the altimeters have been tested in varying noise levels. Especially during ground testing and the test launch when the rocket is fully assembled to replicate launch day conditions. Through each test the altimeter beeping could be heard with our heads at a minimum of 200 mm away.

b. Flight Computer Design Verification and Analysis

To ensure the v1.4 layout met the team's design standards and was suitable for manufacturing, the Hardware subteam performed three key analyses: a power distribution analysis, and a crosstalk analysis. These analyses helped validate the redesigned layout before committing to fabrication.

The power analysis was performed using Keysight Power Delivery Network Analyzer, which verified that the PCB layout can support the expected current demands. As shown in Figure 31, all power nets passed the full report with voltage drops and current densities within acceptable limits. The analysis confirmed that the power pours in the layout were sized appropriately to handle the current flow through them.

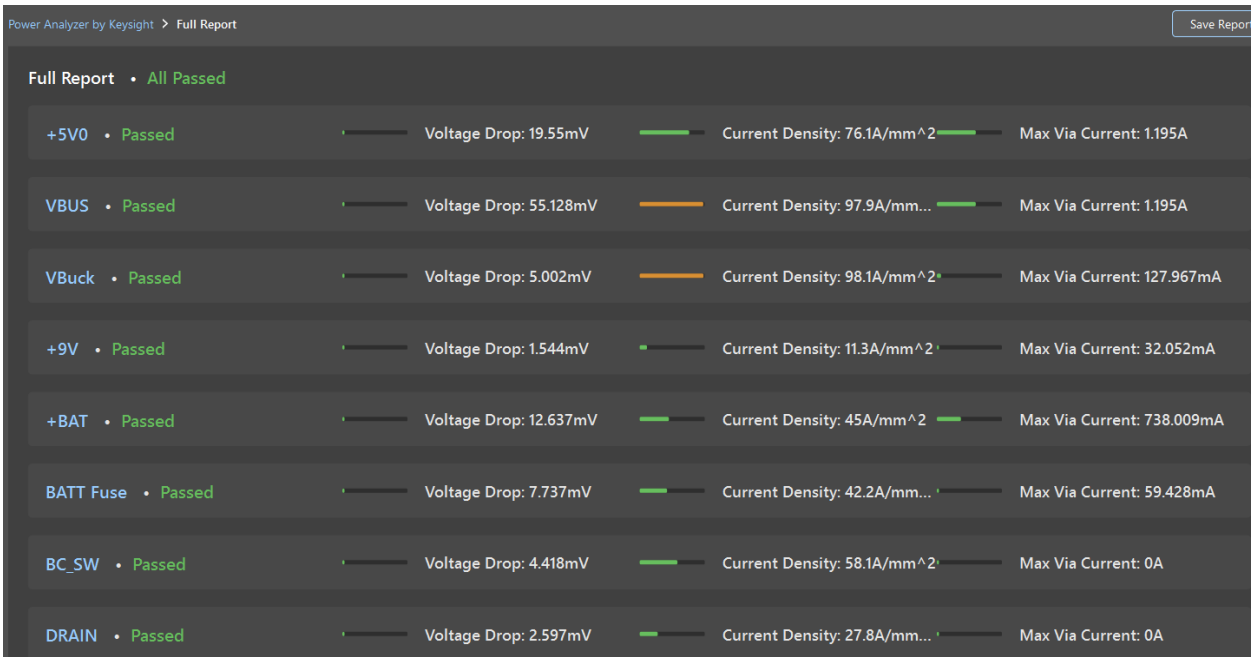


Figure 31 Power distribution analysis results for MARTHA v1.4

The crosstalk analysis examined coupling between adjacent signal traces, particularly on the UART bus where high-speed data lines run in close proximity. The redesigned layout increased trace spacing and applied the 3W rule more consistently than v1.3, resulting in crosstalk levels well below thresholds that could cause data corruption. As shown in Figure 32, the simulated waveforms on nets USART_1RX and USART_1TX confirm this; the noise observed on J2-5 and U2-42 (TX lines) during RX signal transitions peak at approximately 50-300 mV. It remains within acceptable noise margins and posing no risk of data corruption. In the simulation, the RX lines (J2-4 and U2-43) act as the aggressors, while the idle TX lines (J2-5 and U2-42) act as the targets, with the induced noise visible on the transition.

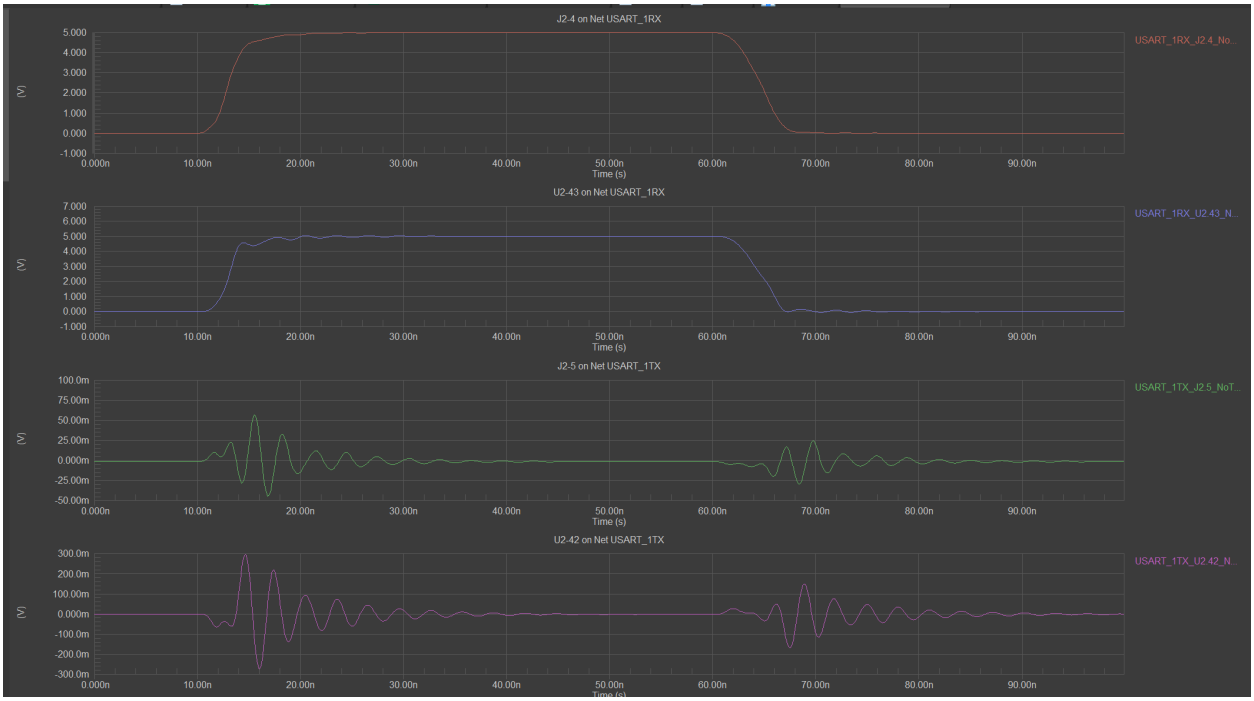


Figure 32 Crosstalk analysis results for UART

c. Software Testing

In addition to the unit tests enforced by our CI/CD pipelines, we test our flight software through data injection and test flights. For data injection, we stream previous flight data into a UART port of MARTHA who is compiled with a build flag to read sensor data from that UART stream rather than from the actual sensors. All of the tooling for this data injection testing is defined in the Avionics library, allowing our other flight computers to be tested this way as well. After every section of flight data is sent from the host machine to the flight computer, the filght computer responds with an acknowledgement and the status of its state machine. We must use a live stream of data because the flight computer does not have enough storage space to both hold an entire previous flight of data while still being allowed to use its flash chip normally so that we can analyze data logging behavior. At the end of the stream, we can analyze the live state machine data as well as dump the flash chip to ensure all data was logged properly and that launch, apogee, and other states were detected at the correct times. Through the data injection test, we exercise the entire software stack except for the sensor drivers (sensor drivers are tested elsewhere using unit tests).

MARTHA flew on two rockets this academic year. Once on an L2 certification rocket in November which validated our live telemetry system and again when we tested this competition vehicle in March. For both launches, launch and apogee detection performed nominally. For the March launch, we validated the new battery voltage monitoring tool.

c. Telemetry Testing

The telemetry system has been tested on two flights. In our initial live-telemetry test flight to approximately 1,300 feet, MARTHA successfully streamed data to the ground station at 2 Hz, and the dashboard provided enough real-time information for the team to identify apogee and parachute events before recovery. In the team's test flight, the telemetry rate was increased to 5 Hz and the transmit power was reduced to 25 dBm, yet performance remained nominal, with the connection only being lost as the rocket neared the ground and broke line of sight. During this second flight, we received a complete 5 Hz telemetry log, which would be highly valuable in the event that the onboard 100 Hz log was not recovered. Both test flights also helped us identify major issues with the ground station GUI, including poor direct-sunlight contrast, graph axis update problems, and slow reconnection behavior. These issues have since been addressed, and many of the fixes have already been validated.

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Appendix C—Hazard Analyses

Appendix C: Hazard Analysis offers an in-depth examination of potential hazards associated with the project, focusing on aspects such as hazardous material handling, transportation, and storage of propellants, among other safety-critical design components. This analysis is designed to ensure the safety of all operating personnel by meticulously detailing the risks involved and presenting effective mitigation strategies for each identified hazard. By methodically addressing both procedural and design-related safety issues, this appendix aims to minimize risks and enhance overall operational safety, adhering to regulatory compliance and engineering best practices. Our hazard matrix can be seen in Table 11, below.

Table 11 Hazard analysis of *CU Up There*

Team 91	***CU Up There***			**5/01/2026**
Hazard	**Possible Causes**	**Risk of Mishap and Rationale**	**Mitigation Approach**	**Risk of Injury after Mitigation**
Epoxy lands in eyes, on skin, or is ingested during manufacturing process	Improper Eye Protection	Medium; Large amounts of epoxy are used throughout manufacturing process, so proper PPE is a must	High attention detail and strict enforcement of PPE measures while handling epoxy resin or hardener	Low.
	Improper glove and lab coat skin coverage			
	Improper face and respirator use			
Explosion of solid-propellant rocket motor during launch with blast or flying debris causing injury	Motor end closures fail to hold	Medium; Visual inspections are the most that can be done, but there may be subsurface cracking that can only be detected with ultrasound.	Inspect grains and ensure they are in proper condition.	Low.
	Motor case unable to contain normal operating pressure		Inspect motor case for damage during final assembly before launch	
	Cracks in propellant grain		Only essential personnel in launch crew will be a minimum 500 feet from rocket at launch, in abidance with TRA guidelines. However, ESRA has a system for personnel to be 2000 feet, decreasing possibility of injury by a factor of 16.	
Vehicle deviates from nominal flight path, intercepts personnel at high speed	Under stable vehicle steering into high wind as it departs from rail	Medium; Payload weighed less than expected during test launch.	Weight has been added to the payload to ensure a stable flight.	Low.
	Over stable vehicle tipping over as it departs from rail		Refrain from launching in winds exceeding 20 *mph*, but design launch vehicle to remain	

			nominal in winds above this limit	
	Coupler failure on motor ignition		Aero-structures have been checked for any crack propagation and replaced as necessary.	
Recovery system fails to deploy, vehicle or payload comes in contact with personnel at an excessive speed	Recovery Systems overheat.	Medium; while battery drain tests have been performed high heat can cause lack of hardware and power function	Thermal protection via deployment bag.	Low.
	Power malfunction.		Ensure recovery systems are flight tested and successful.	
	Wet black powder.		Follow proper checklists, ensuring all steps are followed.	
Recovery system partially deploys, launch vehicle comes in contact with personnel	Recovery Systems overheat.	High; due to previous launches not including in-depth checklists, drogue or main non-deployment has occurred.	Thermal protection via deployment bag.	Medium.
	Power malfunction.		Ensure recovery systems are flight tested and successful.	
	Wet black powder.		Follow proper checklists, ensuring all steps are followed.	
Black powder charges prematurely ignite outside the launch vehicle.	Improper handling of black powder	Low; black powder is stored and transported in airtight, AC-controlled locations. Black powder is only handled by experienced personnel.	Ensure black powder is storage container is correctly sealed.	Low.
	Improper storage of black powder		Ensure black powder is secured during transportation.	
Black powder charges prematurely ignite inside the vehicle.	Improper handling of black powder	Medium; High heat may cause electronic failure or other hazards. Yanking of e-matches can cause premature activation.	Ensure black powder is storage container is correctly sealed.	Low.
	Improper storage of black powder		Ensure cautious handling of e-matches.	
	E-Matches prematurely go off.		Ensure following of proper checklists.	

Rocket motor does not ignite when given command, but ignites during investigation.	Delay of e-match signal.	Medium; risk of accident can always occur.	Proceed with extra caution when approaching a failed motor ignition.	Medium.
			Wait minute after attempted motor ignition, as required by TRA guidelines.	
			Encourage competition officials to maintain proper handling of their Wilson FX system which has a double relay safety mechanism.	
Recovery system deploys during assembly or prelaunch, causing injury.	Flight computers are prematurely powered and engage.	Medium; Due to the nature of black powder and energetics, danger is always there.	Ensure electronics bay does not stop beeping.	Low
	E-match prematurely activated, causing deployment.		Independent switches for flight computer power on and arming charges.	
			Provide enough slack in e-matches and do not pull on them.	

Appendix D—Risk Assessment

Appendix D: Risk Assessment provides a comprehensive overview of the risks and reliability concerns pertinent to the mission. This section meticulously catalogs all identified failure modes that could jeopardize mission success. It presents these in a structured matrix aligned with the mission phases outlined by the CONOPS. A detailed mitigation strategy is elaborated for each risk identified, addressing potential issues through specific process improvements and design modifications. This appendix aims to ensure that participants are fully informed of the potential challenges and the proactive measures to mitigate these risks, thereby enhancing the project's overall reliability and success rate. The values for likelihood are dependent on historical data of all teams flying at the IREC. The varying levels of likelihood and impact can be seen in Table 12.

Table 12 Risk level designations

Level	Likelihood	Impact
1	Minimal Chance (1-2%)	No effect on the mission
2	Possible (2-5%)	Minor impact on technical performance
3	Probable (5-10%)	Affects mission success
4	Highly Possible (10-20%)	Significant risk to mission critical systems
5	Definite Chance (>20%)	Human Risk

Table 13 presents a detailed Risk Assessment for the project. It outlines the potential failure modes of various components and systems, their effects, inherent risk levels, and the respective mitigation plans to reduce these risks. Each entry in the table identifies a specific potential failure, describes the possible consequences of such a failure, assesses the risk in terms of likelihood and impact, and details the actions taken to mitigate the risk to more manageable

levels. The mitigation strategies encompass procedural adjustments, enhanced manufacturing practices, rigorous testing, and quality control measures. This structured approach ensures that risks are systematically identified, assessed, and mitigated to support the stability and safety of the project. The table also illustrates the effectiveness of the mitigation strategies by comparing initial and mitigated risk levels, thereby highlighting the tangible benefits of the risk management processes employed.

Table 13 Risk assessment table for *CU Up There*

Potential Failure Mode	**Potential Effects of Failure**	**Risk Levels**	**Mitigation Plan**	**Mitigated Risk**
Improperly cut Body Tube	Higher Drag, non-rigid body joints	Likelihood: 2 Impact: 1	Sanding of Body Tube edges, smooth surface finish	Likelihood: 1 Impact: 1
Crack propagation from holes in airframe	Loss of rigidity in body tube, structural failure of body tube	Likelihood: 2 Impact: 2	Reinforce holes with epoxy	Likelihood: 1 Impact: 2
Improperly secured fins	Loss of fin, catastrophic loss of vehicle	Likelihood: 2 Impact: 4	High attention to detail throughout manufacturing process, through-wall and tip to tip layup	Likelihood: 2 Impact: 4
Improperly secured bulkheads	Shift of Avionics or Payload, loss of stability	Likelihood: 4 Impact: 2	High attention to detail, testing of each bulkhead	Likelihood: 1 Impact: 1
Fins bent or damaged	Erratic flight behavior	Likelihood: 4 Impact: 3	Use of fin jig and level, and careful storage during transport. Fin covers made from pool noodles to protect tips.	Likelihood: 2 Impact: 3
Centering Rings out of alignment	Erratic flight behavior	Likelihood: 3 Impact: 2	Quality control of manufactured angles during construction. Verification of proper alignment with test flight.	Likelihood: 1 Impact: 2
Avionics wires caught or snagged during integration	Damage to avionics systems	Likelihood: 4 Impact: 1	Use of integration lists and visual inspection throughout integration process. Ensure properly lacked wire runs and quick connectors.	Likelihood: 2 Impact: 1
Payload detaches during descent	Loss or damage to payload	Likelihood: 2 Impact: 1	Flight tested to ensure connection.	Likelihood: 1 Impact: 1
Motor retention fails	Motor falls out of vehicle after boost	Likelihood: 3 Impact: 3	Secure forward and aft retention with epoxy and screws, forward eye bolt fully screwed in	Likelihood: 1 Impact: 3
Drogue parachute becomes entangled	Higher than expected descent velocity	Likelihood: 1 Impact: 2	Careful packing of drogue parachute, deploy main parachute prior to excessive	Likelihood: 1 Impact: 2

			velocity and flight test the software.	
Main parachute fails to fully deploy	Higher than expected descent velocity	Likelihood: 4 Impact: 3	Careful packing of main parachute using properly stored packing bag, checklist of line connections	Likelihood: 3 Impact: 2
Main parachute deploys at apogee	Excessive drift of vehicle	Likelihood: 4 Impact: 1	Careful packing of main parachute, ejection testing, testing of shear pins. Ensure properly sized ejection charges to prevent hammering. Perform realistic ground tests followed by flight tests.	Likelihood: 3 Impact: 1
Arming switch fails and a flight computer turns off.	Minor loss of data, due to two other redundant flight computers.	Likelihood: 2 Impact: 1	Ensure switches are fully armed in place and are connected to avionics bay correctly.	Likelihood: 1 Impact: 1
Recovery Systems Overheat on the Pad.	Potential loss of vehicle, no chute deployment.	Likelihood: 3 Impact: 4	Ensure recovery systems are flight tested and there is temperature mitigating finish to the vehicle. Arrive and launch early.	Likelihood: 2 Impact: 3
Loss of GPS signal from trackers within vehicle during flight.	Potential loss of vehicle.	Likelihood: 3 Impact: 3	Dual redundancy of trackers, eliminating single point of failure. Monitor battery voltage of GPS throughout preflight and launch. Use of RF transparent materials in line of the antenna's toroidal propagation pattern. Screen record phone GPS interface during flight to record path of vehicle throughout flight. Use easily distinguishable colors to increase vehicle's visual footprint on landing	Likelihood: 1 Impact: 2
No ignition from e-match charges.	No motor ignition.	Likelihood: 3 Impact: 3	Ensure e-matches are correctly attached. Follow pre-launch checklists.	Likelihood: 2 Impact: 2
Premature motor ignition.	Potential motor explosion and human injury.	Likelihood: 2 Impact: 5	Ensure e-matches are correctly attached. Follow pre-launch checklists. Coordinate with ESRA Solid PAS	Likelihood: 1 Impact: 4

			personnel to ensure the Wilson FX is disarmed prior to installing igniter.	
Parachute deploys at motor burnout.	Destruction of vehicle.	Likelihood: 1 Impact: 4	Ensure Avionics correctly detects apogee, using applicable filters and delays Ensure barometric holes are properly sized and positioned. Ensure forward retainer on motor is secure. Avionics are flight tested.	Likelihood: 1 Impact:3
Motor fails to burn all fuel during lift-off.	Potential failure to apogee.	Likelihood: 3 Impact: 2	Ensure COTS motor is structurally sound and grain is good.	Likelihood: 2 Impact: 2
Vehicle rail guide breaks during ignition.	Potential human risk and vehicle deviation.	Likelihood: 4 Impact: 5	Ensure rail guide is properly installed and follow preflight checklists. Ensure rail and rail guide are properly aligned before launch.	Likelihood: 2 Impact: 4
Motor does not produce enough thrust for lift-off.	Potential failure to apogee or failure to launch.	Likelihood: 2 Impact: 3	Ensure COTS motor is structurally sound and grain is good. Ensure motor simulations are correct. Use a trusted motor vendor and a certified M-class motor.	Likelihood: 1 Impact: 2
Vehicle deviates from nominal flight path.	Potential human risk.	Likelihood: 4 Impact: 5	Ensure rail is properly fastened and follow preflight checklists. Angle rails away from flight line and in accordance with TRA guidelines.	Likelihood: 2 Impact: 3
Trailer involved in crash.	Potential human risk. Potential risk to launch vehicle.	Likelihood: 3 Impact: 5	Ensure to travel with a diverse Subteam pool. Follow all driving laws and regulations. Ensure trailer and towing vehicle are properly fitted for travel.	Likelihood: 2 Impact: 4
Team transportation is involved in an accident.	Potential human risk.	Likelihood: 2 Impact: 5	Ensure to travel with a diverse Subteam pool. Follow all driving laws and regulations. Ensure travel vehicle is properly fitted for travel.	Likelihood: 2 Impact: 4
Organization funds run out.	Potential risk of mission failure.	Likelihood: 5 Impact: 4	Team properly budgets each fiscal year and	Likelihood: 1 Impact: 1

	Potential to have no launch vehicle.		adheres to a strict budget. Enough funds are in excess to allow for emergency funds as necessary.	
Team runs out of time for vehicle manufacturing.	Potential risk of mission failure via nonflight.	Likelihood: 4 Impact: 4	Ensure vehicle is assembled and undergo a successful test flight with enough time for manufacturing and assembly after flight.	Likelihood: 1 Impact: 1

Appendix E—Checklists

Final Assembly

Completed by	
Date & Time	

Assembly Step Procedure:

A.1 Ejection Charge Preparation ☐

1.a. Drogue Primary ☐

i Apply proper PPE per ESRA specifications ☐

ii Prepare rubber glove finger, tape, two e-matches, and precision scale ☐

iii Check continuity of each e-match, visually inspect for damage ☐

iv Measure out 2.5 *grams* of 4F black powder, close black powder container ☐

v Pour black powder into glove finger ☐

vi Insert both primary drogue e-matches into powder inside the glove finger ☐

vii Tape glove finger tightly closed and wrapped, ensuring e-matches are secured, and black powder is sealed and can hold until the BP is burnt ☐

viii Place assembled ejection charge into charge well ☐

ix Clearly label charge as "DROGUE PRIMARY CHARGE" ☐

1.b. Drogue Secondary ☐

i Apply proper PPE per ESRA specifications ☐

ii Prepare rubber glove finger, tape, two e-matches, and precision scale ☐

iii Check continuity of each e-match, visually inspect for damage ☐

iv Measure out 5 *grams* of 4F black powder, close black powder container ☐

v Pour black powder into glove finger ☐

vi Insert both secondary drogue e-matches into powder inside the glove finger ☐

vii Tape tightly closed glove finger and wrapped, ensuring e-matches are secured, and black powder is sealed and can hold until the BP is burnt ☐

viii Place assembled ejection charge into charge well ☐

ix Clearly label charge as "DROGUE SECONDARY CHARGE" ☐

1.c. Main Primary ☐

i Apply proper PPE per ESRA specifications ☐

ii Prepare rubber glove finger, tape, two e-matches, and precision scale ☐

iii Check continuity of both e-matches, visually inspect for damage ☐

iv Measure out 3 *grams* of 4F black powder, close black powder container ☐

v Pour black powder into glove finger ☐

vi Insert both primary main e-matches into powder inside the glove finger. ☐

vii Tape glove finger closed, ensuring e-matches are secured, and black powder is sealed ☐

viii Clearly label the charge as “MAIN PRIMARY CHARGE” ☐

1.d. Main Secondary ☐

i Apply proper PPE per ESRA specifications ☐

ii Prepare rubber glove finger, tape, two e-matches, and precision scale ☐

iii Check continuity of e-matches, visually inspect for damage ☐

iv Measure out 6 *grams* of 4F black powder, close black powder container ☐

v Pour black powder into glove finger ☐

vi Insert both secondary main e-matches into powder inside the glove finger ☐

vii Tape glove finger closed, ensuring e-matches are secured, and black powder is sealed ☐

viii Clearly label charge as “MAIN SECONDARY CHARGE” ☐

A.2 Parachute Preparation ☐

2.a. Drogue Parachute Preparation ☐

i Visually inspect each panel of drogue parachute for rips, tears, or punctures ☐

ii Visually inspect full length of each shroud line and shock cord for severed, frayed, torn, or brittle sections ☐

iii Lay drogue parachute flat on a stable, clean working surface sheltered from wind and dust ☐

iv Following manufacturer guidelines, fold drogue parachute in half such that the shroud lines align cleanly, and there are no large wrinkles in the parachute ☐

v Fold towards the center panel by panel, such that the shroud lines remain organized, until it meets the shape of a triangle with a rounded bottom edge ☐

vi Perform a Z-fold such that the top point of the triangle remains on top of the folded parachute ☐

vii Perform a Z-Fold of the shroud lines, ensuring lines remain untangled ☐

viii Wrap final length of shroud lines once, and no more, around folded drogue parachute such that the parachute exits the parachute bay prior to unfolding ☐

ix Group drogue parachute and lines together, and set in sheltered, clean place ☐

2.b. Main Parachute Preparation ☐

i Visually inspect each panel of main parachute for rips, tears, or punctures ☐

ii Visually inspect the full length of each shroud line and shock cord for severed, frayed, torn, or brittle sections ☐

iii Lay main parachute flat on a stable, clean working surface sheltered from wind and dust ☐

iv Following manufacturer instructions, fold main parachute ☐

v Carefully insert main parachute into deployment bag ☐

vi Following manufacturer instructions, organize shroud lines onto deployment bag ☐

vii Set packed main parachute in sheltered, clean place ☐

2.c. Shock Cord Preparation ☐

i Insert a quick link through loops placed at the ends of the shock cord and in the knot ☐

ii Place the swivel at both ends of the shock cords with an additional quick link to eventually attach to the eyebolts ☐

iii Fold the shock cords in a Z fold and wrap one wrap of painter's tape around the fold to hold it in place ☐

iv Repeat again for the second shock cord ☐

A.3 Ballast Preparation ☐

3.a. Pre-Assembly ☐

i Gather 4 ¼-20 x ¼ *in* hex drive screws ☐

ii Gather 3D printed Ballast box ☐

iii Gather 3D printed Ballast lid ☐

iv Gather rolls of pennies ☐

3.b. Assembly ☐

i Slide the necessary amount of penny rolls into ballast box circular slots ☐

ii Fasten ballast lid on utilizing 4 ¼-20 x ¼ *in* hex drive screws ☐

3.c. Post-Assembly Checklist ☐

i Ensure ballast lid is intact to ballast box ☐

A.4 Recovery Electronics Bay Preparation ☐

4.a. Pre-Assembly ☐

i Gather recovery electronics internals assembly ☐

ii Gather lower recovery bulkhead ☐

iii Gather Upper recovery bulkhead ☐

iv Gather coupler, airframe, and hardstop ring assembly ☐

v Gather six ¼-20 washers ☐

vi Gather two 3/8 *in* threaded rod ☐

vii Gather four charge wells ☐

viii Gather four eyebolts ☐

ix Gather fourteen 3/8 *in* nut ☐

x Gather four thin-headed ¼-20 x 1 *in* bolt ☐

xi Gather four ¼-20 lock nuts ☐

4.b. Assembly ☐

i Fasten two charge wells to each upper & lower bulkhead utilizing ¼-20 x 1 *in* bolt and lock nuts ☐

ii Fasten two eyebolts to each upper and lower bulkheads utilizing 3/8 *in* nuts ☐

iii Slide both threaded rods through designated holes in lower recovery bulkhead, fasten a 3/8 *in* nut on each side of the bulkhead ☐

iv Slide recovery electronics internals onto threaded rods, connect aviation connectors to bulkhead

v Fasten 3/8 *in* nuts onto threaded rods to hold internals in place ☐

vi Slide assembled pieces into coupler-airframe-ring assembly where the bulkhead is resting on the inside coupler ☐

vii Connect aviation connectors to upper recovery bulkhead, slide bulkhead onto threaded rods from opposite side of coupler-airframe-ring assembly; bulkhead will be resting on ring ☐

viii Fasten two 3/8 *in* bolts to protruding threaded rod from the upper bulkhead, once fastened, bay will be locked into coupler-airframe-ring assembly ☐

4.c. Post-Assembly ☐

i Ensure key switch faces line up with pre-cut holes in the airframe ☐

ii Ensure eye bolts are securely fastened ☐

iii Ensure recovery internals bay cannot move along he threaded rods ☐

iv Ensure charge wells are tightly attached ☐

A.5 Internal Spars and Hard-Stop Section Preparation ☐

5.a. Pre-Assembly ☐

i Gather one pre-drilled air frame ☐

ii Gather one pre-drilled coupler ☐

iii Gather two carbon fiber spars ☐

iv Gather two spar slot bulkheads ☐

v Gather one hard stop bulkhead ☐

vi Gather one nose cone bulkhead ☐

vii Gather eight aluminum 1*in* x 1*in* gussets ☐

viii Gather two aluminum, hard stop rings ☐

ix Gather one hard stop gasket ☐

x Gather one 3/8 *in* threaded rod ☐

xi Gather 3/8 *in – 1/4 in* thread adapter ☐

xii Gather 16 3/4 *in* ¼-20 bolt ☐
xiii Gather 16 ¼-20 nut ☐
xiv Gather eight 3/4 *in* ¼-20 hex-drive bolts ☐
xv Gather two eyebolts ☐
xvi Gather three 3/8 *in* nuts ☐
xvii Gather one 3/8 *in* threaded flange ☐

5.b. Assembly ☐

i Fasten two eyebolts to hard stop bulkhead with 3/8 in nuts. ☐
ii Slide pre-drilled coupler into predrilled airframe ensuring holes line up. ☐
iii Slide in first (lower) hard stop ring, line up radial holes and bolt in utilizing four 3/4 *in* ¼-20 hex-drive bolts ☐
iv Place gasket and hard stop bulkhead down sequentially resting on the fastened aluminum ring. ☐
v Fasten second hard stop in above hard stop bulkhead utilizing four 3/4 *in* ¼-20 hex-drive bolts ☐
vi Holding two spars evenly and parallel to each other, slide on two spar slotted bulkheads, slide bulkheads to the correct length for specified bay heights. ☐
vii For each spar slotted bulkheads, fasten two gussets at each spar to bulkhead junction on one side of the bulkheads, this will be done utilizing two 3/4 *in* ¼-20 bolts and nuts for each gusset. ☐
viii At the top of spar assembly, fasten the nose cone bulkhead utilizing gussets as done in step *vii* ☐
ix At bottom of spar assembly, fasten downward facing gussets utilizing two 3/4 *in* ¼-20 bolts and nuts ☐
x Slide spar assembly into coupler of hard stop section, fasten downward facing gussets to hard stop bulkhead utilizing two 3/4 *in* ¼-20 bolts and nuts ☐
xi Fasten 3/8 *in* threaded flange to top of nosecone bulkhead ☐
xii Insert 3/8 *in* threaded rod into flange, fasten 3/8 *in* nut to the threaded rod on the opposite side of the bulkhead thus locking in threaded rod to nose cone bulkhead ☐
xiii Apply Loctite to 3/8 *in – 1/4 in* thread adapter, fasten to top of threaded rod ☐

5.c. Post-Assembly ☐

i Ensure all mechanically fastened pieces are tight ☐
ii Confirm no motion of aluminum rings in the hard stop section ☐
iii Ensure spar slotted bulkheads are in correct locations on spar system. ☐

A.6 Payload Experiment Assembly ☐

6.a. Assemble vials with porcine blood ☐

i Sterilize the vials with isopropyl alcohol in preparation for blood transfer ☐
ii Sterilize all syringes with isopropyl alcohol to be used for blood transfer ☐
iii Use syringes to fill 5mL vials with porcine blood ☐

6.b. Assemble Styrofoam container with blood vials and ice packs ☐

i Place vials into designated circular slots in Styrofoam cube ☐
ii Place ice packs into designated rectangular slots in Styrofoam cube ☐
iii Seal lid on Styrofoam cube using the built-in dowl system, ensuring that the fit is tight ☐

6.c. Set up monitoring and powering devices in payload container ☐

i Place thermocouples into openings on Styrofoam lid so that they are adjacent to blood vials ☐
ii Secure battery in holding stand and screw stand into the wall of the payload container ☐
iii Secure analog-to-digital converter to wall of payload container ☐
iv Ensure that thermocouples and the analog-to-digital converter are connected to MARTHA ☐

6.d. Check that the Styrofoam cube, thermocouples, battery, and analog-todigital converter are secured in the payload container before screwing the container shut. ☐

A.7 Payload Integration ☐

7.a. Pre-Assembly ☐

i Gather assembled spar system with hard stop section ☐

ii Gather assembled SRAD electronics bay ☐
iii Gather assembled 3U CubeSat experiment ☐
iv Gather assembled ballast bay ☐
v Gather 4:1 Ogive nose cone ☐
vi Gather aluminum threaded nose cone tip ☐
vii Gather main body airframe ☐
viii Gather twelve 3/4 *in* ¼-20 hex-drive bolts ☐
ix Gather twelve, 1 *in* diameter washers ☐
x Gather Loctite ☐

7.b. Assembly ☐

i Slide 3U CubeSat into the lowest bay, fasten to spars with four 3/4 *in* ¼-20 hex-drive bolts and four washers ☐

ii Slide ballast assembly into the middle bay, fasten to spars with four 3/4 *in* ¼-20 hex-drive bolts and four washers ☐

iii Slide SRAD electronics bay into the lowest bay, fasten to spars with four 3/4 *in* ¼-20 hex-drive bolts and four washers. ☐

iv Slide main body tube over the thread rod and down over the spar system ☐
v Slide nose cone over threaded rod and mate with main body airframe ☐
vi Apply Loctite to nose cone tip ☐
vii Fasten nosecone tip to lock down nose cone and main body airframe ☐

7.c. Post-Assembly ☐

i Ensure nose cone tip is tightly fastened. ☐

Preflight Checklist

Completed by	
Date & Time	

Lower Rail to horizontal position ☐

a. Remove vehicle from transport truck and carry to pad ☐
b. Remove aluminum blanket wrap protection from vehicle ☐
c. Slide lower rail guide onto rail, ensuring no undue stress on the rail guide, until the upper rail guide slides onto rail ☐
d. Slide vehicle fully down rail ☐
e. Remove nose cone tip protection ☐
f. Turn on and connect to featherweight GPS trackers ☐ ☐
g. Turn on SRAD electronics ☐ ☐

Arming

Completed by	
Date & Time	

Vehicle is vertical and locked in place on pad ☐

a. Arm COTS flight computers ☐
   i. Primary altimeter ☐
      1. Turn key switch to enable altimeter ☐
      2. Listen for successful startup sequence of primary altimeter, RRC3+ ☐
      a. Observe beep pattern for main deployment of 900 *ft* AGL ☐
   3. Listen for beeps indicating continuity on both primary deployment lines ☐
   ii. Secondary altimeter ☐

1. Turn key switch to enable altimeter ☐
2. Listen for successful sequence on secondary altimeter, StratoLogger ☐
   a. Observe beep pattern for main deployment of 950 *ft* AGL ☐
3. Listen for a second set of beeps at a new tone, distinct from RRC3+, indicating continuity on both secondary deployment lines. ☐

iii. Tertiary altimeter ☐
1. Turn key switch to enable altimeter ☐
2. Listen for successful sequence on tertiary altimeter, Blue Raven ☐
3. Listen for a third set of beeps at a new tone, distinct from StratoLogger, indicating continuity on all RRC3+ deployment lines. ☐

iv. Verify all flight computers are on, in ready to launch mode, and have continuity across all charges ☐

b. Arm Motor ☐
i. Insert dual-headed igniter, which is wrapped around a wooden dowel for stability, up the motor ☐
ii. Clip the ends of the igniter to the alligator clips supplied by the competition ☐
iii. Using duct tape, tape down the ends of the igniter to ensure it does not become disconnected due to weather conditions ☐
iv. Ensure continuity in the igniter ☐

Launch Check List

Completed by	
Date & Time	

1. GPS signal loss ☐
   1.a. Have two people quickly move to opposite ends of the launch site to get different vantage points and follow the vehicle's descent closely ☐
   1.b. Use a phone's GPS to mark area where the vehicle was seen by the two members and focus search efforts on the area where each line of sight crosses ☐
2. Motor chuff ☐
   2.a. Replace igniter ☐
   2.b. If second failure, remove vehicle from rails and return to base camp ☐
3. Replace motor ☐

Recovery Checklist

Checked by	
Date & Time	

1. Flight completed:
   a. Include off nominal checklist? **Yes** ☐ **No** ☐
      i. Off nominal points: _____
      _____
      _____
   b. Record vehicle GPS location from primary/secondary Featherweight GPS trackers
      i. Primary GPS location: _____
      ii. Secondary GPS location: _____
   c. Record any flight anomalies:
      i. Unfired black powder charges …….....☐
      ii. Unexpected vehicle fragmentation ……...☐
      iii. Ruptured batteries ……☐
      iv. Miscellaneous: _____
      _____

d. Picked up GPS + radio pack from recovery tent & complete MCC check-in .....☐
e. Route to vehicle approved by team's Safety Officer .....☐
   i. Route to vehicle confirmed by MCC .....☐
f. Recovery team personnel deemed fit by team's Safety Officer .....☐
g. All clear to depart received from MCC .....☐
h. Misc. notes: _____
_____
_____
_____

2. Recovery team deployed:
   a. Include off nominal checklist? **Yes** ☐ **No** ☐
      i. Off nominal points: _____
      _____
      _____
   b. Arrived at planned vehicle exit location .....☐
   c. Confirm GPS locations are within same area as previously.....☐ ☐
      i. Primary GPS location: _____
      ii. Secondary GPS location: _____
   d. Exited recovery vehicle at planned location:
   e. Complete check-in with MCC ……...☐
      i. Time of vehicle exit:
      _____
   f. Begin on foot search for vehicle ……...☐
      i. MCC check-ins every 30 minutes: _____
      ii. Notable events during search: _____
      _____
   g. Misc. notes: _____
   _____
   _____
   _____

3. Vehicle Located:
   a. Include off nominal checklist? **Yes** ☐ **No** ☐
      i. Off nominal points: _____
      _____
      _____
   b. Vehicle spotted …….☐
      i. Time of vehicle spotting: _____
      ii. Visual assessment of possible flight anomalies: _____
      _____
      iii. All parts of vehicle are present ……..☐
         1. Parts not found with vehicle: _____
      iv. All battery systems are intact ……☐
         1. Ruptured batteries: _____
      v. Confirmation by Safety Officer of depletion of all energetics ……..☐
         1. Unspent energetics: _____
         2. All hazards disarmed and removed from vehicle by Safety Officer ……..☐
      vi. All clear to approach vehicle given by team's Safety Officer ……..☐
   c. GPS location of vehicle: _____
   d. Record apogee of primary/secondary recovery altimeters through beep readings …….☐ ☐
      i. Primary altimeter apogee [ft]: _____
      ii. Secondary altimeter apogee [ft]: _____

iii. Recovery altimeters powered off …….☐ ☐

e. If applicable: Removal of vehicle sections from flora …….☐

f. SRAD avionics powered off ……☐

i. Record state indication of primary/secondary SRAD flight computers through LED blinks …..☐

1. Primary final state: _____
2. Secondary final state: _____
3. MARTHA boards powered off ……☐

g. Vehicle fully powered off ……..☐

h. MCC check-ins: _____
_____

i. Misc. notes: _____
_____
_____
_____

4. Return to base camp:

a. Include off nominal checklist? **Yes** ☐ **No** ☐

i. Off nominal points: _____
_____
_____

b. All vehicle sections present at landing area collected **……..**☐

i. Time of departure from vehicle landing area: _____

c. Arrived at recovery vehicle ……..☐

i. Time of arrival to recovery vehicle: _____

ii. All clear from Safety Officer to begin vehicular return to base camp ……..☐

d. Arrival at base camp ……..☐

i. Time of arrival to base camp: _____

ii. Complete final MCC check-in & return GPS + radio pack ……..☐

ii. Brought vehicle to post flight inspection ……..☐

e. Notable events during return: _____
_____

f. Misc. notes: _____
_____
_____

g. Completed recovery of vehicle ……☐

Off-Nominal Checklist

Checked by	
Date & Time	

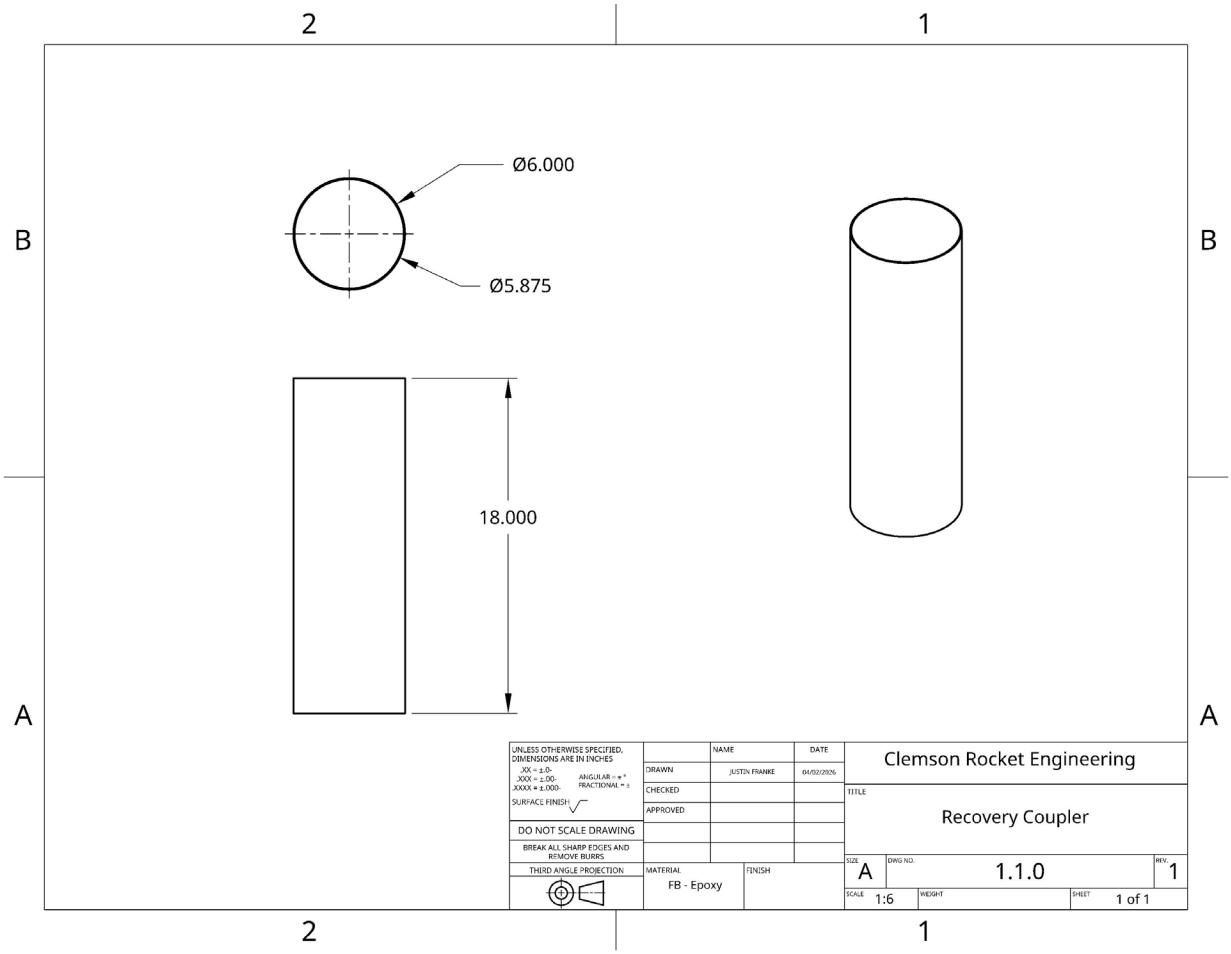
Recovery:

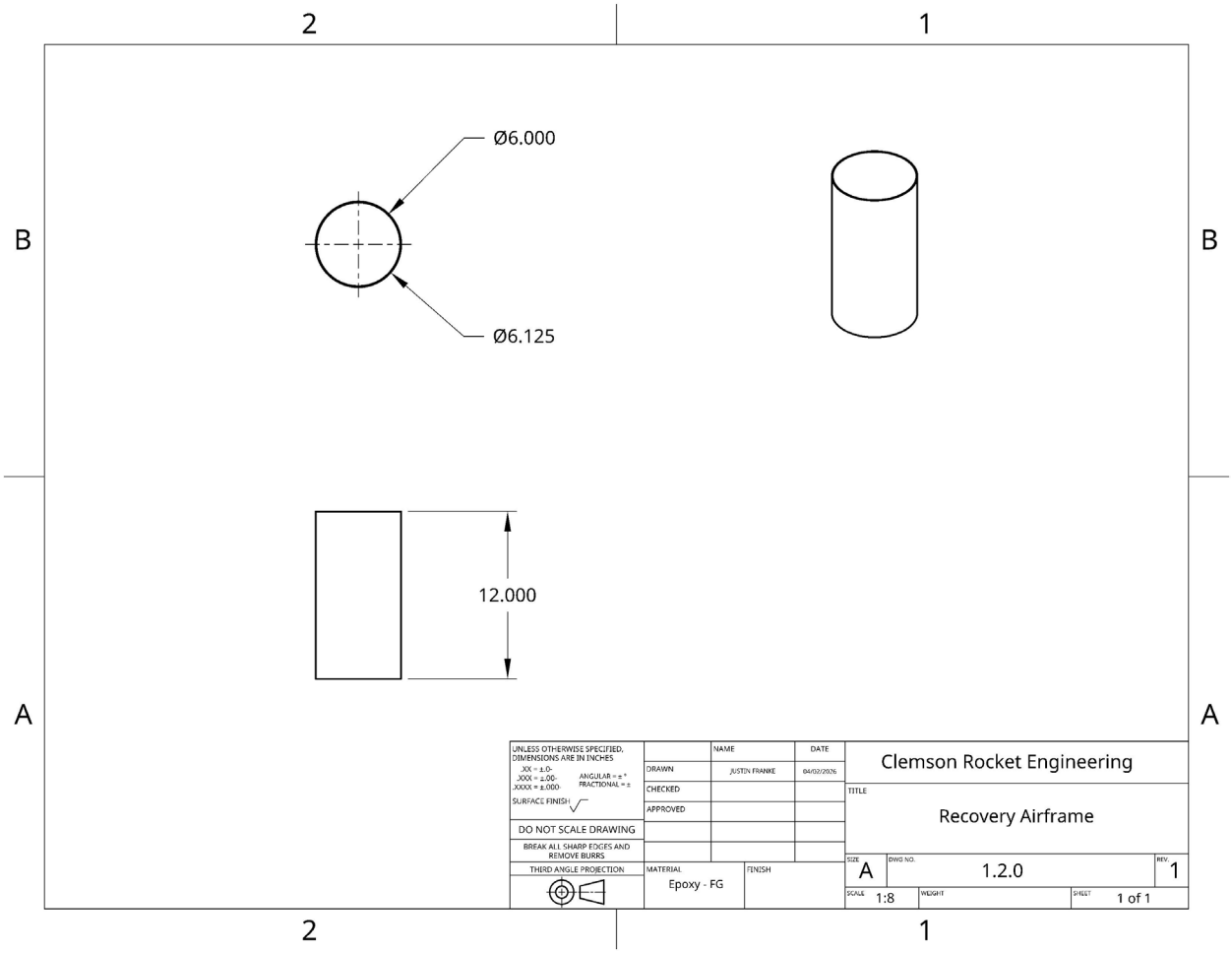
1. Unspent energetics ☐
   a. Safety Officer inspects vehicle ☐
      i. Others remain at a safe distance ☐
      ii. Photograph unspent charge for failure analysis ☐
   b. Disconnect charges from wiring ☐
   c. Open unspent charge, pour unused black powder into designated water bottle to make inert ☐
   d. Repeat for all charges ☐
2. Unexpected vehicle fragmentation ☐
   a. Identify debris field ☐
   b. Photograph sections of debris field ☐
   c. Analyze any missing sections of the vehicle ☐

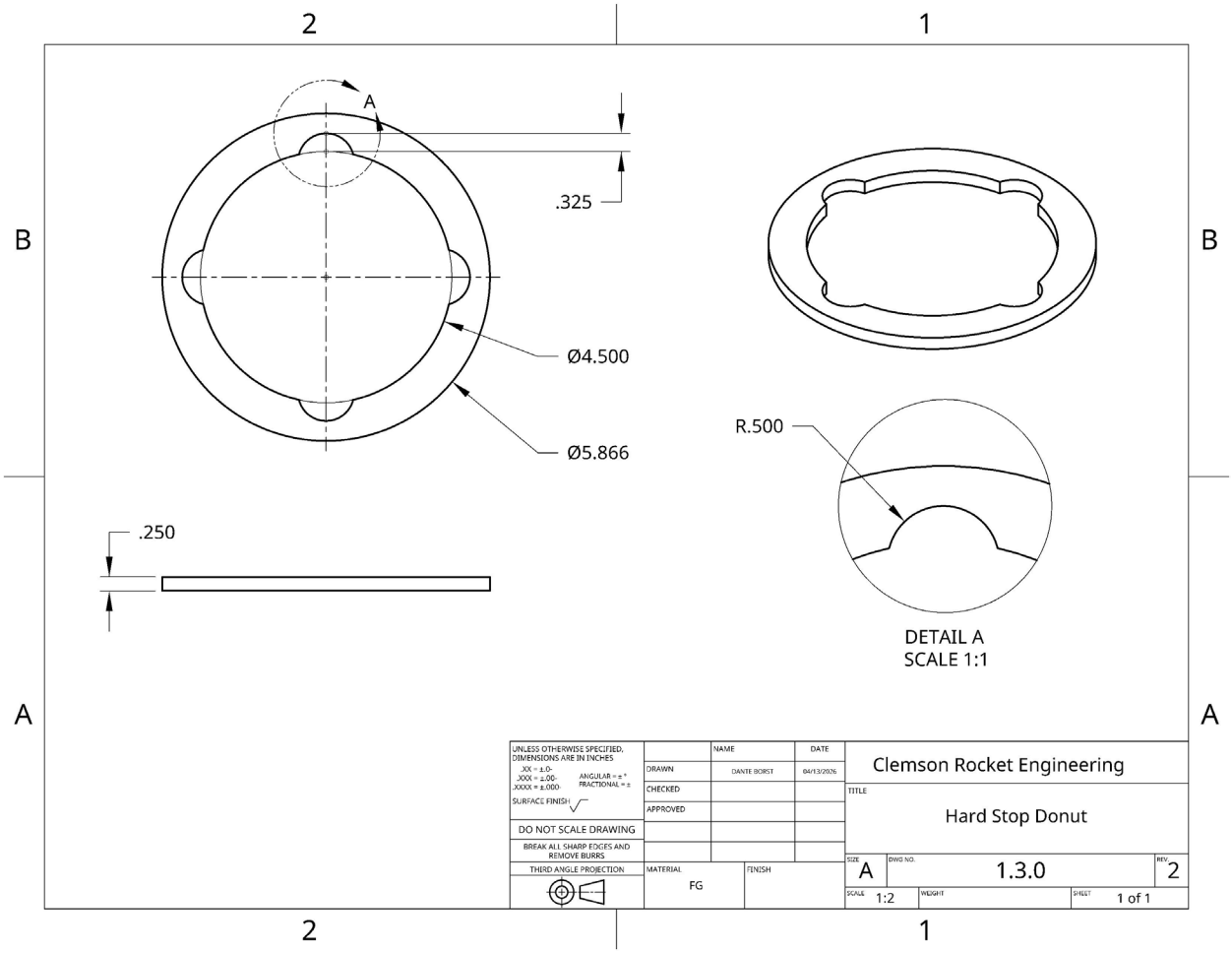
d. Collect fragments by hand or in bag depending on size and quantity of debris ☐

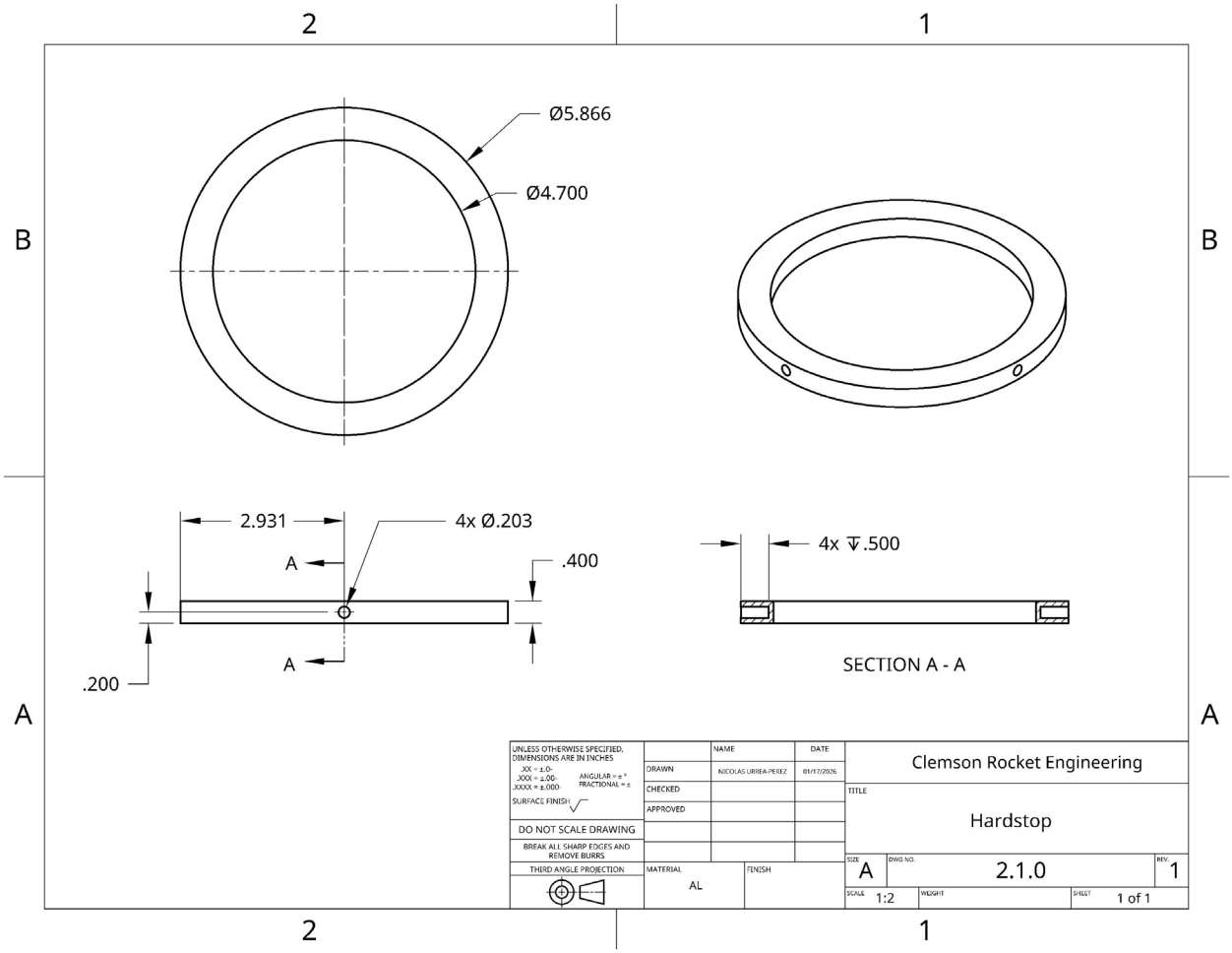
3. Ruptured battery(s) ☐
   a. Report to MCC battery rupture and possible fire ☐
   b. Retreat from the vehicle ☐
      i. Maintain safe yet observable distance in case hazard increases ☐
   c. Once the fire is extinguished, assess the damage and determine the vehicle's condition ☐
   d. If safe to do so, remove other batteries and energetics from vehicle immediately ☐
4. Loss of GPS signal ☐
   a. Attempt to regain connection by changing location ☐
   b. Proceed to last known location as documented in checklist and begin search ☐
   c. If not found at the last known location/immediate area, contact MCC, giving description ☐
5. Heat stroke/medical situation ☐
   a. Mitigate heat issues through proper attire and hydration ☐
   b. Contact MCC relaying situation and urgent need for medical assistance ☐
   c. If the situation has caused incapacitation, remain with the affected recovery team member ☐
   d. If movement is possible and travel feasible/not a risk to affected member or other recovery team members, return to recovery vehicle and relay as such to MCC ☐
      i. This is only in the extreme case that an immediate need to enter a climate-controlled environment is required, otherwise a shelter in place policy is enforced.
6. Radio pack failure ☐
   a. Return to recovery vehicle/base camp immediately ☐
   b. Return radio pack and explain failure ☐
   c. Receive new pack and begin recovery again ☐
7. Vehicle is irretrievable ☐
   a. Take photos of vehicle location and condition ☐
   b. Mark GPS location of vehicle ☐
   c. Return to base camp and assess proper tools for recovery ☐

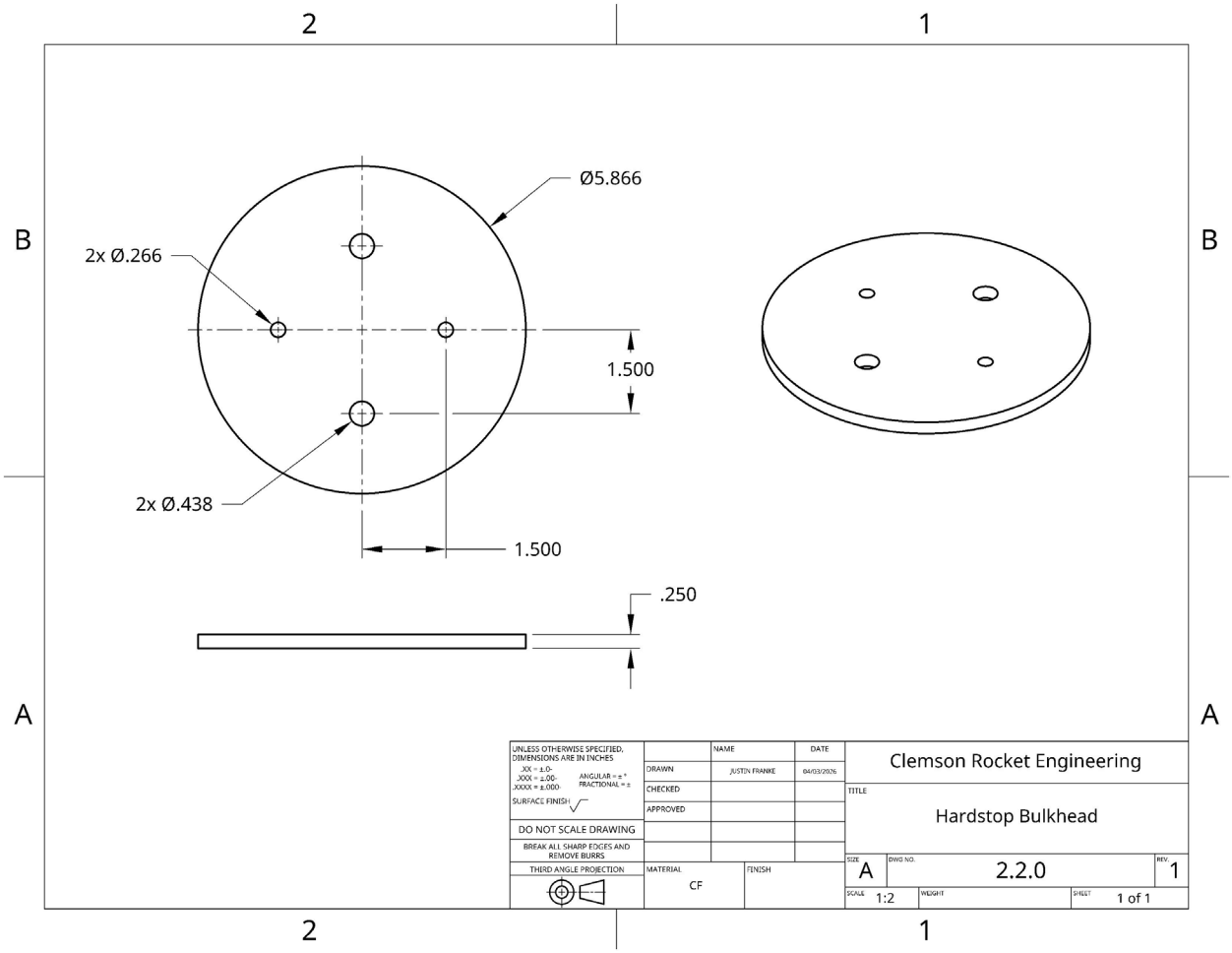
Appendix F—Engineering Drawings

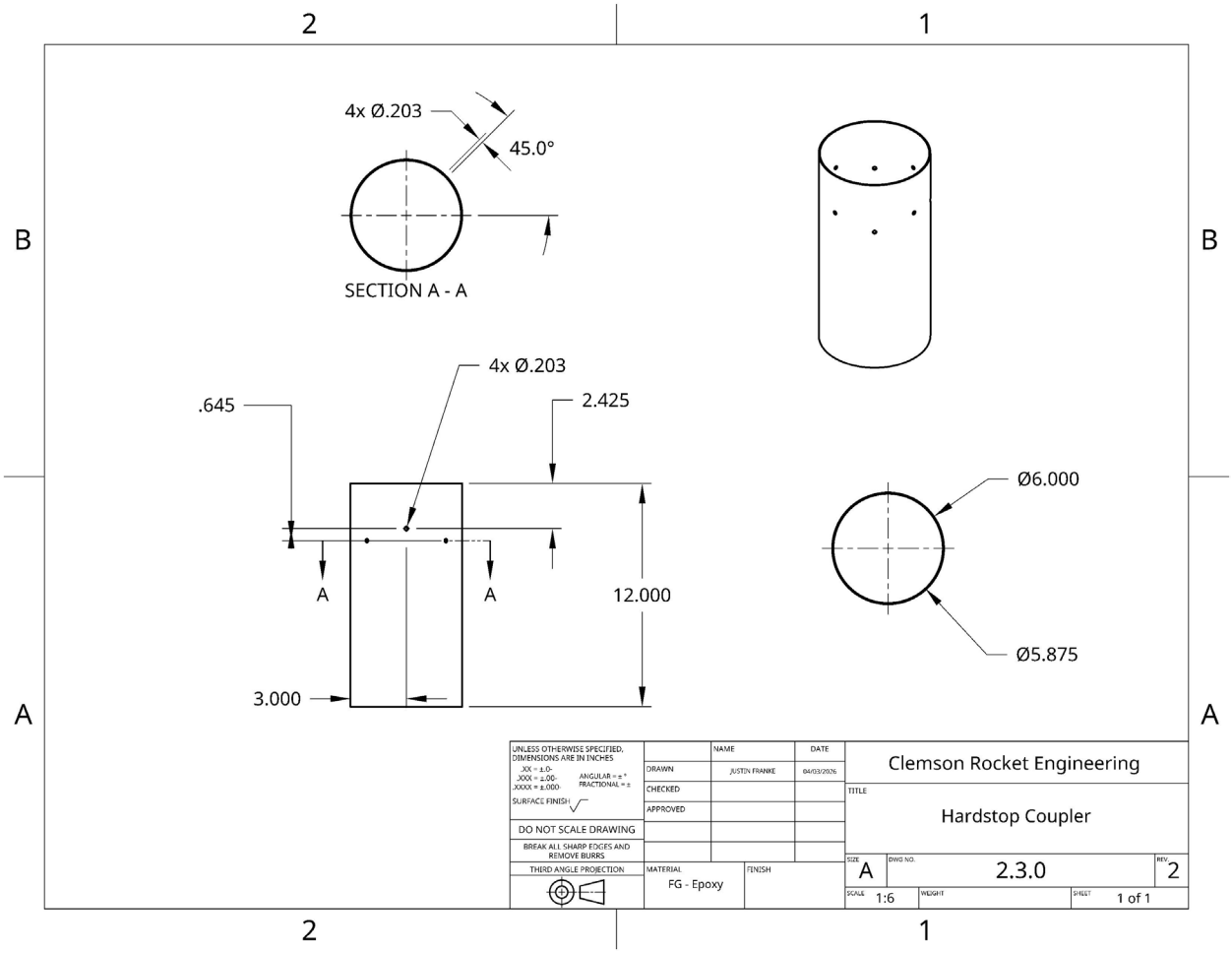


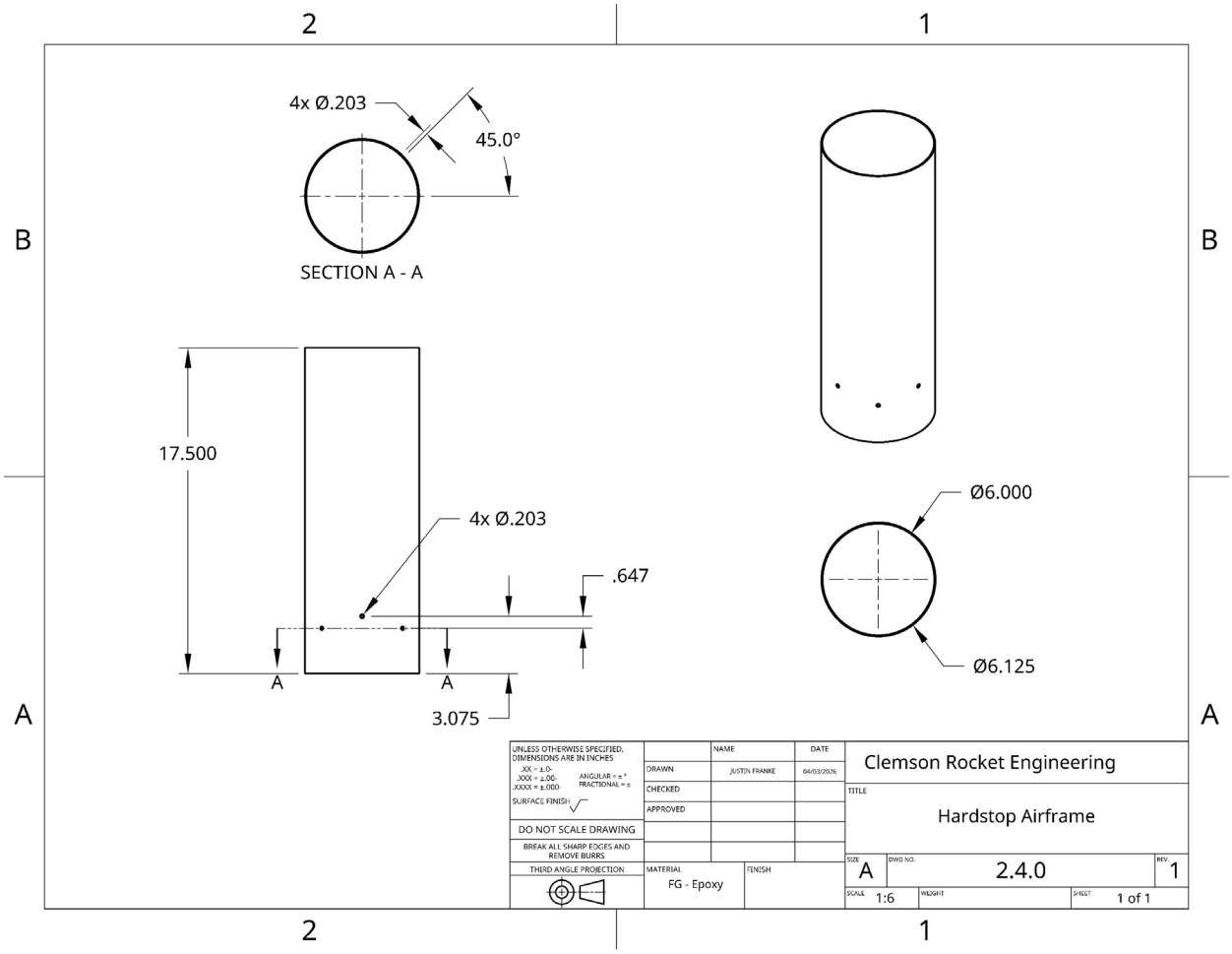

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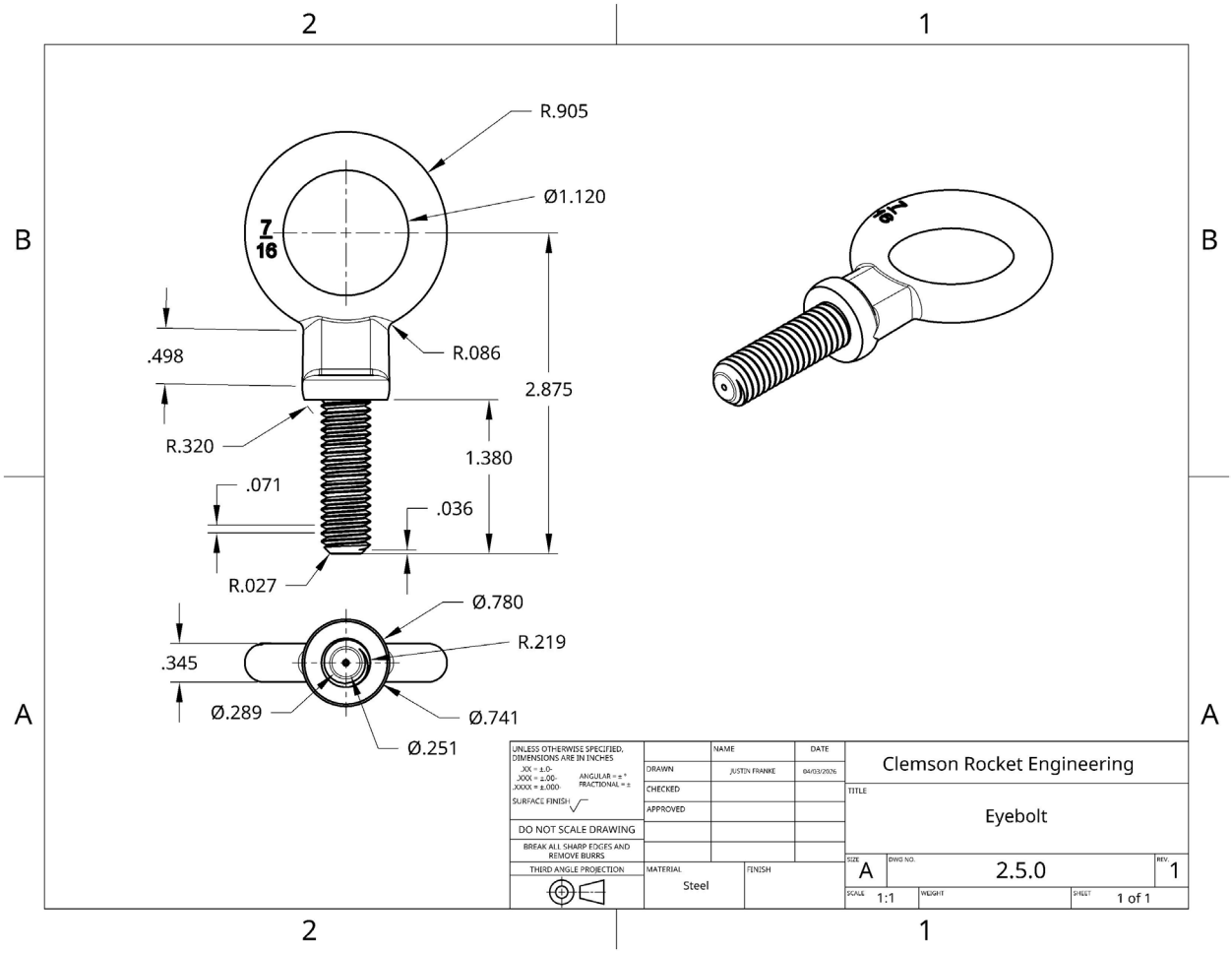

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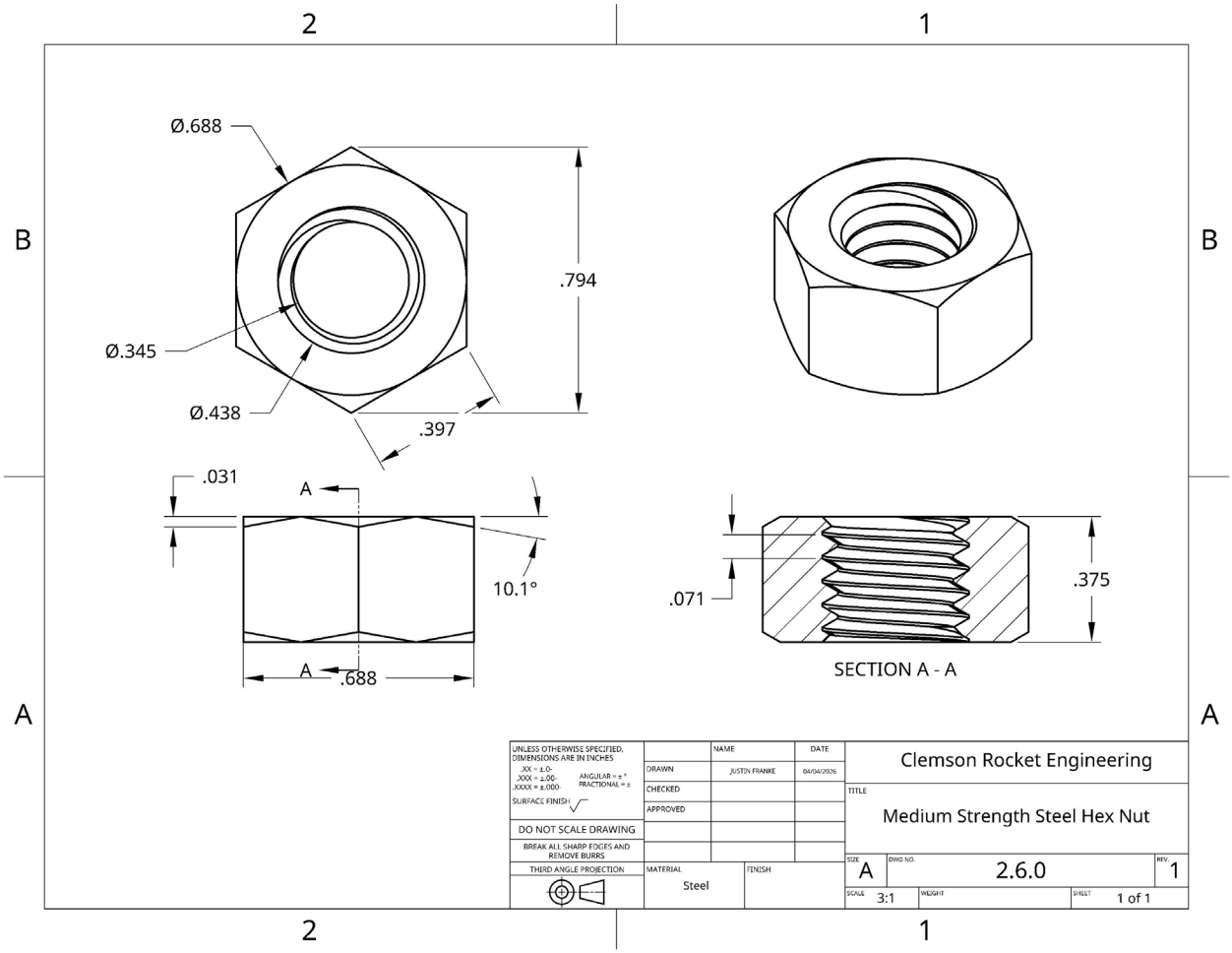

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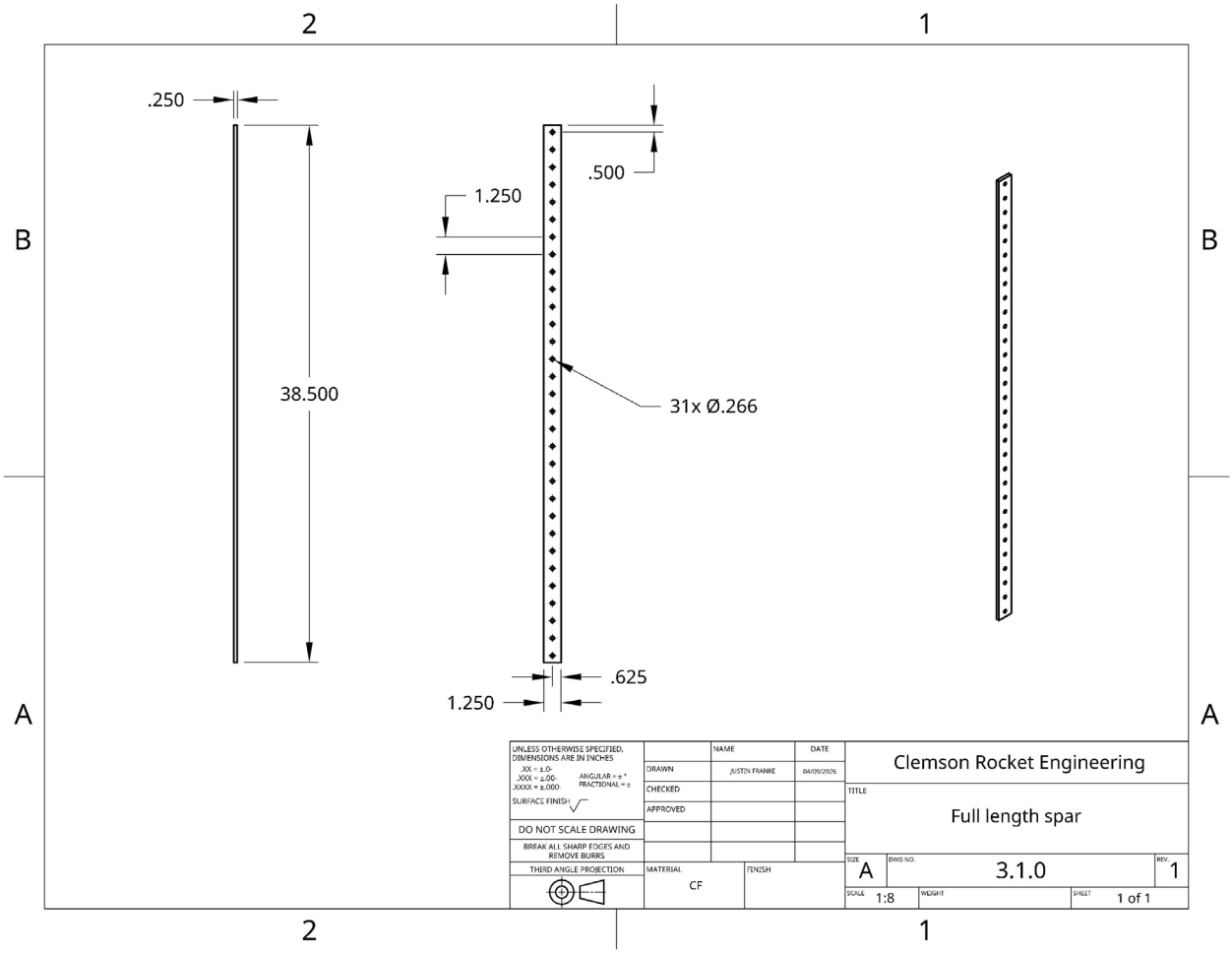

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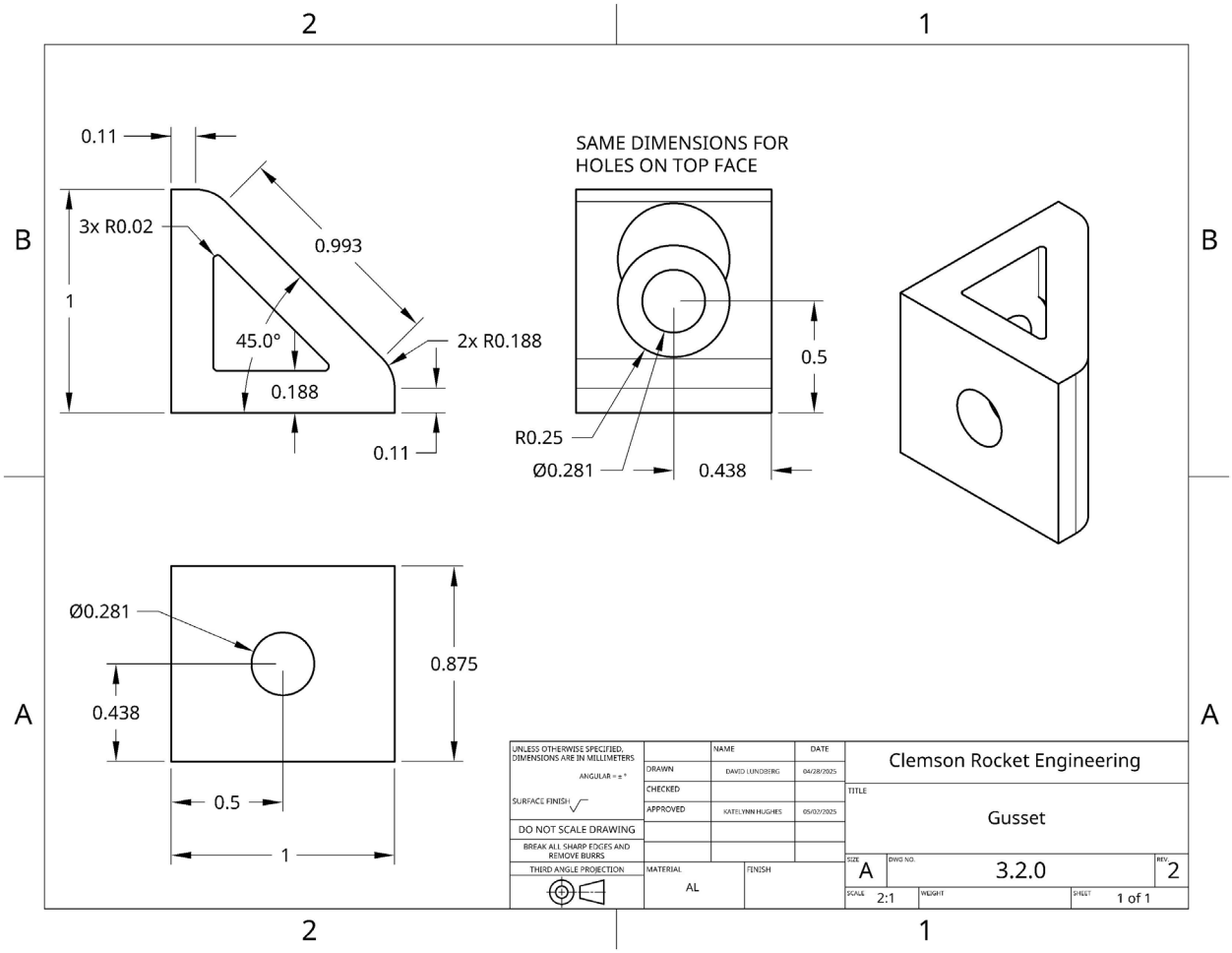

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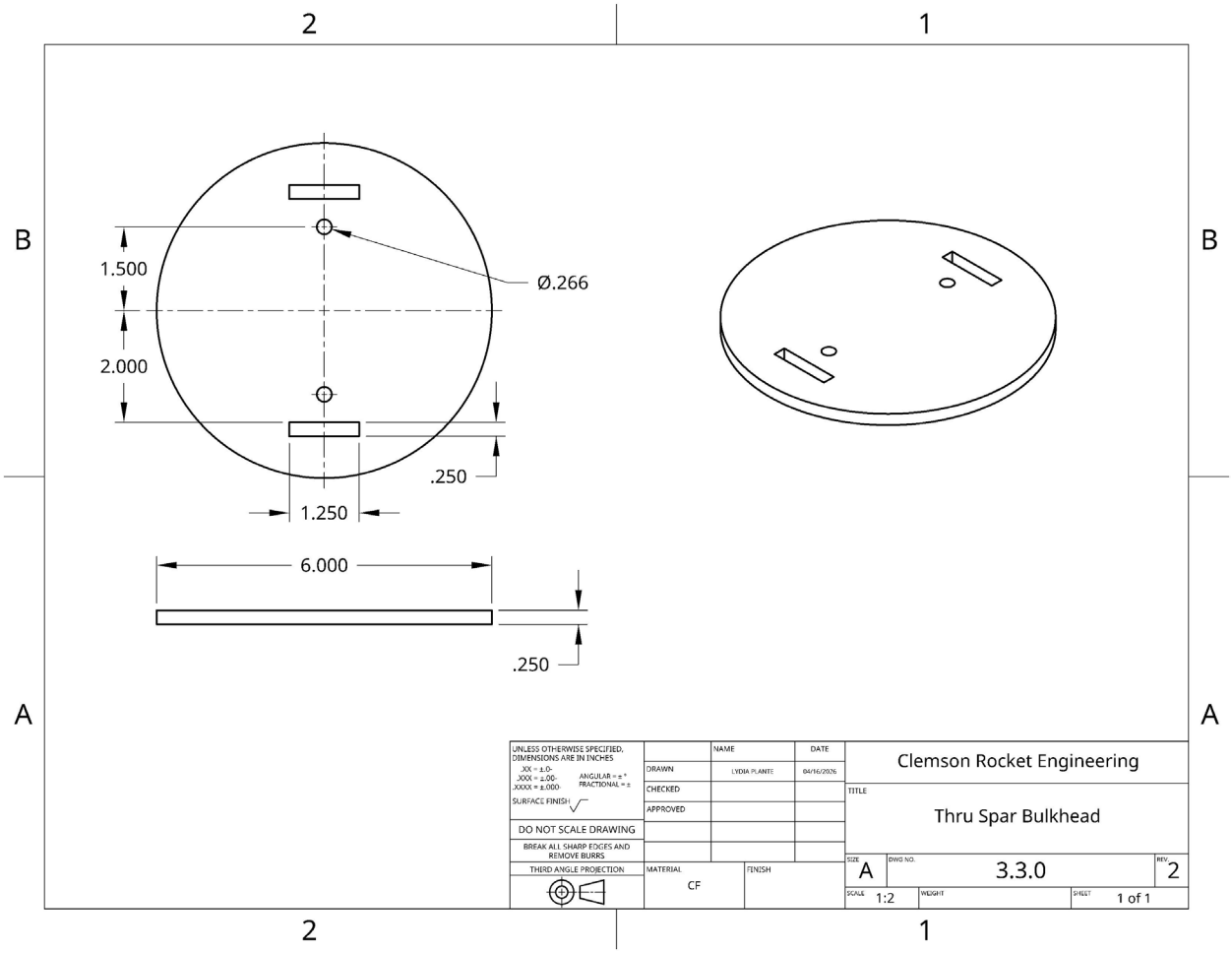

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Ø6.000
Ø6.125
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/03/2026
CHECKED
APPROVED
MATERIAL
FG - Epoxy
FINISH
Clemson Rocket Engineering
TITLE
Hardstop Airframe
SIZE
A
DWG NO.
2.4.0
REV.
1
SCALE
1:6
WEIGHT
SHEET
1 of 1

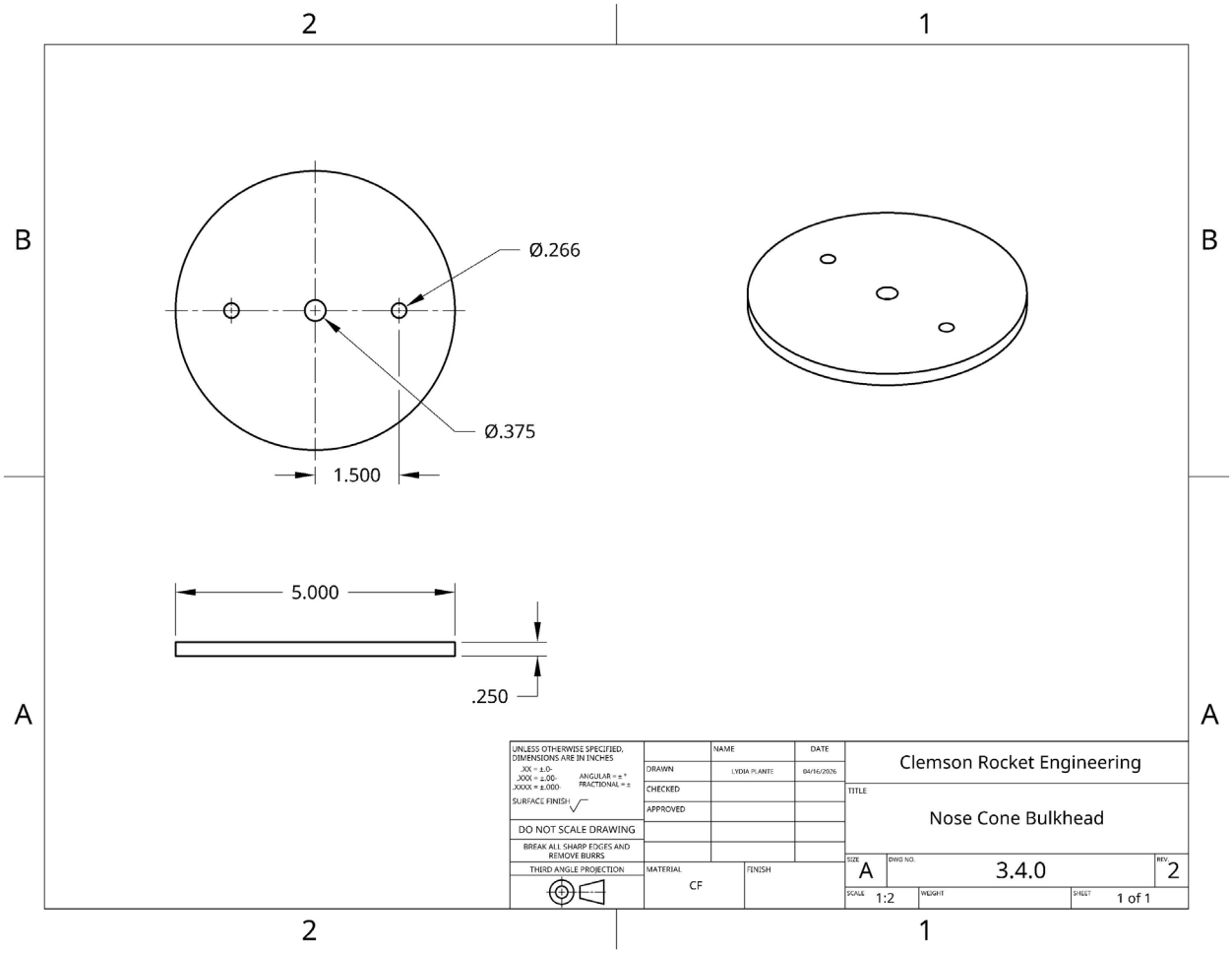

2
1
B
A
R.905
Ø1.120
7/16
.498
R.086
2.875
R.320
1.380
.071
.036
R.027
Ø.780
R.219
.345
Ø.289
Ø.741
Ø.251
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/03/2026
CHECKED
APPROVED
MATERIAL
Steel
FINISH
Clemson Rocket Engineering
TITLE
Eyebolt
SIZE
A
DWG NO.
2.5.0
REV
1
SCALE
1:1
WEIGHT
SHEET
1 of 1

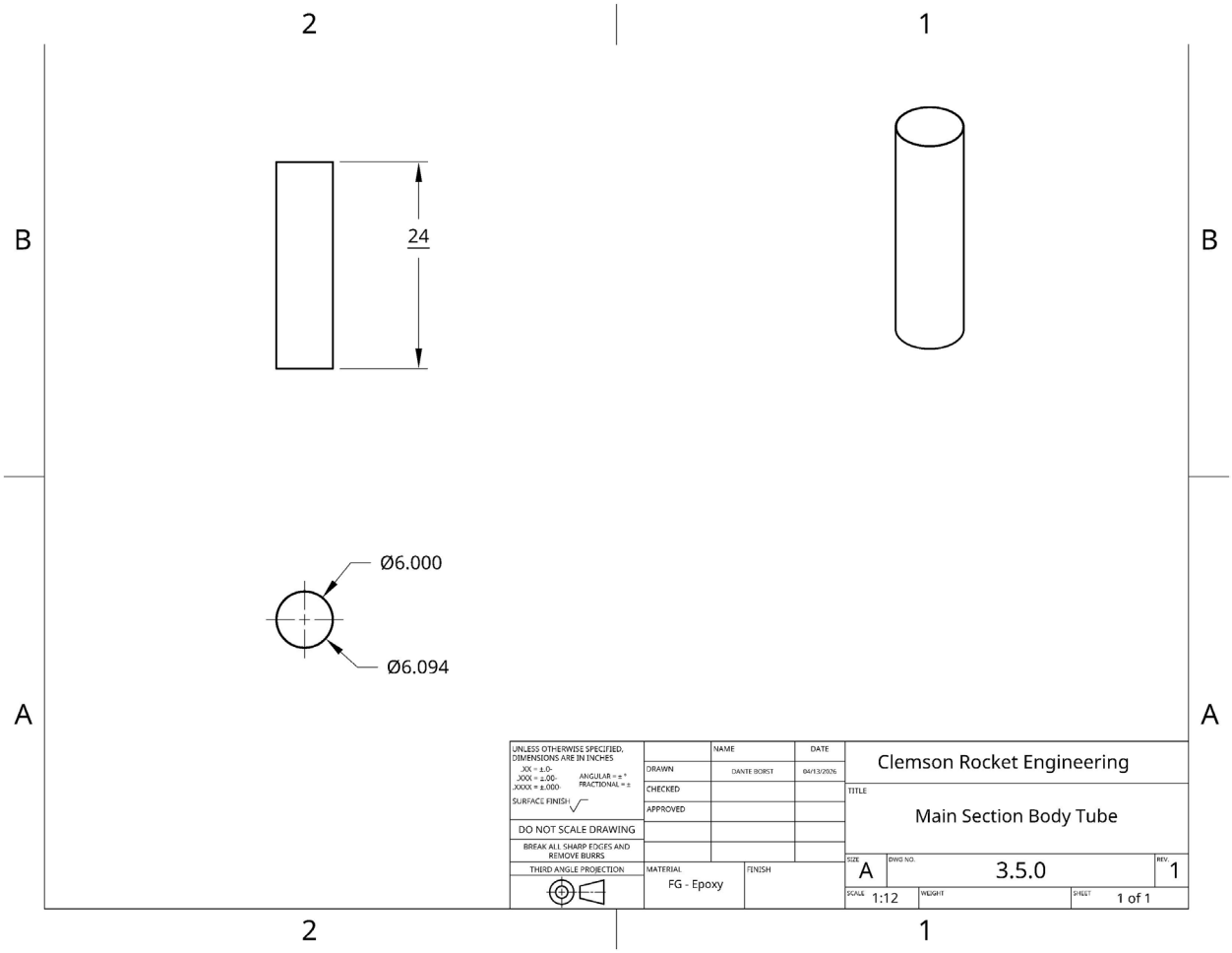

2
1
B
A
Ø.688
.794
Ø.345
Ø.438
.397
.031
A
10.1°
A
.688
.071
.375
SECTION A - A
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/04/2026
CHECKED
APPROVED
MATERIAL
Steel
FINISH
Clemson Rocket Engineering
TITLE
Medium Strength Steel Hex Nut
SIZE
A
DWG NO.
2.6.0
REV
1
SCALE
3:1
WEIGHT
SHEET
1 of 1

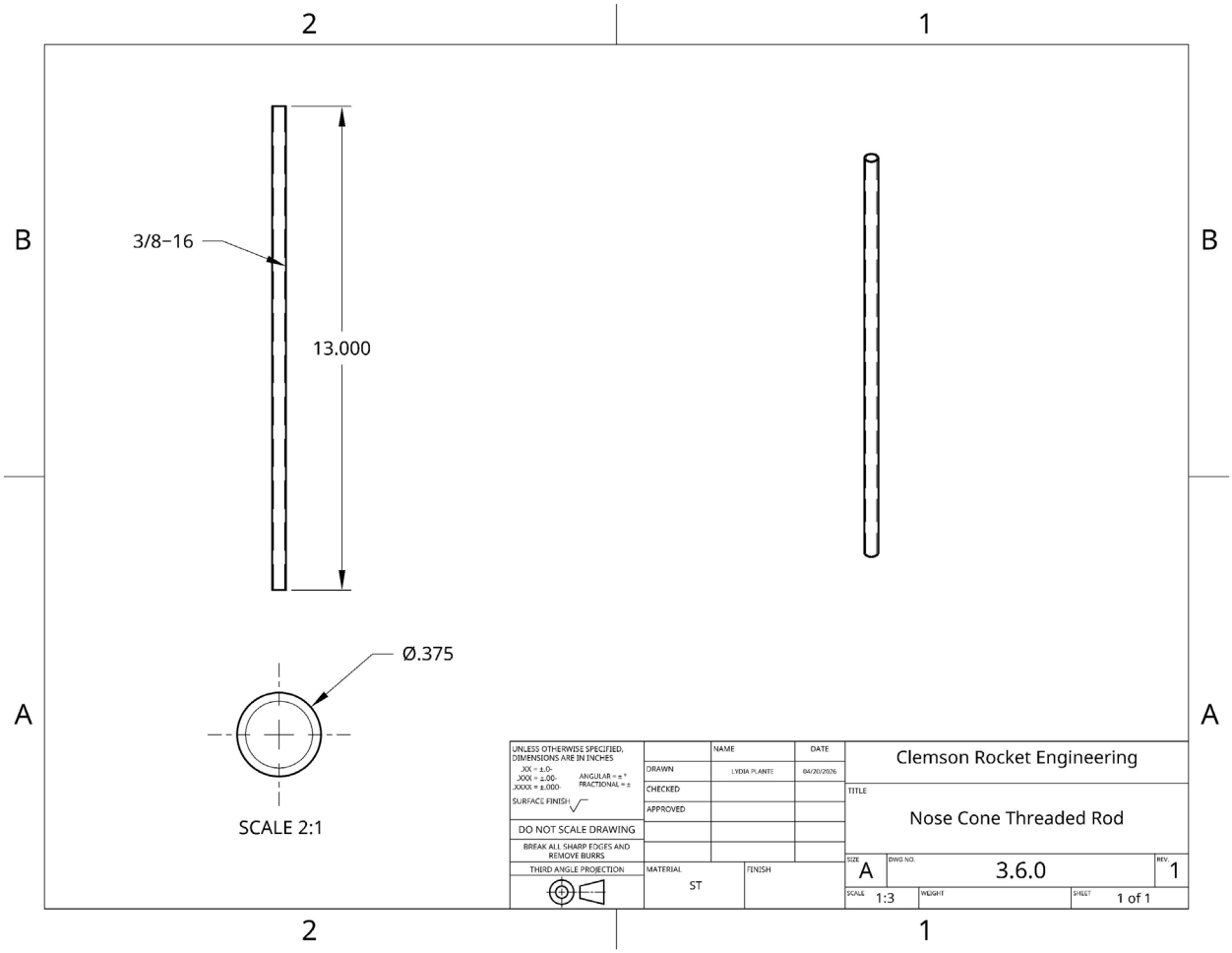

2
1
B
A
.250
38.500
1.250
.500
31x Ø.266
.625
1.250
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/09/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Full length spar
SIZE
A
DWG NO.
3.1.0
REV.
1
SCALE
1:8
WEIGHT
SHEET
1 of 1

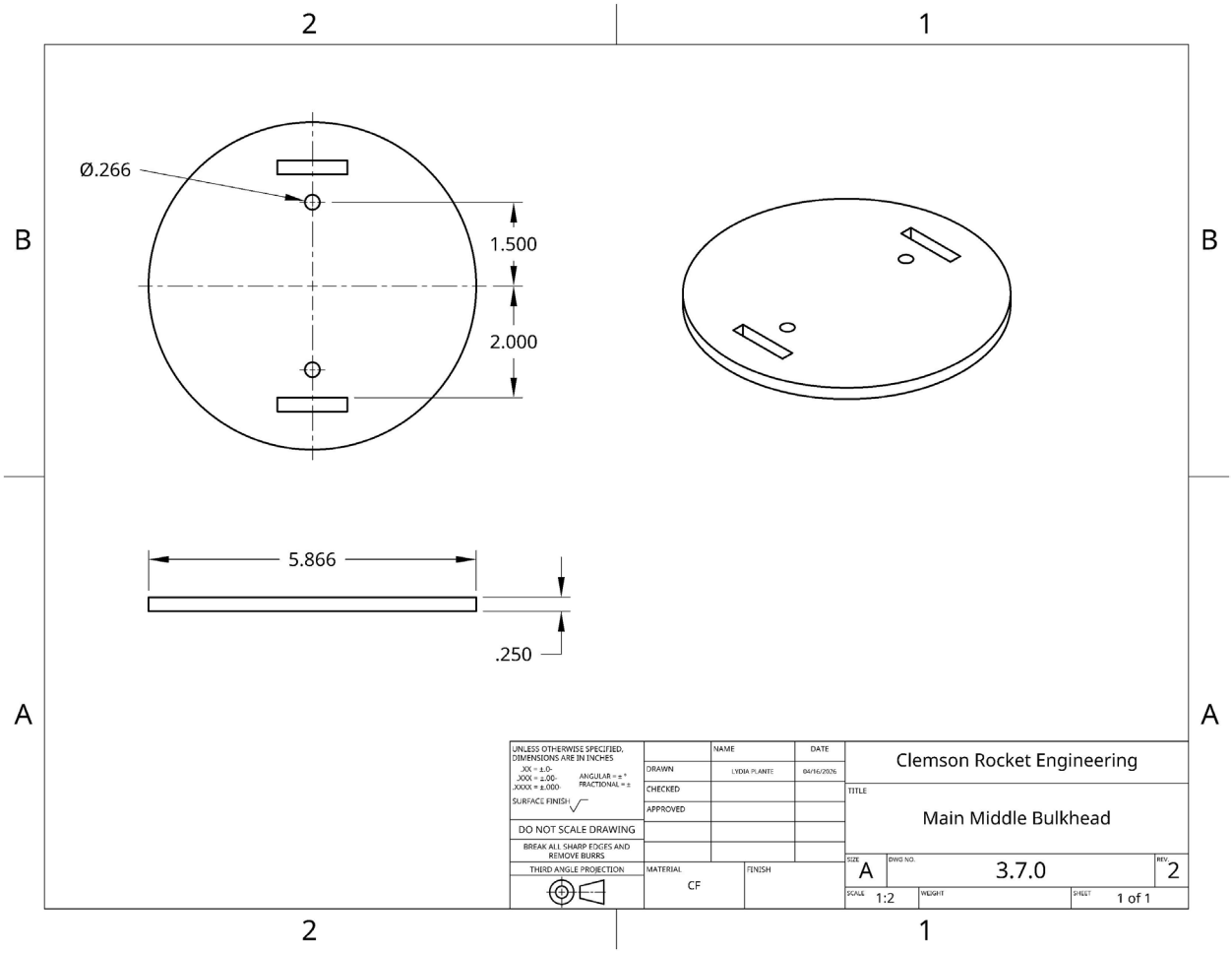

2
1
B
A
0.11
3x R0.02
0.993
1
45.0°
2x R0.188
0.188
0.11
SAME DIMENSIONS FOR
HOLES ON TOP FACE
0.5
R0.25
Ø0.281
0.438
Ø0.281
0.875
0.438
0.5
1
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS
ANGULAR = ± °
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
04/28/2025
CHECKED
APPROVED
KATELYNN HUGHES
05/02/2025
MATERIAL
AL
FINISH
Clemson Rocket Engineering
TITLE
Gusset
SIZE
A
DWG NO.
3.2.0
REV
2
SCALE
2:1
WEIGHT
SHEET
1 of 1

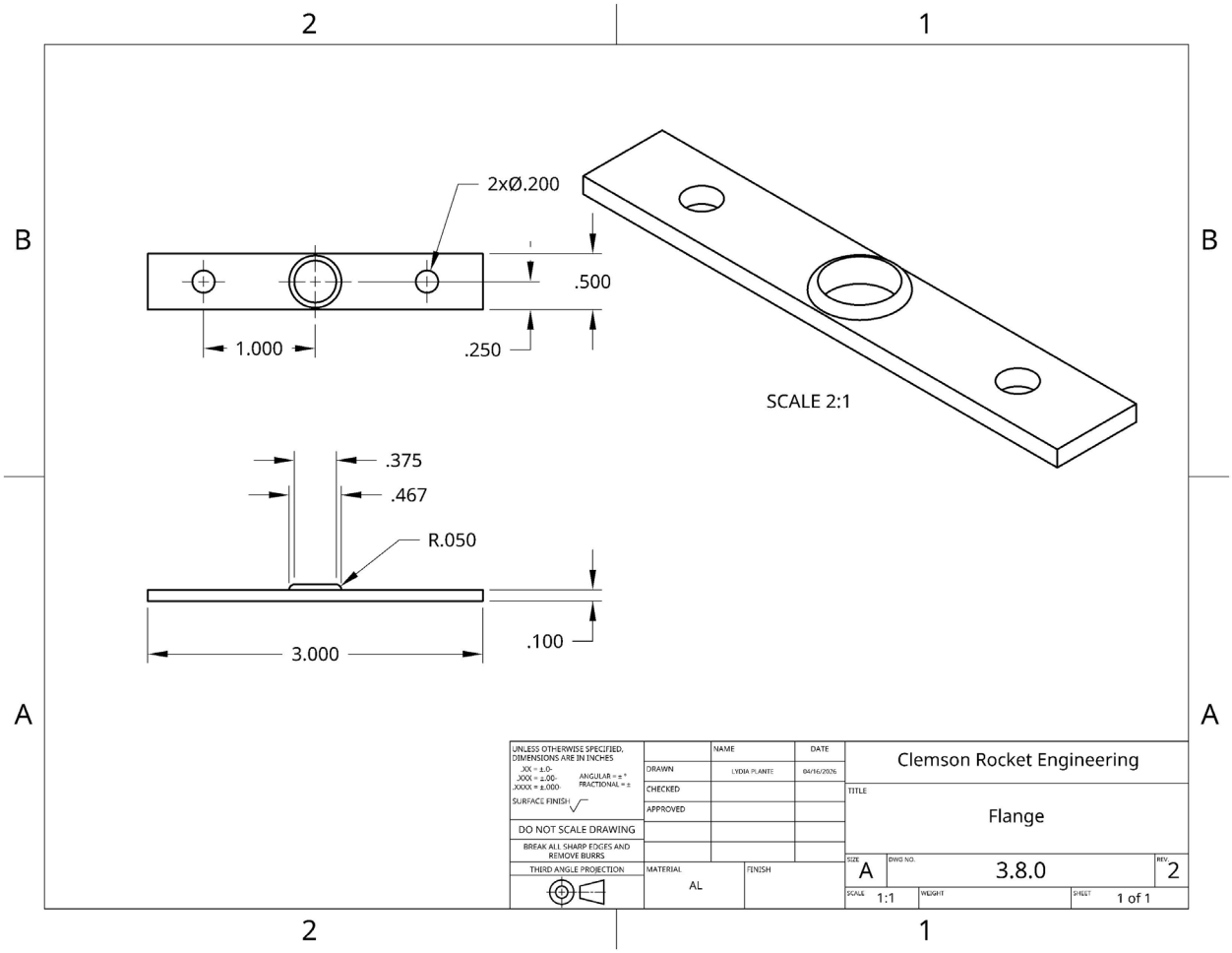

2
1
B
A
1.500
2.000
Ø.266
.250
1.250
6.000
.250
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/16/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Thru Spar Bulkhead
SIZE
A
DWG NO.
3.3.0
REV
2
SCALE
1:2
WEIGHT
SHEET
1 of 1

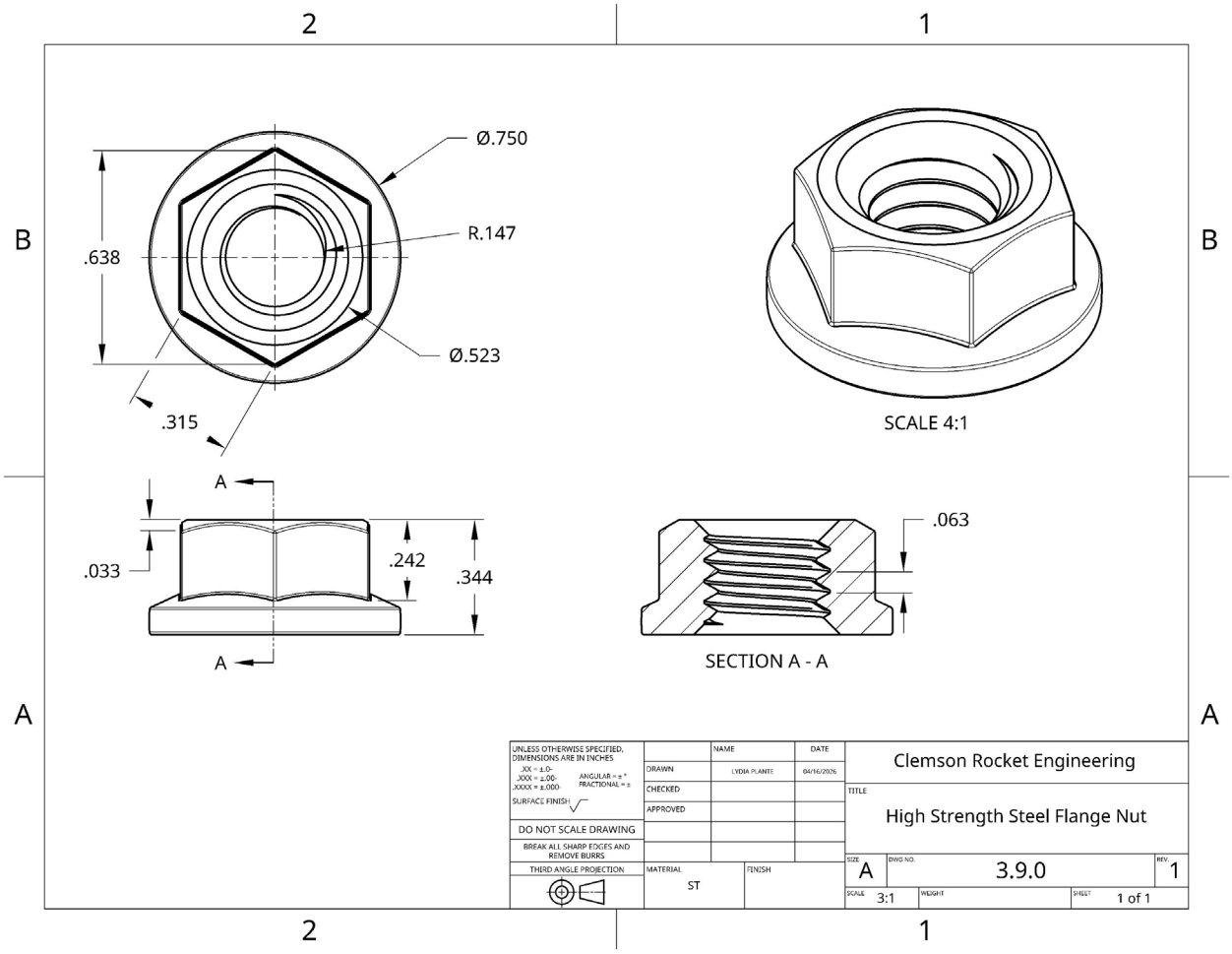

2
1
B
B
Ø.266
Ø.375
1.500
5.000
.250
A
A
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/16/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Nose Cone Bulkhead
SIZE
A
DWG NO.
3.4.0
REV
2
SCALE
1:2
WEIGHT
SHEET
1 of 1
2
1

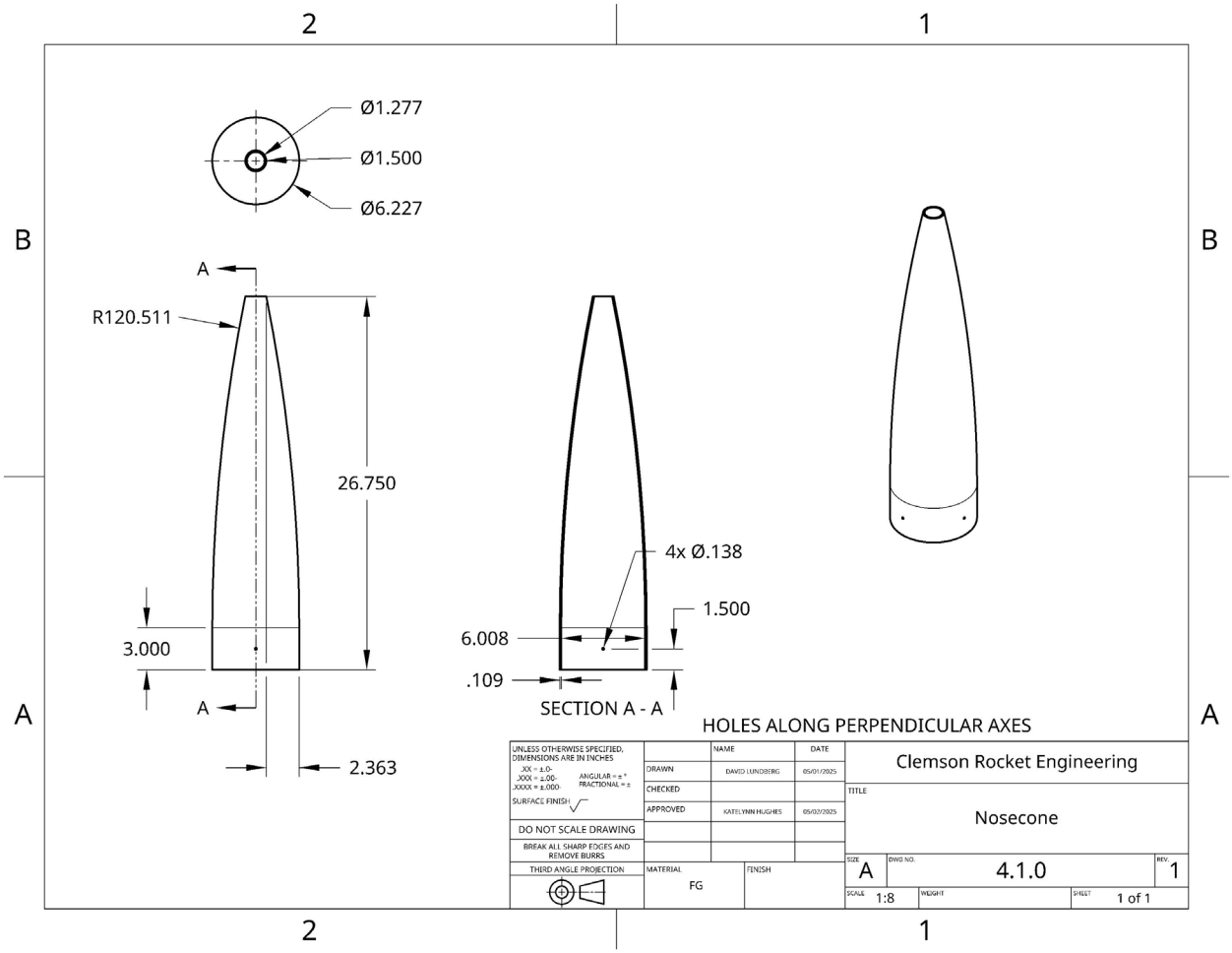

2
1
B
B
24
Ø6.000
Ø6.094
A
A
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DANTE BOIRST
04/13/2026
CHECKED
APPROVED
MATERIAL
FG - Epoxy
FINISH
Clemson Rocket Engineering
TITLE
Main Section Body Tube
SIZE
A
DWG NO.
3.5.0
REV
1
SCALE
1:12
WEIGHT
SHEET
1 of 1
2
1

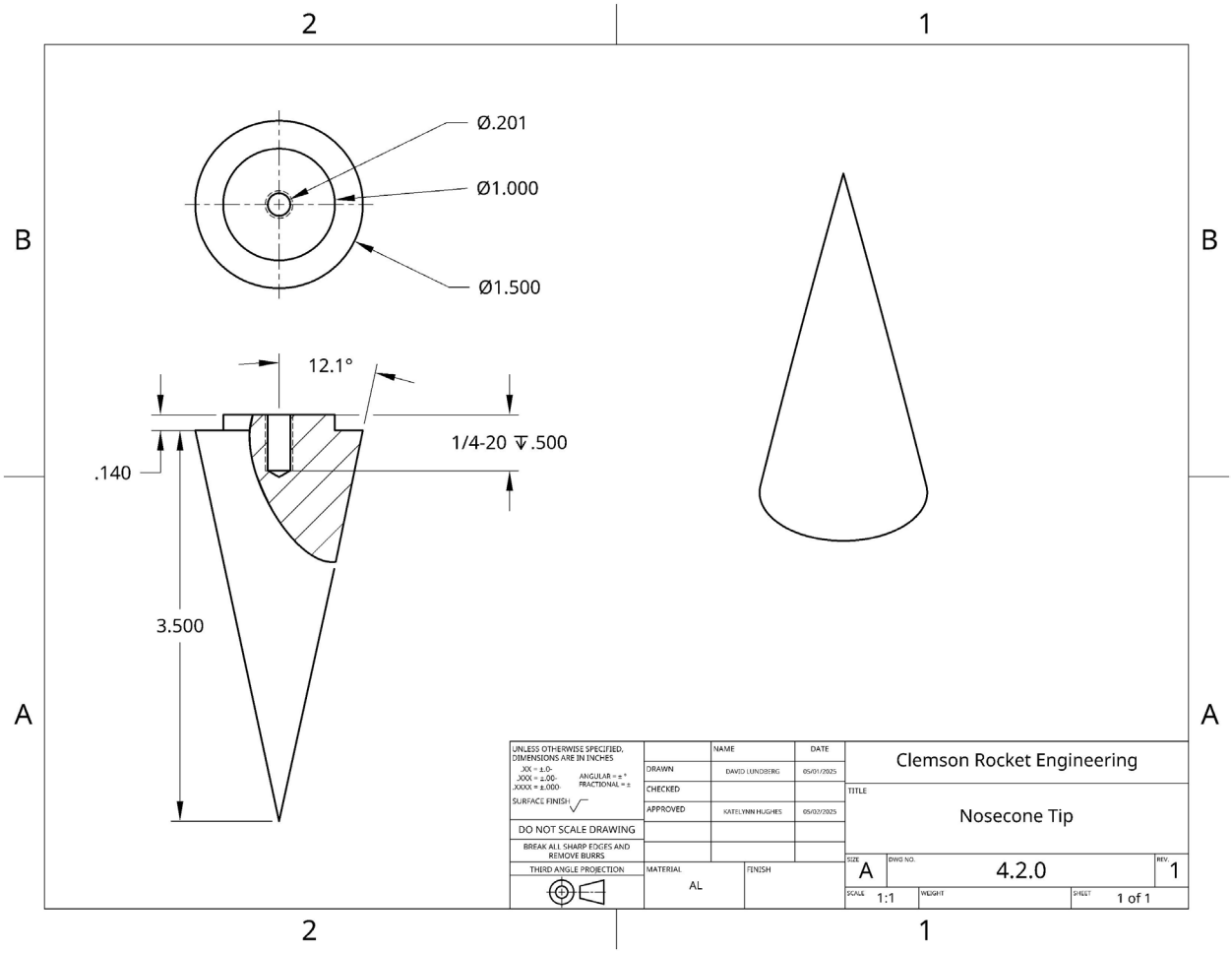

2
1
B
A
3/8−16
13.000
Ø.375
SCALE 2:1
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/20/2026
CHECKED
APPROVED
MATERIAL
ST
FINISH
Clemson Rocket Engineering
TITLE
Nose Cone Threaded Rod
SIZE
A
DWG NO.
3.6.0
REV.
1
SCALE
1:3
WEIGHT
SHEET
1 of 1

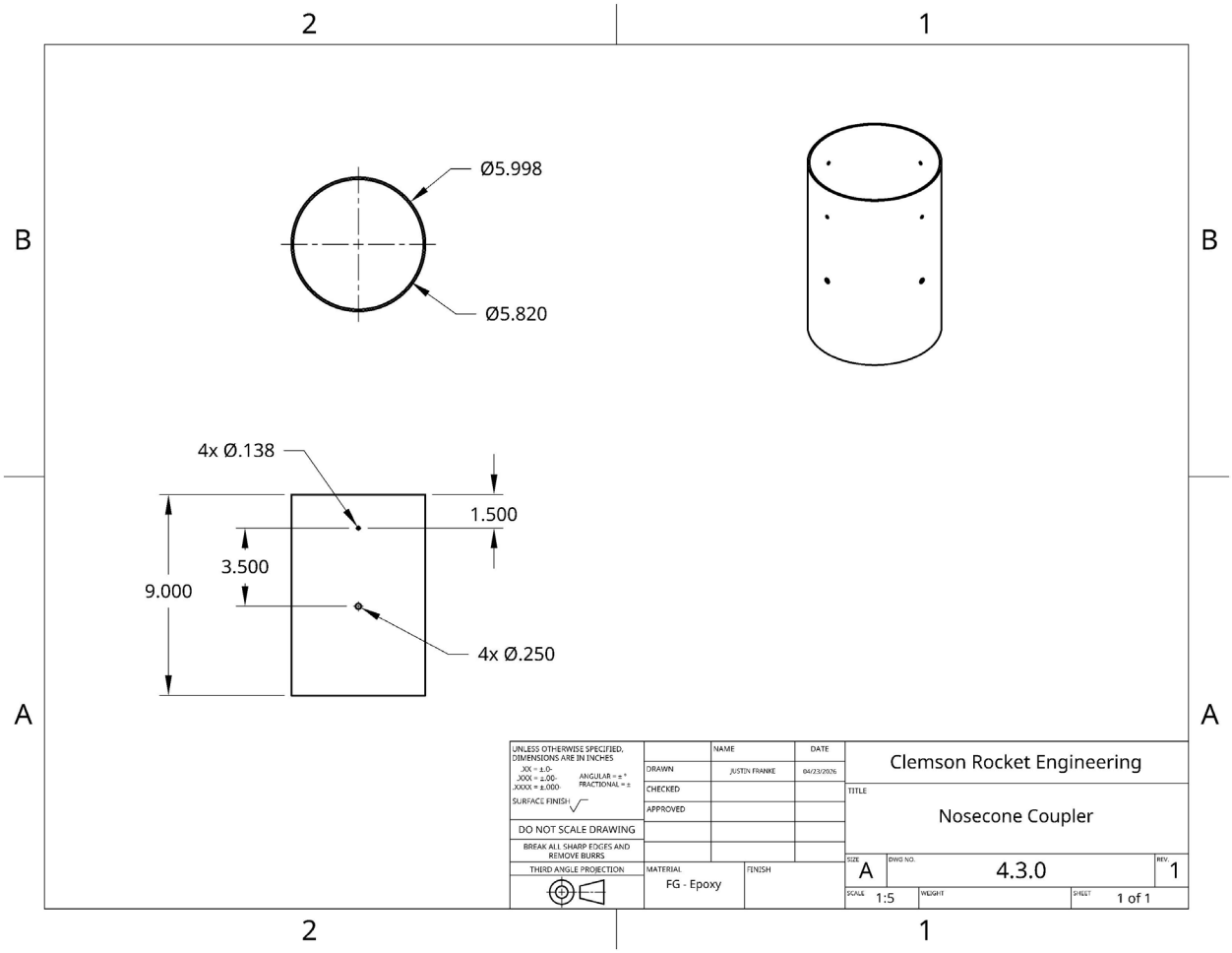

2
1
B
A
Ø.266
1.500
2.000
5.866
.250
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/16/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Main Middle Bulkhead
SIZE
A
DWG NO.
3.7.0
REV
2
SCALE
1:2
WEIGHT
SHEET
1 of 1

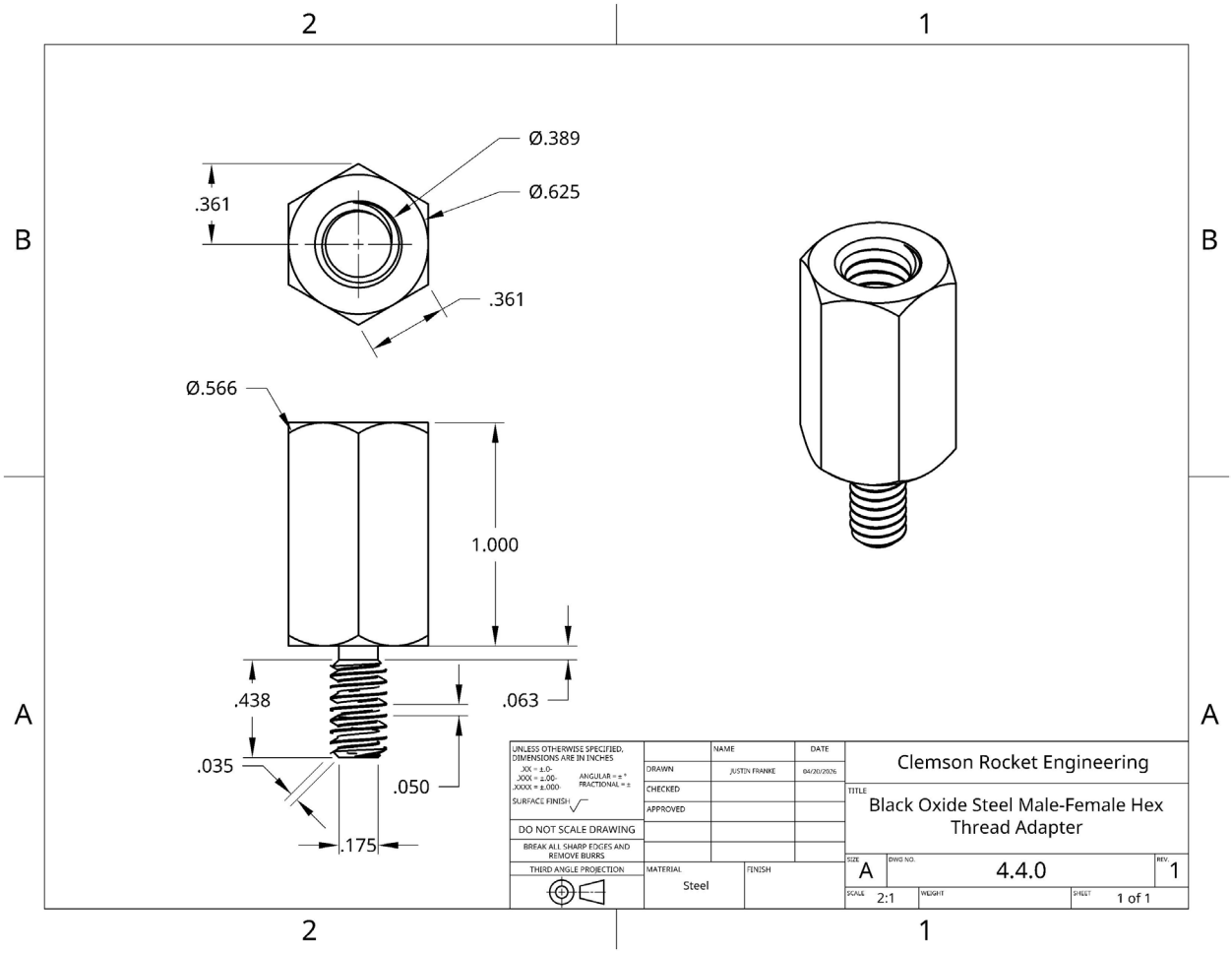

2
1
B
A
2xØ.200
.500
.250
1.000
SCALE 2:1
.375
.467
R.050
.100
3.000
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/16/2026
CHECKED
APPROVED
MATERIAL
AL
FINISH
Clemson Rocket Engineering
TITLE
Flange
SIZE
A
DWG NO.
3.8.0
REV
2
SCALE
1:1
WEIGHT
SHEET
1 of 1

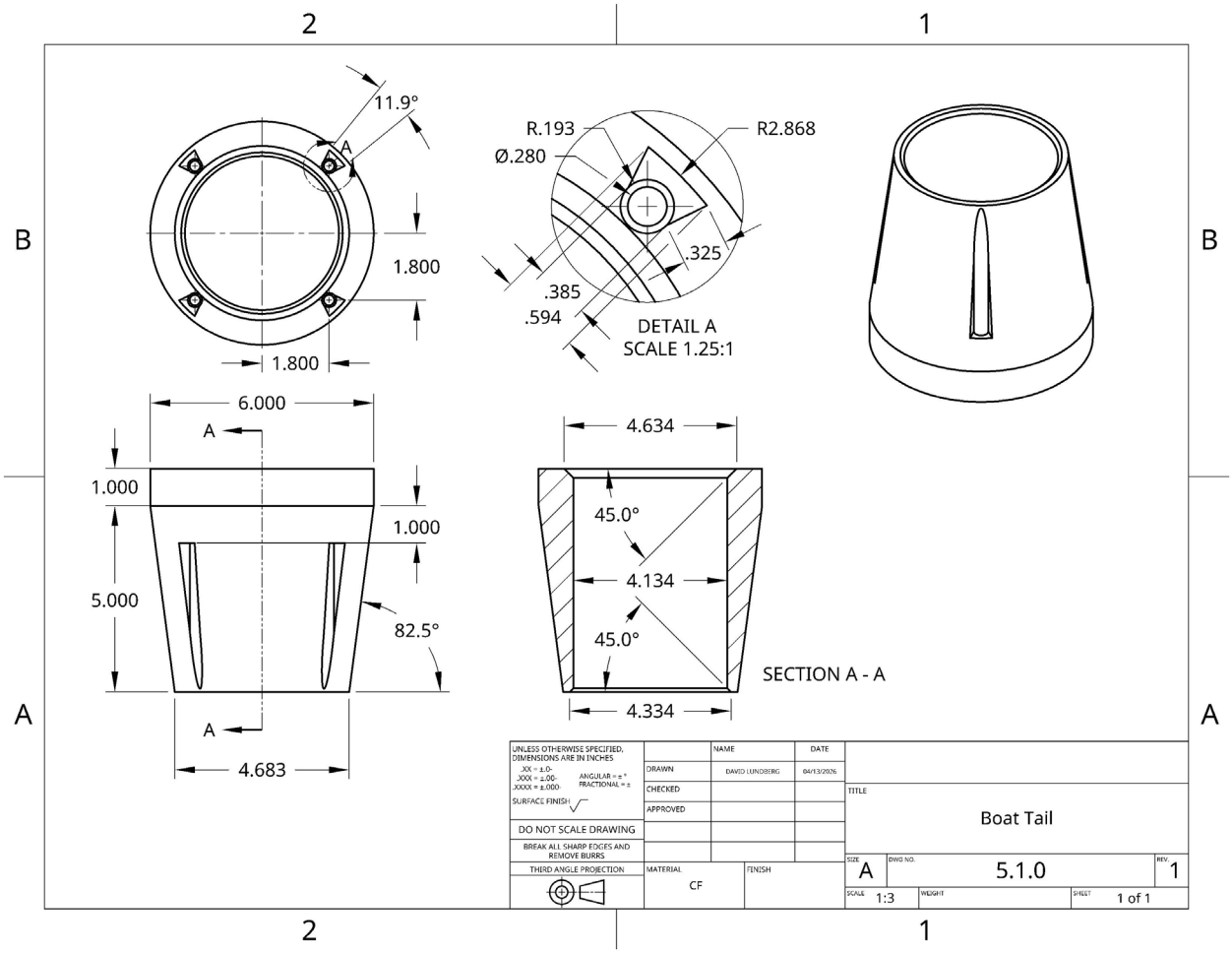

2
1
B
A
Ø.750
R.147
.638
Ø.523
.315
SCALE 4:1
A
.033
.242
.344
A
.063
SECTION A - A
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/16/2026
CHECKED
APPROVED
MATERIAL
ST
FINISH
Clemson Rocket Engineering
TITLE
High Strength Steel Flange Nut
SIZE
A
DWG NO.
3.9.0
REV
1
SCALE
3:1
WEIGHT
SHEET
1 of 1

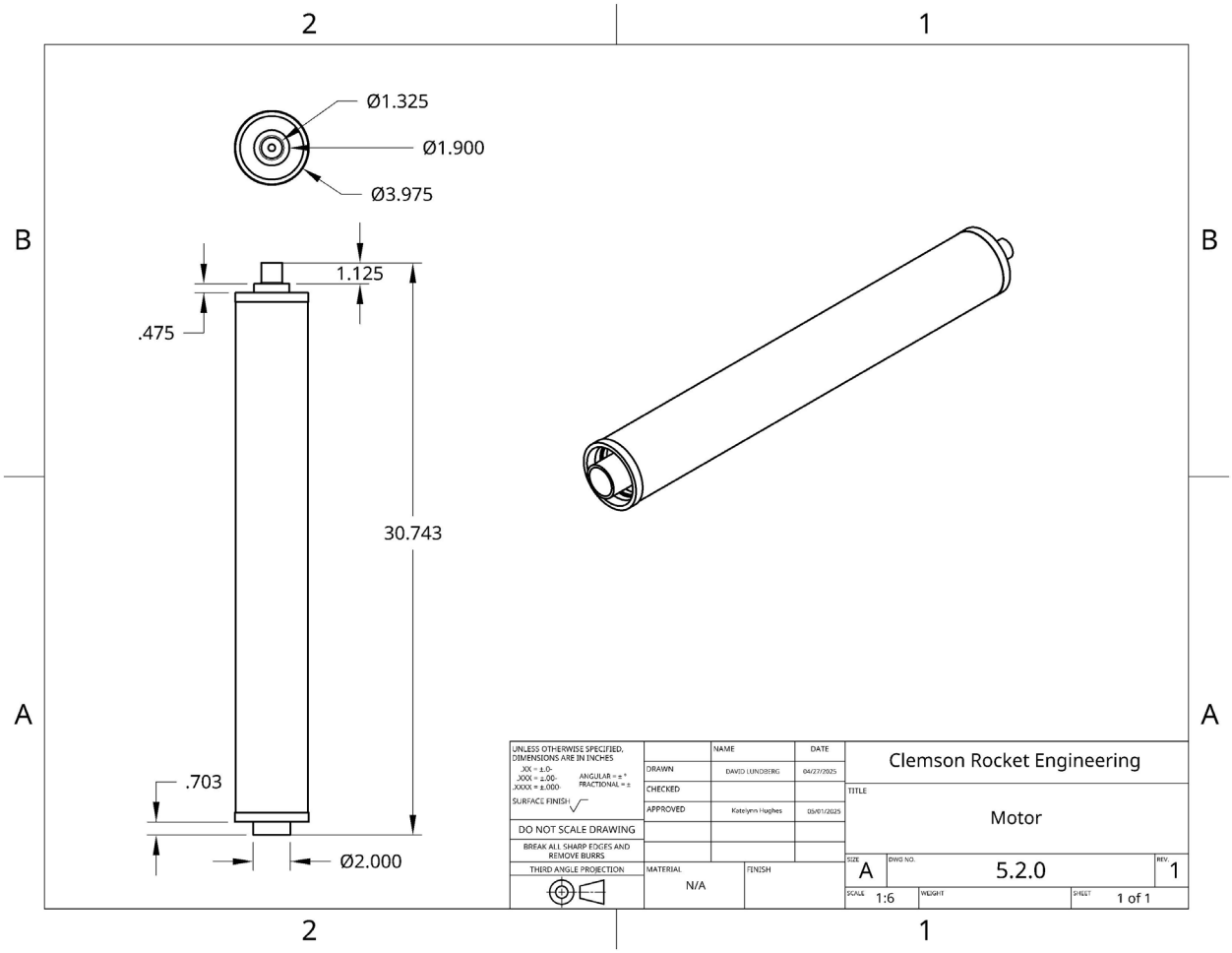

2
1
B
A
Ø1.277
Ø1.500
Ø6.227
A
R120.511
26.750
3.000
A
2.363
4x Ø.138
1.500
6.008
.109
SECTION A - A
HOLES ALONG PERPENDICULAR AXES
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
05/01/2025
CHECKED
APPROVED
KATELYNN HUGHES
05/02/2025
MATERIAL
FG
FINISH
Clemson Rocket Engineering
TITLE
Nosecone
SIZE
A
DWG NO.
4.1.0
REV.
1
SCALE
1:8
WEIGHT
SHEET
1 of 1

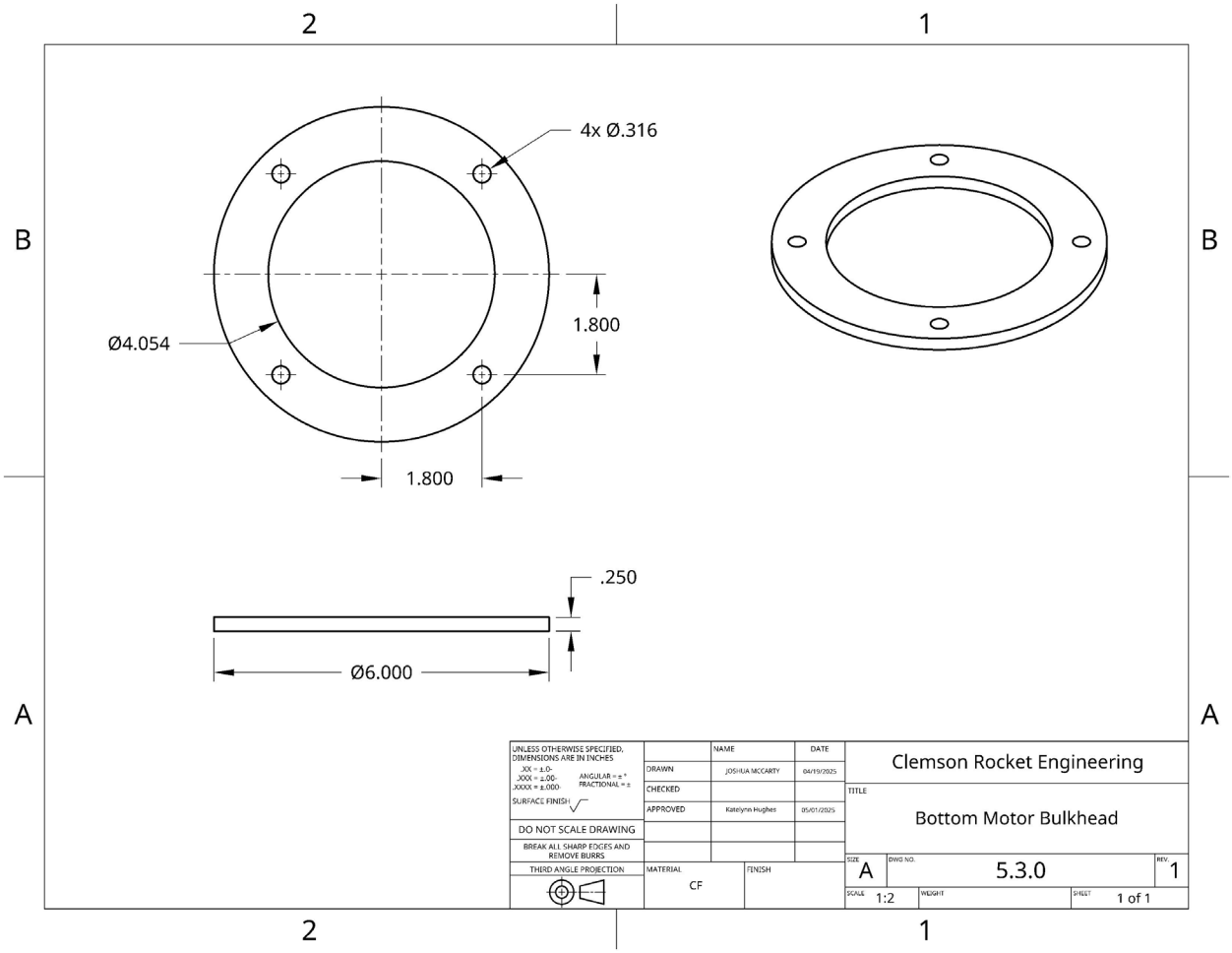

2
1
B
A
Ø.201
Ø1.000
Ø1.500
12.1°
1/4-20 ↧.500
.140
3.500
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
05/01/2025
CHECKED
APPROVED
KATELYNN HUGHES
05/02/2025
MATERIAL
AL
FINISH
Clemson Rocket Engineering
TITLE
Nosecone Tip
SIZE
A
DWG NO.
4.2.0
REV.
1
SCALE
1:1
WEIGHT
SHEET
1 of 1

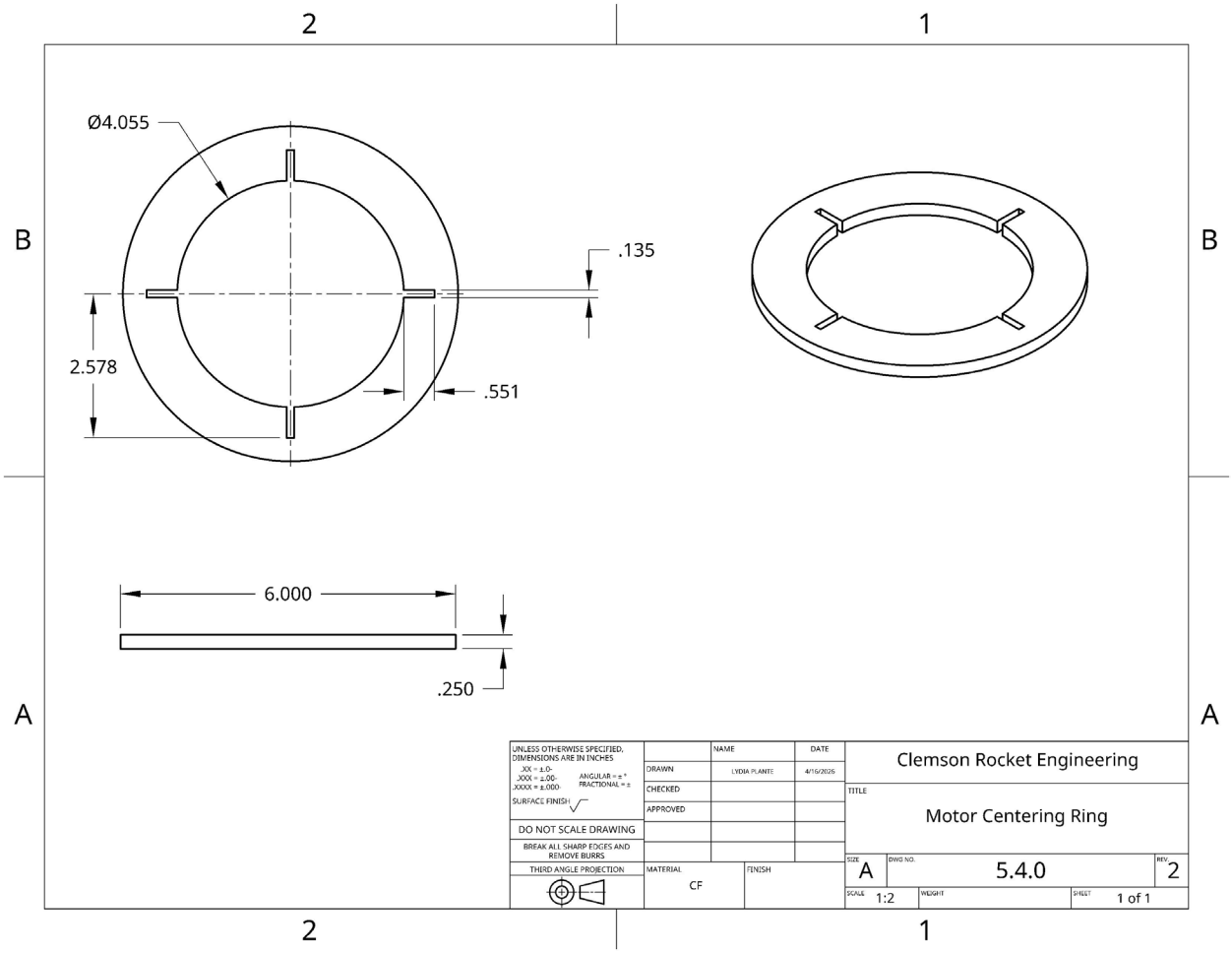

2
1
B
B
Ø5.998
Ø5.820
4x Ø.138
1.500
3.500
9.000
4x Ø.250
A
A
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/23/2026
CHECKED
APPROVED
MATERIAL
FG - Epoxy
FINISH
Clemson Rocket Engineering
TITLE
Nosecone Coupler
SIZE
A
DWG NO.
4.3.0
REV
1
SCALE
1:5
WEIGHT
SHEET
1 of 1
2
1

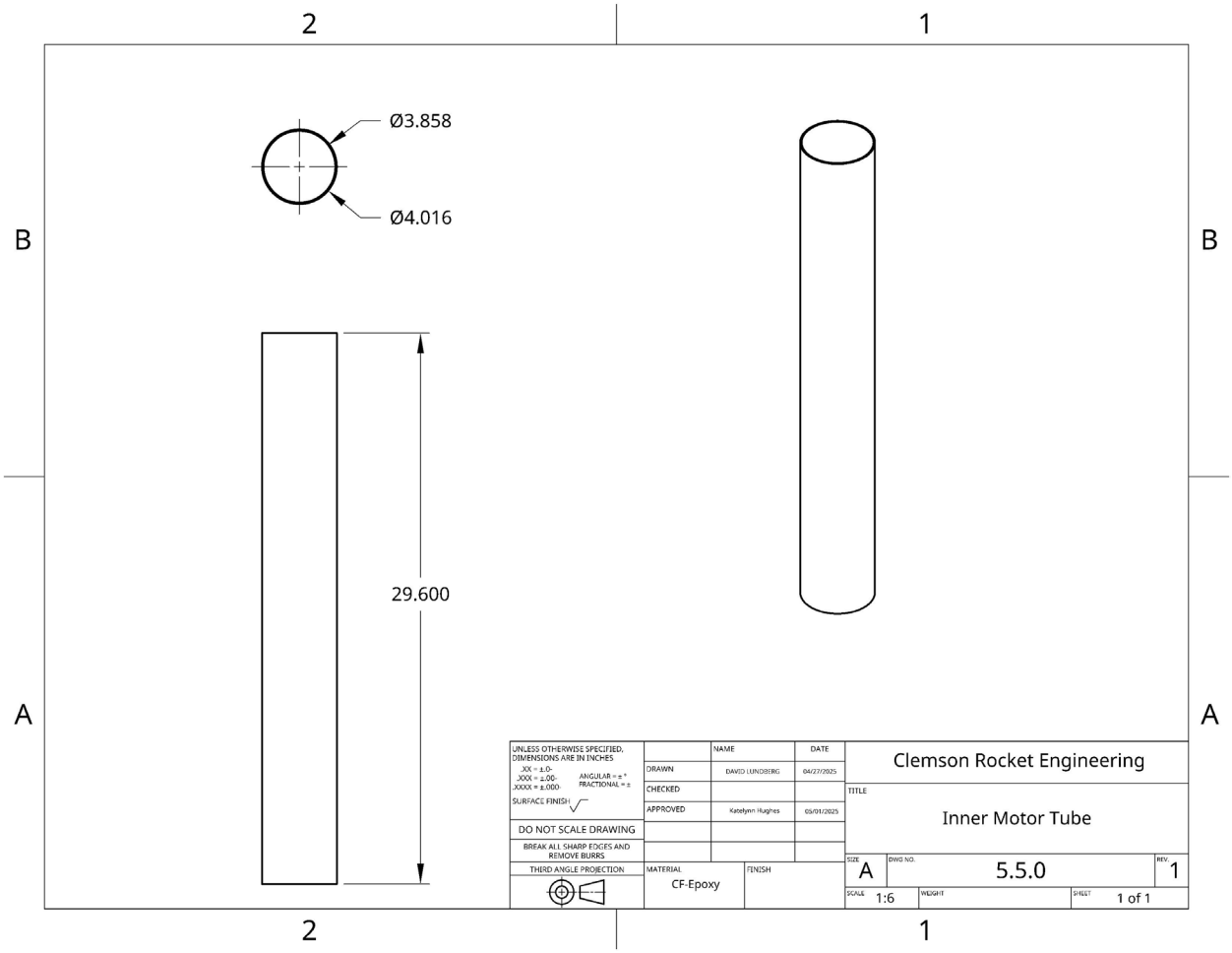

2
1
B
A
Ø.389
Ø.625
.361
.361
Ø.566
1.000
.438
.063
.035
.050
.175
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/20/2026
CHECKED
APPROVED
MATERIAL
Steel
FINISH
Clemson Rocket Engineering
TITLE
Black Oxide Steel Male-Female Hex Thread Adapter
SIZE
A
DWG NO.
4.4.0
REV
1
SCALE
2:1
WEIGHT
SHEET
1 of 1

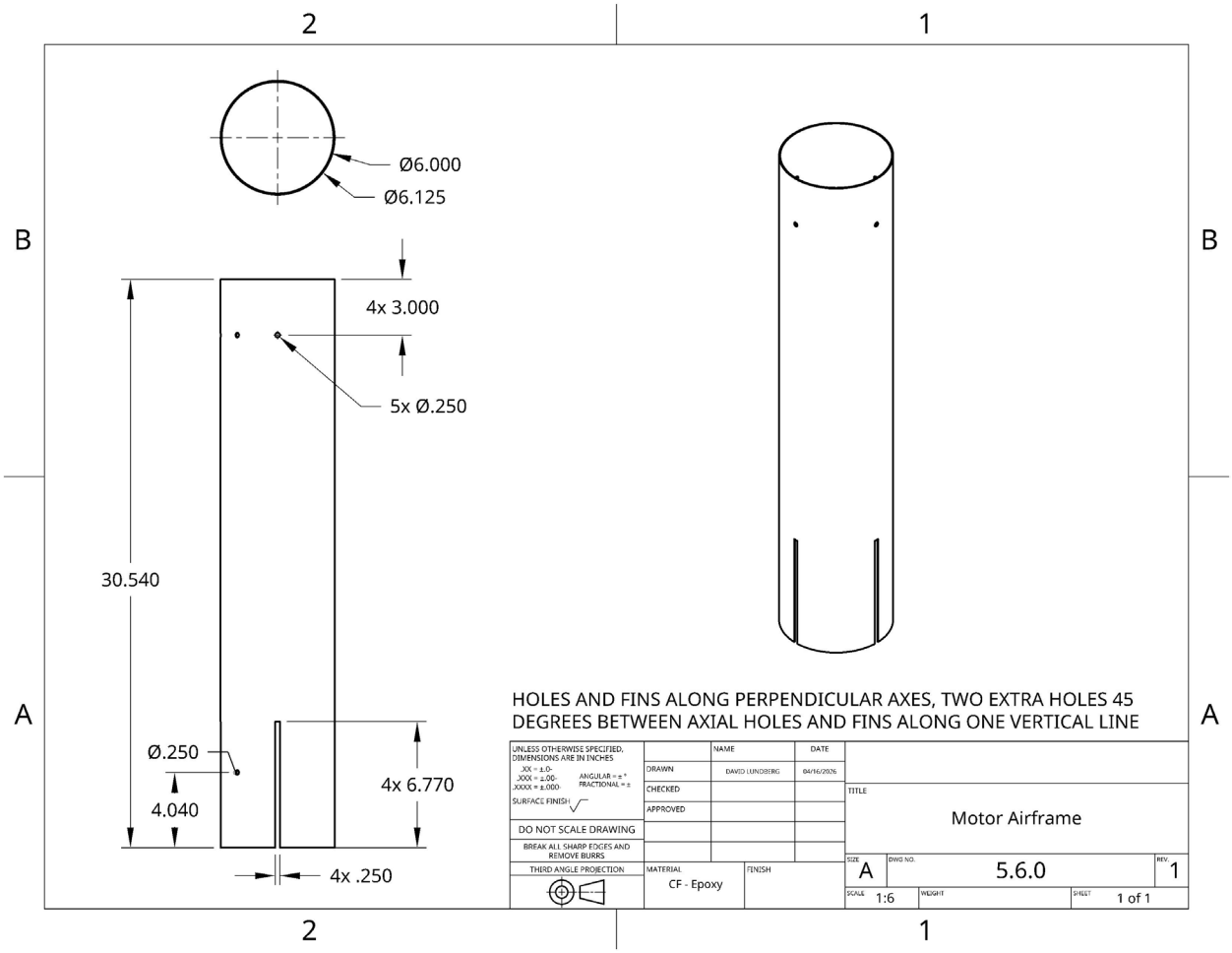

2
1
B
A
11.9°
A
1.800
1.800
R.193
R2.868
Ø.280
.325
.385
.594
DETAIL A
SCALE 1.25:1
6.000
A
1.000
1.000
5.000
82.5°
A
4.683
4.634
45.0°
4.134
45.0°
4.334
SECTION A - A
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
04/13/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
TITLE
Boat Tail
SIZE
A
DWG NO.
5.1.0
REV
1
SCALE
1:3
WEIGHT
SHEET
1 of 1

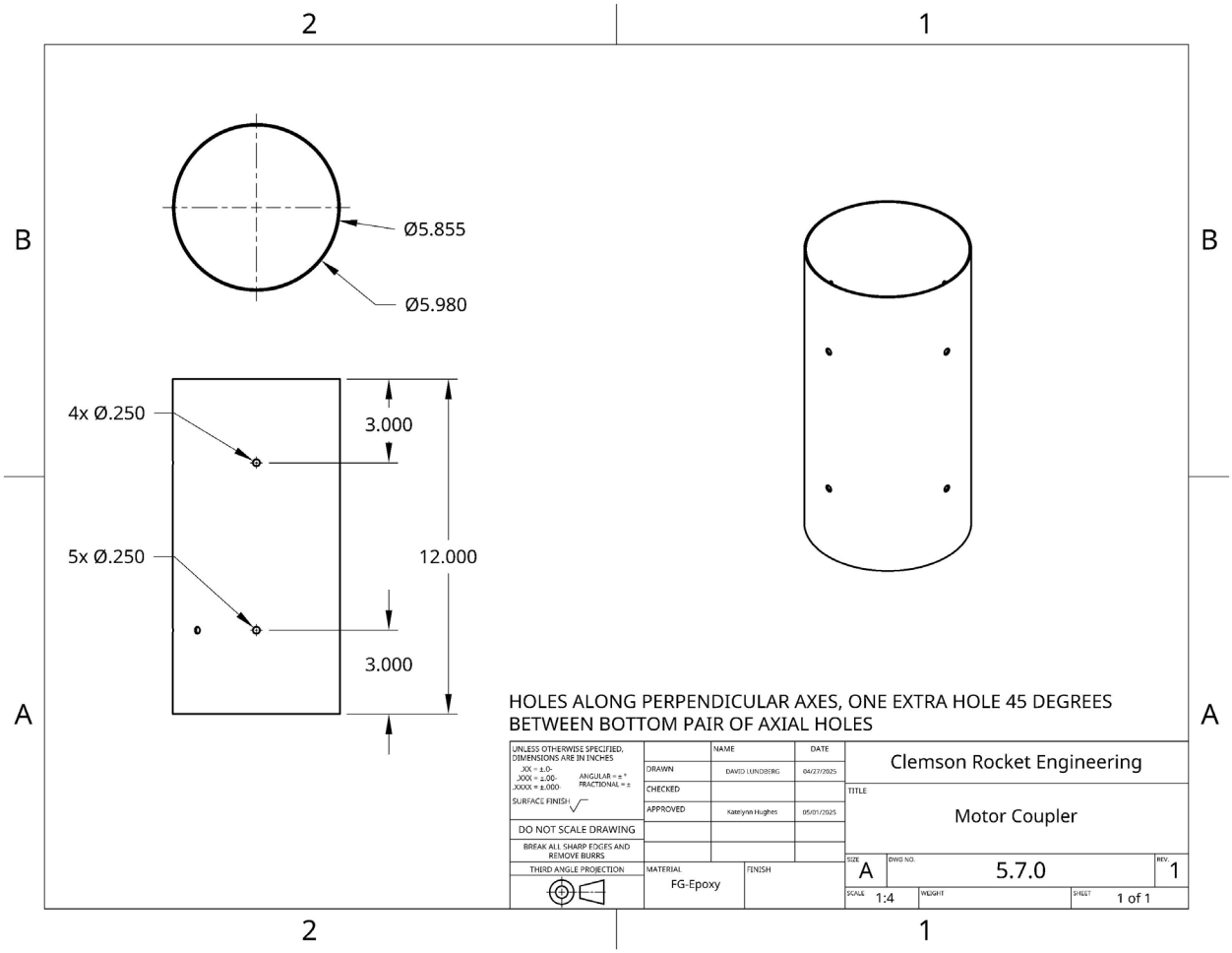

2
1
B
A
Ø1.325
Ø1.900
Ø3.975
1.125
.475
30.743
.703
Ø2.000
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
04/27/2025
CHECKED
APPROVED
Katelynn Hughes
05/01/2025
MATERIAL
N/A
FINISH
Clemson Rocket Engineering
TITLE
Motor
SIZE
A
DWG NO.
5.2.0
REV
1
SCALE
1:6
WEIGHT
SHEET
1 of 1

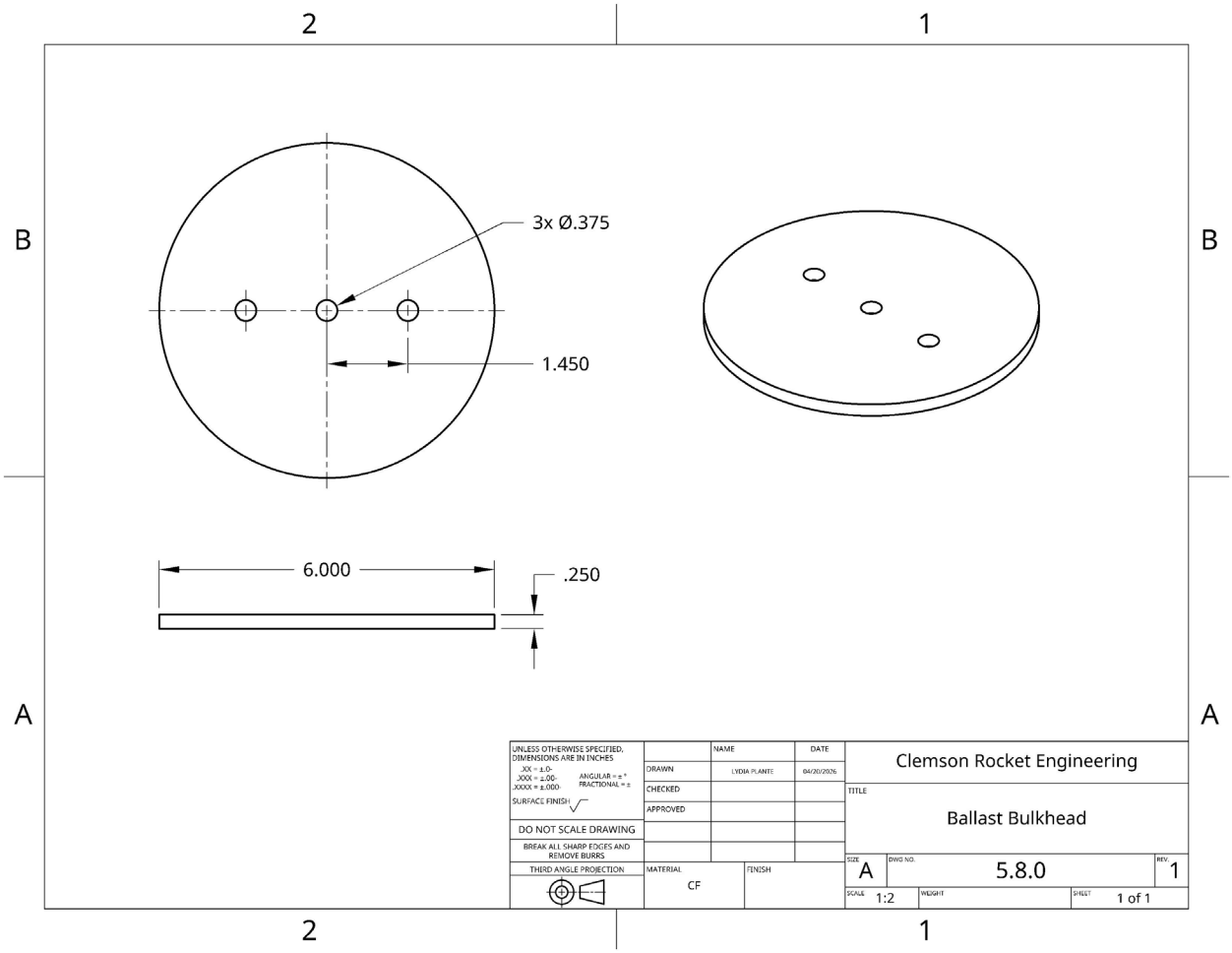

2
1
B
A
4x Ø.316
Ø4.054
1.800
1.800
.250
Ø6.000
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JOSHUA MCCARTY
04/19/2025
CHECKED
APPROVED
Katelynn Hughes
05/01/2025
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Bottom Motor Bulkhead
SIZE
A
DWG NO.
5.3.0
REV.
1
SCALE
1:2
WEIGHT
SHEET
1 of 1

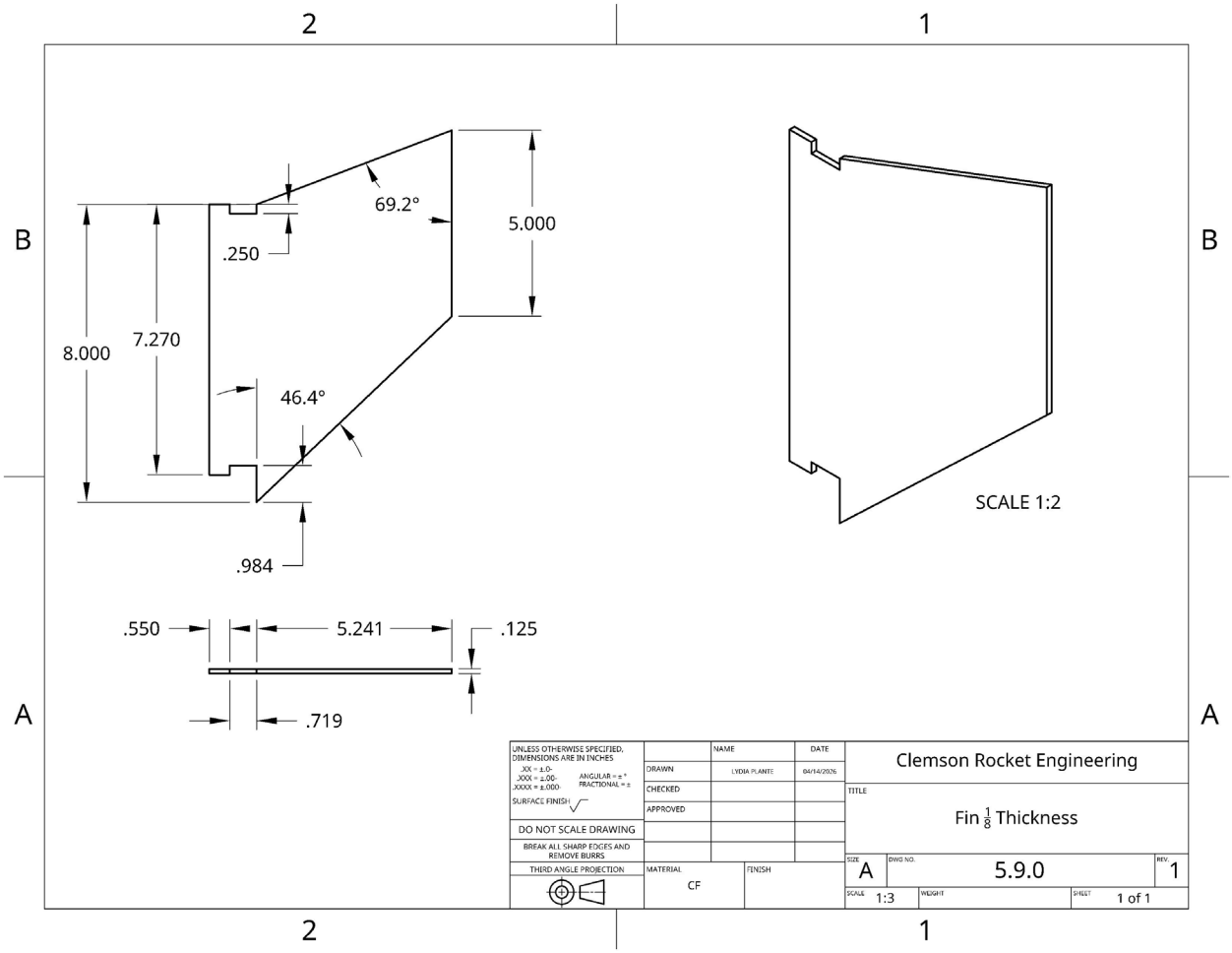

2
1
B
A
Ø4.055
.135
2.578
.551
6.000
.250
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
4/16/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Motor Centering Ring
SIZE
A
DWG NO.
5.4.0
REV
2
SCALE
1:2
WEIGHT
SHEET
1 of 1

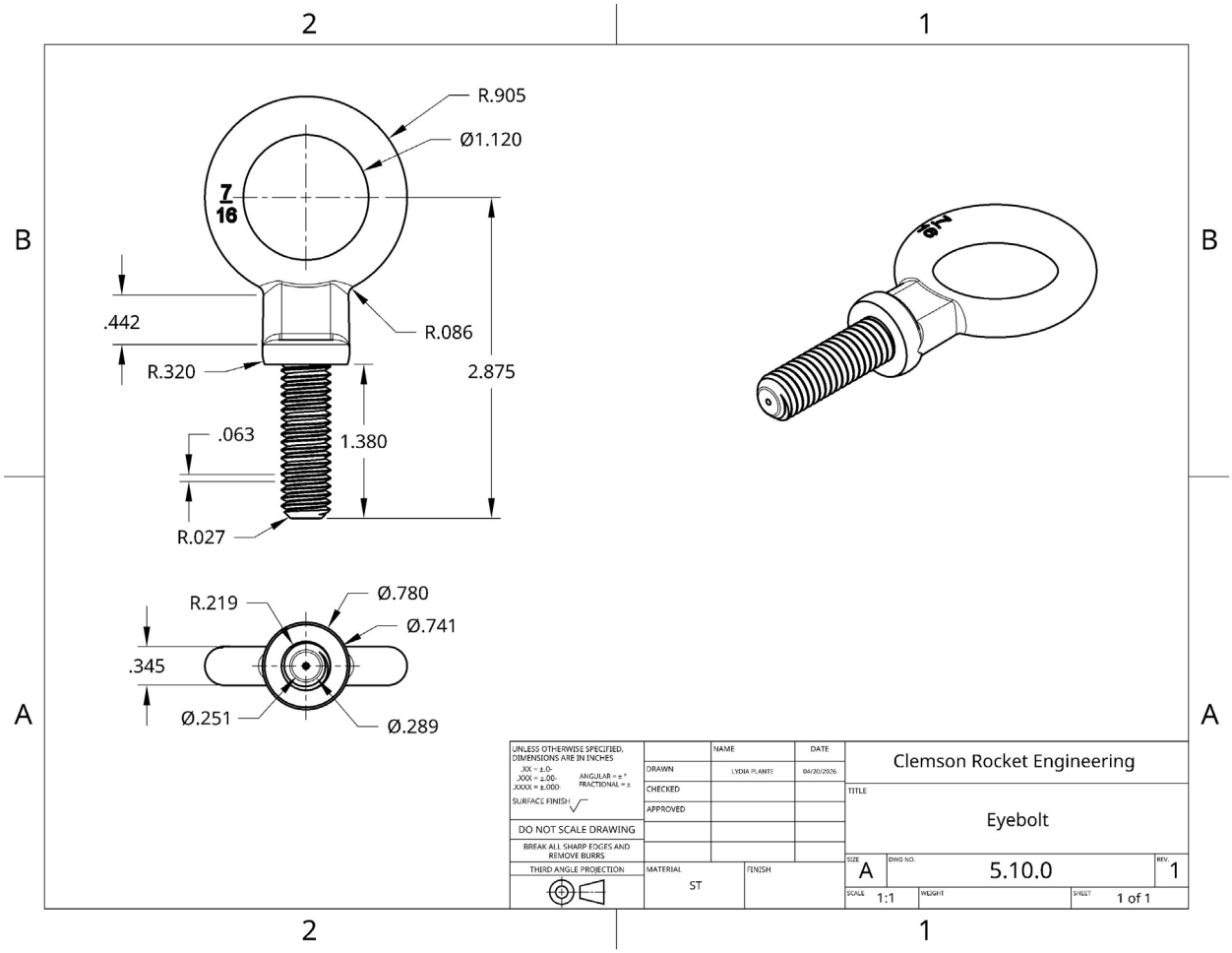

2
1
B
A
Ø3.858
Ø4.016
29.600
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
04/27/2025
CHECKED
APPROVED
Katelynn Hughes
05/01/2025
MATERIAL
CF-Epoxy
FINISH
Clemson Rocket Engineering
TITLE
Inner Motor Tube
SIZE
A
DWG NO.
5.5.0
REV
1
SCALE
1:6
WEIGHT
SHEET
1 of 1

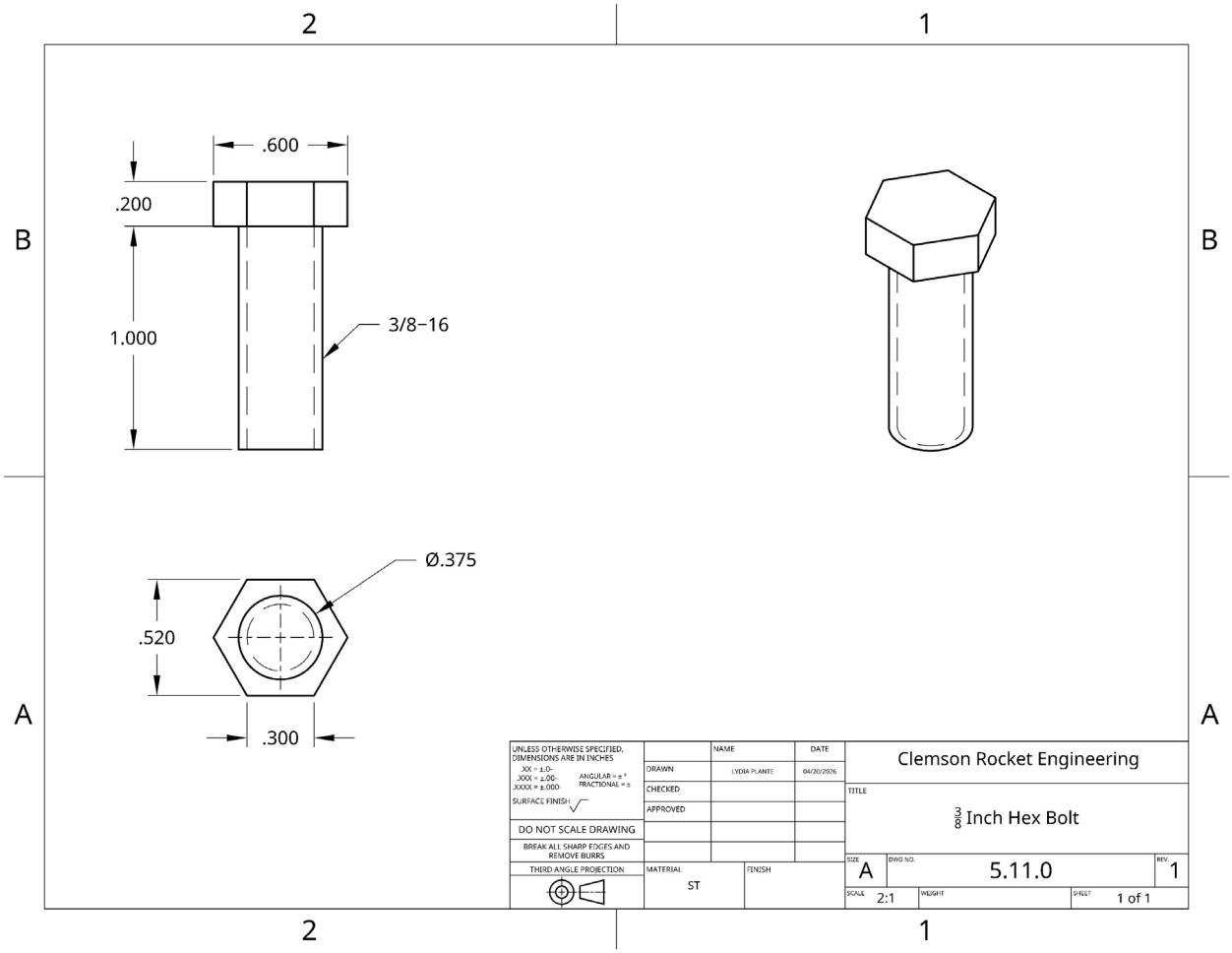

2
1
B
A
Ø6.000
Ø6.125
4x 3.000
5x Ø.250
30.540
Ø.250
4.040
4x 6.770
4x .250
HOLES AND FINS ALONG PERPENDICULAR AXES, TWO EXTRA HOLES 45 DEGREES BETWEEN AXIAL HOLES AND FINS ALONG ONE VERTICAL LINE
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
04/16/2026
CHECKED
APPROVED
MATERIAL
CF - Epoxy
FINISH
TITLE
Motor Airframe
SIZE
A
DWG NO.
5.6.0
REV
1
SCALE
1:6
WEIGHT
SHEET
1 of 1

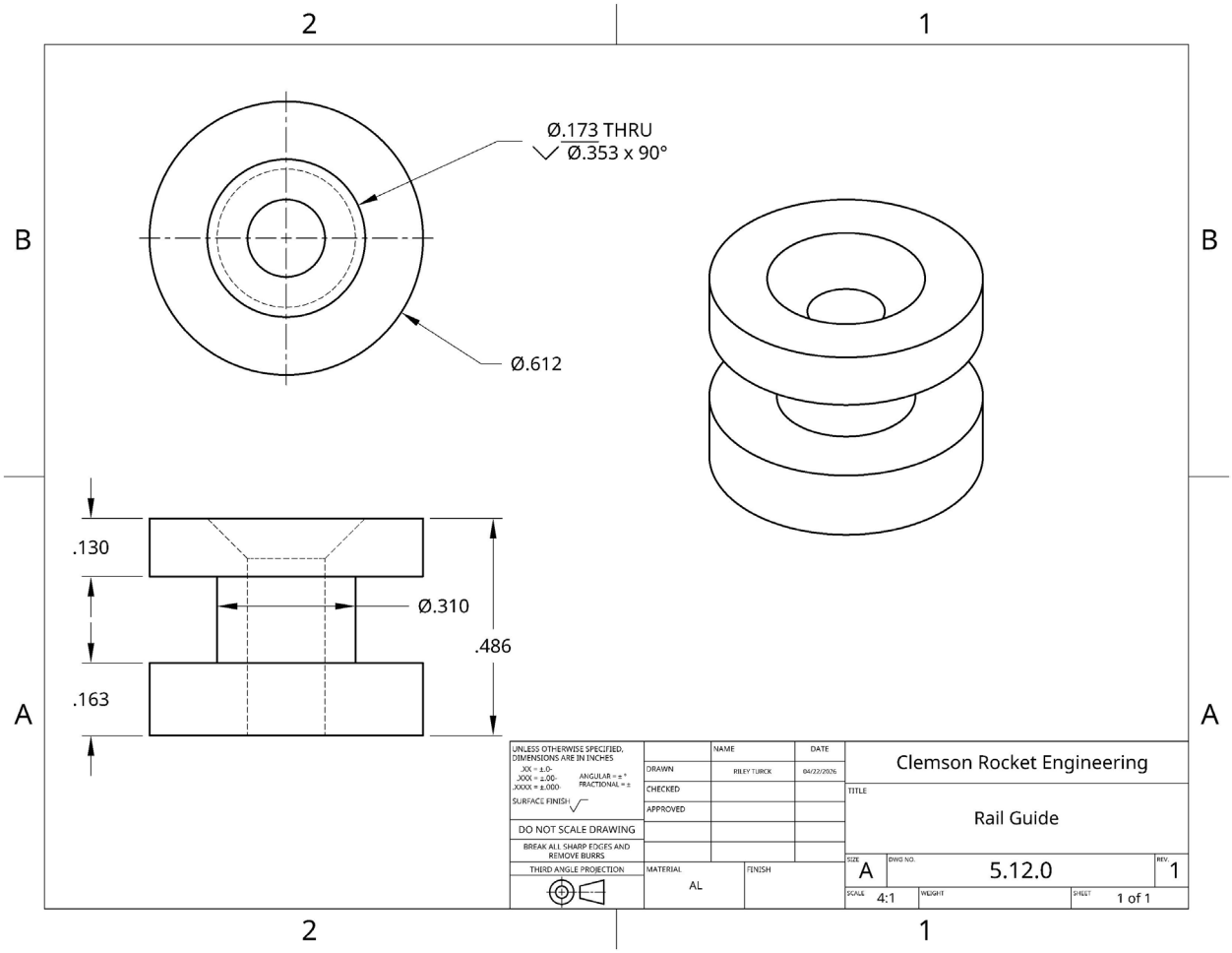

2
1
B
A
Ø5.855
Ø5.980
4x Ø.250
5x Ø.250
3.000
12.000
3.000
HOLES ALONG PERPENDICULAR AXES, ONE EXTRA HOLE 45 DEGREES BETWEEN BOTTOM PAIR OF AXIAL HOLES
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
DAVID LUNDBERG
04/27/2025
CHECKED
APPROVED
Katelynn Hughes
05/01/2025
MATERIAL
FG-Epoxy
FINISH
Clemson Rocket Engineering
TITLE
Motor Coupler
SIZE
A
DWG NO.
5.7.0
REV
1
SCALE
1:4
WEIGHT
SHEET
1 of 1

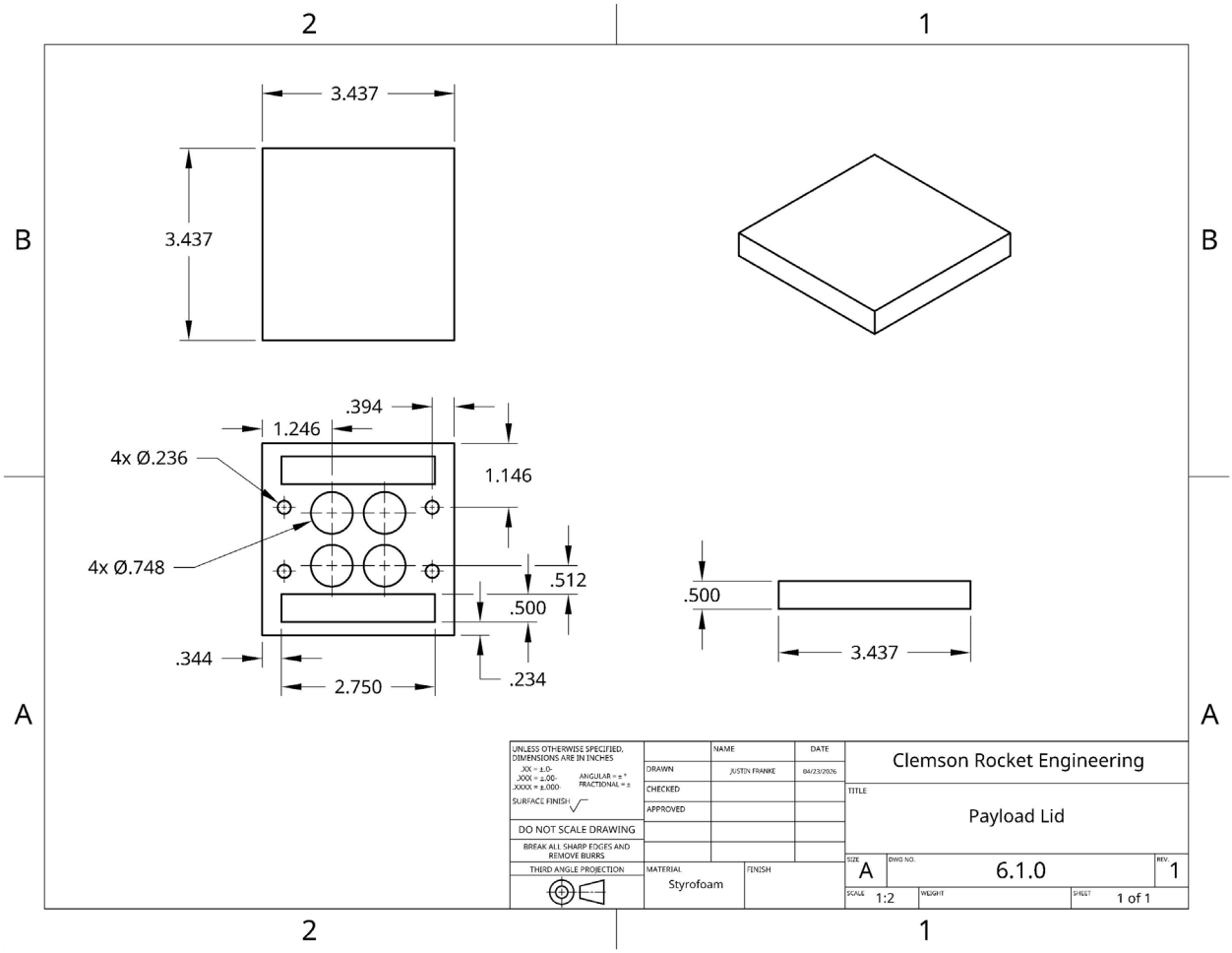

2
1
B
A
3x Ø.375
1.450
6.000
.250
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/20/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Ballast Bulkhead
SIZE
A
DWG NO.
5.8.0
REV
1
SCALE
1:2
WEIGHT
SHEET
1 of 1

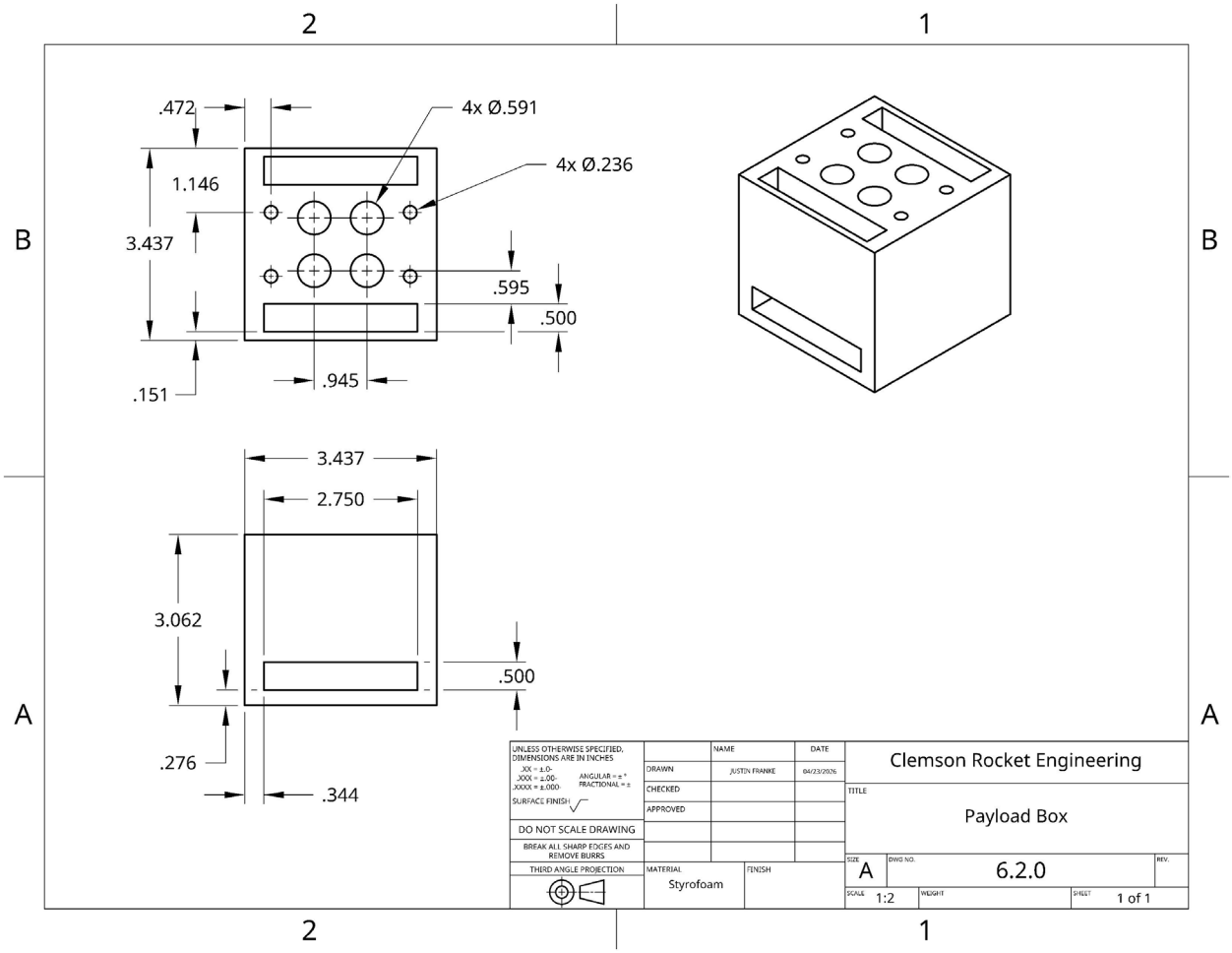

2
1
B
A
69.2°
5.000
.250
8.000
7.270
46.4°
.984
.550
5.241
.125
.719
SCALE 1:2
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/14/2026
CHECKED
APPROVED
MATERIAL
CF
FINISH
Clemson Rocket Engineering
TITLE
Fin 1/8 Thickness
SIZE
A
DWG NO.
5.9.0
REV
1
SCALE
1:3
WEIGHT
SHEET
1 of 1


2
1
B
A
R.905
Ø1.120
7/16
.442
R.086
R.320
2.875
.063
1.380
R.027
R.219
Ø.780
Ø.741
.345
Ø.251
Ø.289
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/20/2026
CHECKED
APPROVED
MATERIAL
ST
FINISH
Clemson Rocket Engineering
TITLE
Eyebolt
SIZE
A
DWG NO.
5.10.0
REV
1
SCALE
1:1
WEIGHT
SHEET
1 of 1


2
1
B
A
.600
.200
1.000
3/8-16
Ø.375
.520
.300
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
LYDIA PLANTE
04/20/2026
CHECKED
APPROVED
MATERIAL
ST
FINISH
Clemson Rocket Engineering
TITLE
DWG
Inch Hex Bolt
SIZE
A
DWG NO.
5.11.0
REV
1
SCALE
2:1
WEIGHT
SHEET
1 of 1


2
1
B
A
Ø.173 THRU
Ø.353 x 90°
Ø.612
.130
Ø.310
.486
.163
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
RILEY TURCK
04/22/2026
CHECKED
APPROVED
MATERIAL
AL
FINISH
Clemson Rocket Engineering
TITLE
Rail Guide
SIZE
A
DWG NO.
5.12.0
REV
1
SCALE
4:1
WEIGHT
SHEET
1 of 1


2
1
B
A
3.437
3.437
.394
1.246
4x Ø.236
4x Ø.748
1.146
.512
.500
.344
2.750
.234
.500
3.437
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/23/2026
CHECKED
APPROVED
MATERIAL
Styrofoam
FINISH
Clemson Rocket Engineering
TITLE
Payload Lid
SIZE
A
DWG NO.
6.1.0
REV
1
SCALE
1:2
WEIGHT
SHEET
1 of 1


2
1
B
A
.472
4x Ø.591
4x Ø.236
1.146
3.437
.595
.500
.151
.945
3.437
2.750
3.062
.500
.276
.344
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES
.XX = ±.0-
.XXX = ±.00-
.XXXX = ±.000-
ANGULAR = ± °
FRACTIONAL = ±
SURFACE FINISH
DO NOT SCALE DRAWING
BREAK ALL SHARP EDGES AND REMOVE BURRS
THIRD ANGLE PROJECTION
NAME
DATE
DRAWN
JUSTIN FRANKE
04/23/2026
CHECKED
APPROVED
MATERIAL
Styrofoam
FINISH
Clemson Rocket Engineering
TITLE
Payload Box
SIZE
A
DWG NO.
6.2.0
REV.
SCALE
1:2
WEIGHT
SHEET
1 of 1